Exchange Dynamics and Coefficient Inequalities

A Consolidated Research Manuscript

Eight studies with complete proofs and executable certificates

KEIO

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Three parts cover nonlinear exchange dynamics, partition-related coefficient inequalities, and partial results on involution log concavity. This edition integrates the full contents of eight source manuscripts; it is not a summary or a concatenation of their PDF files.

Abstract

This manuscript brings together eight studies of nonlinear exchange dynamics and coefficient inequalities. The exchange chapters establish quantitative convergence and entropy estimates in their stated parameter regions, and separately exhibit a tail obstruction to an infinite-state exponential-convergence conjecture. The partition chapters present complete manuscript claims concerning reciprocal-subsum denominator log concavity, unimodality of shifted Nekrasov–Okounkov polynomials, and uniform log concavity of regular partition numbers. Their analytic proofs, finite coverage arguments and executable certificates are retained in full. The involution chapter proves infinite edge strips and eventual fixed linear-window log concavity, while retaining the unresolved status of the full Dukes conjecture. The eight studies have related probabilistic, analytic and coefficient methods, but they do not constitute one overarching theorem. A common structure, uniquely namespaced references, a consolidated bibliography and an exact recoverable source archive make their results and qualifications available in one standalone document.

Preface and Scope

How to read this edition

The chapters preserve all theorem statements, proofs, substantive discussions, limitations, numerical certificate specifications and code listings in the eight source manuscripts. Each original abstract appears as a chapter overview. Repeated title pages and local tables of contents are replaced by common front matter, while the source bibliographies are gathered at the end with reference labels of the form chapter.entry. Repeated works retain their source descriptions. Mathematical notation remains local to its chapter: identical symbols in different chapters need not denote identical objects. Section, equation and theorem references are made unique for this edition. Appendices remain attached to their originating chapters and are numbered with that chapter prefix.

The organization does not claim a new theorem connecting all eight subjects. Part I concerns exchange dynamics, Part II concerns partition-related coefficient inequalities, and Part III contains partial involution results. The commonality is methodological and editorial, not a claim that one proof settles all the questions treated here.

Status of the results

Chapters 1–3 are quantitative convergence studies with different hypotheses and different levels of explicitness. Chapter 1 retains a nonexplicit discrete-time prefactor and separates continuous-time entropy decay from discrete-time convergence. Chapter 2 fixes a closed marginal or density region strictly below $1/(\Delta + 1)$; it does not assert uniformity up to that endpoint. Chapter 3 gives an explicit entropy contraction under its stated local condition, including the uniform range $\beta \leq 1/(2\Delta + 1)$, and does not settle the remaining conjectured region.

Chapter 4 is an explicit application of a known slow-tail mechanism, not a claim to have discovered that mechanism. Its obstruction targets a concluding conjecture under broad hypotheses, not the stronger proved exponential-decay theorem discussed in the chapter. Chapters 5–7 retain the original manuscripts' full claims in their stated scopes, with computer-assisted finite components. Chapter 5 classifies reciprocal-subsum denominator log concavity; Chapter 6 claims unimodality, not global coefficient log concavity; Chapter 7 includes the sharp $k \geq 30$, $n \geq 26$ regular-partition assertion and its stated extensions. Chapter 8 remains explicitly partial: the full all-n, all-k Dukes conjecture is unresolved, the linear-window threshold is existential, and the Bernstein and four-step candidates are not promoted to proved statements.

These are the claims and qualifications of the source research manuscripts, not an assertion of independent peer review, publication priority or formal proof-assistant verification. Finite certificate execution checks only the specified computational components. Infinite conclusions still depend on the analytic and algebraic arguments and the exact coverage of finite and infinite regions given in the chapters. This consolidation performs no new proof search and does not enlarge any theorem's domain.

Source and reproducibility policy

All eight manuscripts are attributed to KEIO. The source manifest below identifies the exact input files by SHA-256. The consolidated `.tex` file is sufficient to typeset the complete manuscript: it requires no external input files, bibliography databases, images or PDFs. The original code payloads are unchanged. Their source-specific extraction markers and original filenames are retained in an exact archive after the document terminator. The recovery instructions below restore those originals, so the chapter-specific replay drivers can be used without modifying their assumptions or mixing similarly named files across studies.

Source manifest

The numbers in this manifest are the chapter numbers. SHA-256 values identify the complete original TEX source bytes, not the generated PDF.

1. **Uniform Convergence and Entropy Bounds for Nonlinear Exchange Dynamics**
   `uniform_nonlinear_exchange.tex`
   `3bfe3f99f3637857fd8f93d2e72fd3cd1044e4ccc9e7158c4dcee3f9b21fb334`

2. **Uniform Exponential Convergence for Nonlinear Independent Set Exchange**
   `hard_core_uniform_convergence.tex`
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3. **Fast Entropy Contraction for Nonlinear Independent Set Exchange**
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4. **A Tail Obstruction to an Exponential Convergence Conjecture**
   `exchange_tail_counterexample.tex`
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5. **Log Concavity of Reciprocal Subsum Denominators**
   `KEIO_BBFM_Denominator_Log_Concavity_2026.tex`
   `429a43c5fb3830e7ef85f1eddd90e233934b3ec405b59d1b0b71d0dd05c158aa`

6. **Unimodality of the Shifted Nekrasov Okounkov Polynomials**
   `KEIO_Shifted_Nekrasov_Okounkov_Unimodality_2026.tex`
   `9d56524e20bc44f42f48fa170fa0f041227ba68ca66e73a36696c60199b27e87`

7. **Uniform Log Concavity of Regular Partition Numbers**
   `KEIO_Regular_Partition_Log_Concavity_2026.tex`
   `155421c460f285432f7d208ca00aaeb557e854f4efa5ecc0096c7eed934d5938`

8. **Partial Results on Dukes Inversion Parity Log Concavity**
   `KEIO_Dukes_Partial_Results_2026.tex`
   `1560709b0d12ba6991fa9cd62e51187e66521906600da167837ae399a2400ddb`

Compiling and recovering the exact sources

Typeset this file with ordinary `pdflatex`; run it three times to settle the contents and cross-references. Compilation does not execute any verification code. To restore the exact source files, save the following listing as `recover_sources.py` and run it in a writable directory:

```
python3 recover_sources.py KEIO_Collected_Research_01_08_2026.tex originals
```

The destination must not already exist. The recovery program validates every source hash before writing files. Then use each original manuscript's replay instructions in its own fresh working directory. In particular, do not run a chapter-specific extractor against the complete consolidated file: the exact originals are the intended inputs to those unchanged drivers.

```
#!/usr/bin/env python3
"""Recover the exact eight source manuscripts from the consolidated TEX.
Usage: python3 recover_sources.py CONSOLIDATED.tex NEW_DIRECTORY
"""
import base64
import hashlib
import json
from pathlib import Path
import re
import sys

if len(sys.argv) != 3:
    raise SystemExit(__doc__)
text = Path(sys.argv[1]).read_text(encoding="utf-8")
pattern = (r"^% BEGIN ORIGINAL SOURCE (\d{2}) (\S+) "
           r"([0-9a-f]{64})\n(.*?)"
           r"^% END ORIGINAL SOURCE \1$")
blocks = re.findall(pattern, text, flags=re.M | re.S)
if [int(b[0]) for b in blocks] != list(range(1, 9)):
    raise SystemExit("Expected exactly eight ordered source blocks")
records = []
decoded = []
for number, name, expected, payload in blocks:
    if Path(name).name != name:
        raise SystemExit("Unsafe source filename")
    lines = payload.splitlines()
    if not lines or any(not line.startswith("% ") for line in lines):
        raise SystemExit("Malformed base64 archive")
    raw = base64.b64decode("".join(line[2:] for line in lines),
                           validate=True)
    digest = hashlib.sha256(raw).hexdigest()
    if digest != expected:
        raise SystemExit("SHA-256 mismatch: " + name)
    decoded.append((name, raw))
    records.append(dict(chapter=int(number), filename=name,
                        bytes=len(raw), sha256=digest))
dest = Path(sys.argv[2])
dest.mkdir(parents=True, exist_ok=False)
for name, raw in decoded:
    (dest / name).write_bytes(raw)
(dest / "manifest.json").write_text(
    json.dumps(records, indent=2) + "\n", encoding="utf-8")
print("PASS: recovered eight exact sources in", dest)
```

Chapters 1–4 contain analytic proofs, not computational certificates. Chapters 5–8 print their executable sources and specify requirements, replay modes and expected outputs. The separate verification bundle is not required to compile or recover this standalone source.

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Part I

Nonlinear Exchange Dynamics

The first three chapters develop quantitative convergence statements for nonlinear exchange maps. The fourth gives a tail obstruction for a different, infinite-state exchange model. The distinctions between discrete and continuous time, explicit rates and nonexplicit prefactors, and finite and infinite state spaces are essential; the obstruction does not contradict the preceding theorems.

Chapter 1

Uniform Convergence and Entropy Bounds for Nonlinear Exchange Dynamics

Chapter overview

For each fixed finite interaction matrix, we establish uniform exponential convergence in total variation for the discrete-time nonlinear Glauber and block dynamics of Caputo–Sinclair. The constants are uniform over every initial law, including laws whose one-site marginals approach or lie on the boundary. A bounded-perturbation argument supplies explicit asymptotic exponents; a separate entropy reweighting argument supplies global continuous-time entropy decay. We rederive their qualitative convergence statement by minorization and entropy monotonicity. The proof uses spectral stability of the linearization on the fixed-marginal tangent space at every equilibrium, including boundary equilibria. Compactness then gives a common local contraction and a uniform entrance time. The discrete-time prefactor remains nonquantitative. The argument is self-contained apart from the cited zero-interaction entropy-production inequalities; its relationship to earlier work is discussed explicitly.

1.1 Background and scope

The dynamics below are exactly the discrete-time maps (2.6) and (2.9) of Caputo–Sinclair [1.1]. Their Theorem 2.7 establishes qualitative convergence for arbitrary finite interactions; Remark 2.11 covers pinned spins. Section 5, question 1 asks for exponential decay for arbitrary interactions, even with a very bad dependence on dimension. The result below proves existence of constants depending only on the fixed interaction matrix. The quantitative section supplies an explicit lower bound on the asymptotic decay exponent, but not an explicit discrete-time prefactor or mixing-time bound.

A closely related 2025 question of Caputo–Morellini asks for uniform covariance bounds for constrained replicas, as a route to continuous-time entropy decay. A 13 October 2026 GitHub manuscript by Kaiwen Jin [1.4] claims that covariance bound; the consulted registry labels it an unverified full-resolution claim. The covariance statement, combined with the Kac-program and linearization arguments of [1.3, 1.2], supplies a prior route to the present qualitative all-interaction conclusion. The present working paper gives an independent proof and explicit perturbation bounds; it does not claim priority for the qualitative conclusion. The covariance preprint does not itself derive entropy-decay estimates.

1.2 Definitions and statement

Fix $n \geq 1$, $\Omega = \{-1, 1\}^n$, and a finite real symmetric matrix J. Put

$$w(x) = \exp\{\tfrac{1}{2} x^{\mathsf{T}} J x\} > 0.$$

For $A \subseteq [n]$, let $S_A(x, y) = (y_A x_{A^c}, x_A y_{A^c})$. We use either of the following Markov kernels on ordered pairs:

1. The block kernel selects A with probability proportional to $w((S_A(x, y))_1) w((S_A(x, y))_2)$ and outputs $S_A(x, y)$.

2. The single-site kernel chooses i uniformly and replaces (x, y) by $S_{\{i\}}(x, y)$ with probability

$$a_i(x, y) = \frac{w((S_i(x, y))_1)w((S_i(x, y))_2)}{w(x)w(y) + w((S_i(x, y))_1)w((S_i(x, y))_2)};$$

otherwise it leaves the pair unchanged.

Let Q be the selected pair kernel and $Q_1(z \mid x, y)$ its first-output marginal. Define the polynomial map

$$T(p)(z) = \sum_{x,y} p(x)p(y)Q_1(z \mid x, y).$$

The kernel commutes with exchange of the two replicas. Consequently the two output marginals agree when the input law is $p \otimes p$. Every one-site marginal of p is conserved by T.

Write $m(p) = (\sum_x p(x)x_i)_{i=1}^n$. For $m \in [-1, 1]^n$, denote by μ_m the unique maximizer, among probabilities of mean vector m, of

$$\mathcal{F}(p) = H(p) + \sum_x p(x) \log w(x).$$

On the subcube allowed by the pinned coordinates $m_i = \pm 1$, this maximizer has full support and the Gibbs form $\mu_m(x) \propto w(x)e^{h\cdot x}$ with finite fields on the unpinned coordinates. It is stationary for T.

Theorem 1.1. *For either kernel above, there exist finite constants $C = C(J) \geq 1$ and $c = c(J) > 0$ such that, simultaneously for all $p \in \mathcal{P}(\Omega)$ and all integers $t \geq 0$,*

$$\|T^t(p) - \mu_{m(p)}\|_{\mathrm{TV}} \leq Ce^{-ct}.$$

In particular, the constants do not depend on the initial marginals or on their distance to deterministic marginals.

1.3 Continuity and the fixed tangent space

Lemma 1.2. *The map $m \mapsto \mu_m$ is continuous on the closed cube. The set $\mathcal{M} = \{\mu_m : m \in [-1, 1]^n\}$ is compact. For every p,*

$$T^t(p) \longrightarrow \mu_{m(p)}.$$

Proof. Existence follows from compactness, and uniqueness from strict concavity of entropy. On the allowed subcube a full-support feasible law exists (the product law). Mixing any putative maximizer with that law shows, by the infinite positive one-sided entropy derivative at a zero cell, that the maximizer is positive. Lagrange multipliers on that subcube give its Gibbs form.

For continuity, let $m^{(k)} \to m$. Any subsequential limit ν of $\mu_{m^{(k)}}$ has mean m. For any law p of mean m, there exist $p^{(k)} \to p$ with means $m^{(k)}$: independently on each coordinate, flip the required fraction of -1 spins to $+1$ or vice versa to correct its marginal. A correction at coordinate i changes total variation by at most $|m_i^{(k)} - m_i|/2$ and preserves every other marginal. Thus all corrections change total variation by at most half the sum of these differences. Optimality and continuity of $\mathcal{F}$ give $\mathcal{F}(\nu) \geq \mathcal{F}(p)$. Uniqueness implies $\nu = \mu_m$, proving continuity. Compactness follows. The last assertion is established in [1.1, Theorem 2.7 and Remark 2.11]. For a self-contained argument, we prove it again below using the same classical positivity and entropy principles. □

A self-contained proof of qualitative convergence. Let Q^0 be the pair kernel that chooses one coordinate uniformly and exchanges its two spins with probability $1/2$. The positive transitions of Q^0 are the identity and the one-site exchanges. Every such transition has positive probability for each of our two kernels. Since the pair space is finite, there exists $a = a(J) > 0$, with $a \le 1$, such that $Q \ge aQ^0$ entrywise. This comparison is made with the original finite weight w on the full cube, so a does not depend on the equilibrium fields or marginals.

Fix the initial one-site marginals and write $\pi = \bigotimes_i p_i$. On this marginal fiber the nonlinear map for Q^0 is the positive linear map

$$L_\pi q = \tfrac{1}{2} q + \frac{1}{2n} \sum_{i=1}^{n} \pi_i \otimes q_{-i}.$$

Consequently $T(q) \ge aL_\pi q$ for every probability q in the fiber. Conservation of marginals and positivity of L_π imply inductively

$$T^k(p) \ge a^k L_\pi^k p.$$

In n steps of L_π, the term that refreshes coordinates $1, 2, \dots, n$ in that order is exactly $(2n)^{-n}\pi$. Thus

$$T^t(p) \ge b\pi \quad (t \ge n), \qquad b = a^n (2n)^{-n} > 0. \tag{1.3.1}$$

For $t > n$, apply the same estimate to $T^{t-n}(p)$, which has the same marginals. On the allowed subcube S, π is strictly positive; therefore the trajectory after time n lies in a compact subset of the strictly positive laws on S.

Write $\mu = \mu_{m(p)}$, which is positive on S, and let D denote relative entropy. For a positive law q on S, pair reversibility and the log-sum inequality give

$$D((q \otimes q)Q \| \mu \otimes \mu) \le 2D(q\|\mu).$$

The two output marginals are both $T(q)$, so entropy subadditivity also gives

$$2D(T(q)\|\mu) \le D((q \otimes q)Q \| \mu \otimes \mu).$$

The first inequality is strict unless $q(x)q(y)/(\mu(x)\mu(y))$ is constant on every communicating class of Q. To see the equality condition directly, apply strict convexity of $u \log u$ to the reversed transition probabilities at each output state. The positive diagonal forces the density to agree across every positive transition, and hence throughout every class. Since all one-site swaps are positive transitions, equality implies that $h = \log(q/\mu)$ is additive, by the coordinate-difference argument in Lemma 1.3. Therefore q is a Gibbs law with the same weight and marginals as μ, and uniqueness of the free-energy maximizer gives $q = \mu$.

It follows that $D(T(q)\|\mu) < D(q\|\mu)$ unless $q = \mu$. The entropy along the trajectory decreases to a limit. If a subsequence $T^{t_j}(p)$ converges to q, then (1.3.1) makes q positive on S. Continuity implies $D(T(q)\|\mu) = D(q\|\mu)$, because both the time-t_j and time-(t_j+1) entropies have the same limit. Hence $q = \mu$. Compactness shows that the entire trajectory converges to μ. □

Let the fixed vector space

$$E = \left\{ v \in \mathbb{R}^\Omega : \sum_x v(x) = 0, \quad \sum_x v(x) x_i = 0 \ (1 \le i \le n) \right\}$$

be the tangent space of each affine fixed-marginal fiber. For $\mu \in \mathcal{M}$, write $A_\mu = DT(\mu)$. Explicitly,

$$(A_\mu v)(z) = \sum_{x,y} \{ v(x)\mu(y) + \mu(x)v(y) \} Q_1(z \mid x, y). \tag{1.3.2}$$

The space E is invariant under A_μ: differentiating mass and marginal conservation along the simplex tangent space proves this, or one can check it directly in (1.3.2).

1.4 Interior spectral stability

The linearization and conserved-mode framework has a direct precursor in [1.2, Section 2.3]; these ingredients are not claimed as new. The following statement also applies to a resident subcube, with the other coordinates pinned. A single-site update at a pinned coordinate is then the identity.

Lemma 1.3. *Suppose μ is a Gibbs equilibrium with full support on a subcube S. On vectors supported on S with zero mass, A_μ is self-adjoint in the density representation $v = \mu h$ and has spectrum in $[0, 1]$. Its eigenspace for eigenvalue 1 consists exactly of the vectors μh for which h is an additive one-site function with $\mu[h] = 0$. Hence on vectors additionally having zero one-site marginals its spectral radius is strictly below 1. On all vectors supported on S, the remaining mass eigenvalue is 2.*

Proof. Work on S, where $\mu > 0$. The pair kernel is reversible for $\mu \otimes \mu$. In the block case it is conditional expectation onto the equivalence classes of coordinatewise pair sums. Each one-site kernel is conditional expectation onto its two-point exchange orbits, and the Glauber kernel is an average of these conditional expectations. Thus in both cases Q is a self-adjoint positive-semidefinite contraction on $L^2(\mu \otimes \mu)$.

Put $Mh(x, y) = h(x) + h(y)$ and let $Lh = (A_\mu(\mu h))/\mu$. Reversibility and replica-exchange symmetry yield

$$\langle h, Lk\rangle_\mu = \tfrac{1}{2}\langle Mh, QMk\rangle_{\mu\otimes\mu}. \tag{1.4.1}$$

Indeed, reversibility expresses $Lk(x)$ as the μ-average in y of $QMk(x, y)$; this function is symmetric in the two replicas, allowing the average of $h(x)$ and $h(y)$. If $\mu[h] = 0$, then $\|Mh\|^2 = 2\|h\|^2$. Formula (1.4.1) proves self-adjointness and the spectral interval $[0, 1]$ on the zero-mean space.

Equality in its upper bound holds precisely when $QMh = Mh$. Since every single-site exchange has positive transition probability, this is equivalent to

$$h(x) + h(y) = h(y_i x_{-i}) + h(x_i y_{-i})$$

for every free i and every $x, y \in S$. Taking $x_i = +1$, $y_i = -1$ shows that $h(+1, u) - h(-1, u)$ is independent of the complementary configuration u. Applying this for each coordinate shows h is additive. Conversely additive h makes Mh collision-invariant and, if centered, gives $Lh = h$.

If $v = \mu h$ has zero mass and all zero one-site marginals, then h is orthogonal to every additive function. It therefore has no eigenvalue-1 component. Finite dimension proves strict spectral stability. Finally, $A_\mu \mu = 2\mu$ by homogeneity, while the zero-mass space is invariant, establishing the final assertion. □

1.5 Boundary spectral stability

Proposition 1.4. *For every $\mu \in \mathcal{M}$, including every boundary equilibrium,*

$$\rho(A_\mu|_E) < 1.$$

Proof. Let R be the pinned coordinates of μ, with pinned values b_i, and let $F = [n] \setminus R$. The support S of μ is the full subcube $x_R = b_R$, and μ is positive on S. For each configuration x, define its mutant set

$$B(x) = \{i \in R : x_i \neq b_i\}.$$

Decompose a vector into blocks v_B according to $B(x) = B \subseteq R$. In the derivative (1.3.2), one parent is a resident drawn from μ and the other has mutant set B. The two offspring partition those mutant sites between them. Consequently each offspring has a mutant set contained in B.

The matrix A_μ is therefore block triangular when subsets are ordered by inclusion (or decreasing cardinality). Write D_B for its diagonal block.

For $B \neq \varnothing$, the column sum of D_B is the expected number of offspring whose mutant set is exactly B when one parent is the specified mutant and the other is an independent resident. This interpretation follows directly from replica symmetry in (1.3.2). This number is at most one.

If $|B| \geq 2$, there is strictly positive probability of exchanging just one site of B. This produces two nonempty proper mutant subsets, so neither offspring retains the full set B. Every input mutant and resident has positive probability for such an exchange, and there are finitely many inputs. Thus all column sums of D_B are strictly below one, and $\rho(D_B) < 1$.

If $B = \{i\}$, exactly one offspring always carries that mutant site, so D_B is column-stochastic. It is in fact strictly positive: for any desired free-coordinate configuration z_F, choose a resident y with $y_F = z_F$, which has positive μ-probability, and exchange only site i. The resulting mutant carrier has free coordinates z_F. This event has positive probability for both kernels. Perron–Frobenius therefore gives a simple eigenvalue 1, a strictly positive eigenvector, and all other eigenvalues strictly inside the unit circle. An eigenvector for eigenvalue 1 necessarily has nonzero total mass.

For $B = \varnothing$, $D_\varnothing$ is the resident linearization of Lemma 1.3: its spectrum is contained in $[0, 1] \cup \{2\}$, its 2-eigenvectors are multiples of μ, and its 1-eigenvectors have centered additive densities.

Suppose now $v \in E$ is a nonzero complex eigenvector of A_μ with $|\lambda| \geq 1$. The triangular equations, starting with maximal mutant sets, give $v_B = 0$ for all $|B| \geq 2$. If $\lambda \neq 1$, they also give $v_B = 0$ for singleton B. If $\lambda = 1$, the constraint that the opposite-to-pinned marginal at site i vanishes is

$$0 = \sum_{B \ni i} \sum_{x : B(x) = B} v_B(x) = \sum_{x : B(x) = \{i\}} v_{\{i\}}(x).$$

The preceding Perron–Frobenius observation again gives $v_{\{i\}} = 0$. Thus in every case v is supported on the resident subcube and has zero mass and zero free-site marginals. Lemma 1.3 contradicts $|\lambda| \geq 1$. Hence the restricted spectral radius is strictly below one. □

1.6 Compactness upgrades spectral stability to uniform exponential decay

Lemma 1.5. *Let K be a compact convex subset of a finite-dimensional affine space, T a polynomial map carrying K into itself, and $r : K \to K$ a continuous retraction onto fixed points. Assume $r(Tp) = r(p)$, $p - r(p) \in E$ for a fixed vector space E, and that the affine fibers $r^{-1}(\mu)$ lie in $\mu + E$. Assume also $T^t(p) \to r(p)$ for every $p \in K$ and $\rho(DT(\mu)|_E) < 1$ at every fixed point $\mu \in r(K)$. Then there are $C < \infty, c > 0$ with*

$$\|T^t(p) - r(p)\| \leq Ce^{-ct}$$

for all $p \in K$ and integers $t \geq 0$, provided E is derivative-invariant along the relevant affine fibers.

Proof. Use a Euclidean norm. The family of restricted derivatives A_μ is compact. Continuity of spectral radius gives $\sup_\mu \rho(A_\mu) < 1$. Their powers converge to zero uniformly: choose a circle of radius $a < 1$ enclosing every spectrum, bound the resolvents $(zI - A_\mu)^{-1}$ uniformly on that circle by compactness, and use the Cauchy integral formula for A_μ^k. Choose an integer s such that $\sup_\mu \|A_\mu^s\| < 1/4$.

Since $T(\mu) = \mu$, $D(T^s)(\mu)|_E = A_\mu^s$. Uniform continuity of the derivative gives $\eta > 0$ such that

$$\|T^s(p) - \mu\| \le \tfrac{1}{2}\|p - \mu\|$$

whenever $p \in K$, $r(p) = \mu$, and $\|p - \mu\| \le \eta$. This follows by integrating the derivative along the segment from μ to p; in the present application K is convex and the segment remains in the fixed affine fiber. More generally convexity of these local segments is sufficient.

Set $U = T^s$. For every p, pointwise convergence gives some integer $k(p)$ with $\|U^{k(p)}p - r(p)\| < \eta/2$. Continuity gives an open neighborhood V_p on which $\|U^{k(p)}q - r(q)\| < \eta$. A finite subcover of K and the local contraction imply, for $k_0 = \max k(p)$ among that subcover,

$$\sup_{q\in K} \|U^{k_0+j}q - r(q)\| \le \eta 2^{-j} \qquad (j \ge 0).$$

Indeed each q enters the local contraction neighborhood at its own selected time no later than k_0, and remains there at all later U-times. Finally, the finitely many maps T^a, $0 \le a < s$, are uniformly Lipschitz on K (or on its compact convex hull). Writing $t = sk + a$ establishes the result with $c = (\log 2)/s$ and a finite prefactor absorbing the entrance time. □

Proof of Theorem 1.1. Take $K = \mathcal{P}(\Omega)$, $r(p) = \mu_{m(p)}$, and the fixed space E defined above. Conservation of marginals gives all fiber properties and derivative invariance. Lemma 1.2 gives continuity and pointwise convergence. Proposition 1.4 verifies strict spectral stability on the entire compact equilibrium set, not merely its interior. Lemma 1.5, followed by equivalence of finite-dimensional norms, proves the asserted total variation bound. □

1.7 An explicit bounded-perturbation spectral estimate

Let $R = \max_x w(x)/\min_x w(x)$. Define $g_0 = 1/n$ for the single-site kernel and $g_0 = 1/2$ for the block kernel. For either kernel, the following bound holds at every Gibbs equilibrium, including boundary equilibria:

$$\rho(DT(\mu)|_E) \le 1 - \varepsilon, \qquad \varepsilon = R^{-2}g_0.$$

Here the spectral radius statement uses the positive semidefiniteness of the heat-bath kernels.

To prove the estimate first suppose $\mu > 0$. Write $\mu = w\gamma/Z$, where γ is the product probability determined by its external fields and $Z = \gamma[w]$. Put $m = \min w$, $M = \max w$, $l = m/Z$, and $u = M/Z$. Then $l\gamma \le \mu \le u\gamma$. The product law γ need not have the same marginals as μ.

For each exchange orbit partition, the heat-bath operator is the orthogonal projection onto the space of orbit-invariant functions. Although the projection depends on the reference weights, this invariant subspace does not. Thus the variational formula for squared distance gives, for every pair function F,

$$\langle F, (I - Q_J)F\rangle_{\mu\otimes\mu} \ge l^2\langle F, (I - Q_0)F\rangle_{\gamma\otimes\gamma}.$$

For the block kernel this is one projection comparison; for the single-site kernel average this comparison over sites.

Let Add denote all additive one-site functions. The orthogonal product decomposition of $L^2(\gamma)$ gives

$$\tfrac{1}{2}\langle Mh, (I - Q_0)Mh\rangle_{\gamma\otimes\gamma} \ge g_0 \inf_{a\in\mathrm{Add}} \|h - a\|_\gamma^2.$$

Indeed, on a product component involving $k \ge 2$ coordinates, the linearized single-site eigenvalue is $1 - k/(2n)$ and the block eigenvalue is 2^{1-k}. Additive components have zero Dirichlet energy. If $\mu h \in E$, then h is μ-orthogonal to Add, so

$$\inf_{a\in\mathrm{Add}} \|h - a\|_\gamma^2 \ge u^{-1} \inf_{a\in\mathrm{Add}} \|h - a\|_\mu^2 = u^{-1}\|h\|_\mu^2.$$

The density compression identity therefore yields

$$\langle h, (I-L)h\rangle_\mu \geq (l^2/u)g_0\|h\|_\mu^2 \geq R^{-2}g_0\|h\|_\mu^2,$$

because $l^2/u = m^2/(ZM) \geq m^2/M^2$. Interior equilibria are dense in the closed equilibrium set, and the restricted matrix depends continuously on μ, so the same spectral-radius bound extends to the boundary.

The perturbation and product spectral arguments are classical in character. In particular the linearization/conserved-mode framework appears in Caputo–Sinclair, arXiv:1609.06897, Section 2.3. This comparison is not represented as a new general mathematical principle.

1.7.1 Explicit asymptotic rate; nonexplicit prefactor

The uniform spectral gap supplies an explicit exponential exponent without an explicit entrance time. Put $a = 1 - \varepsilon/2$ and, on E, define

$$N_\mu(v) = \sum_{k\geq 0} a^{-k}\|A_\mu^k v\|_1.$$

Uniform resolvent estimates on the compact family $\{A_\mu|_E\}$ show that there is $B < \infty$ with

$$\|v\|_1 \leq N_\mu(v) \leq B\|v\|_1, \qquad N_\mu(A_\mu v) \leq aN_\mu(v)$$

uniformly in μ. Since T is quadratic with a stochastic pair kernel,

$$T(\mu+v) - \mu = A_\mu v + T(v), \qquad \|T(v)\|_1 \leq \|v\|_1^2.$$

The remainder belongs to E. If $N_\mu(v) \leq \eta = \varepsilon/(4B)$, then

$$N_\mu(T(\mu+v)-\mu) \leq (1-\varepsilon/4)N_\mu(v).$$

Every trajectory eventually enters the smaller ordinary ball $\|v\|_1 < \eta/B$. Continuity of the equilibrium section and a finite-cover argument give a common entrance time on the simplex, exactly as in the compact-upgrade lemma. Consequently, for some finite $C(w)$,

$$\|T^t(p) - \mu_{m(p)}\|_{\mathrm{TV}} \leq C(w)e^{-\varepsilon t/4}.$$

Thus one can take $c = R^{-2}/(4n)$ for single-site dynamics and $c = R^{-2}/8$ for block dynamics. For the Ising weight, if $B_J = \sum_{i<j}|J_{ij}|$, then $R \leq e^{2B_J}$; the respective rates are at least $e^{-4B_J}/(4n)$ and $e^{-4B_J}/8$. These are explicit asymptotic exponents, not explicit mixing-time bounds: the prefactor $C(w)$ remains nonquantitative.

1.8 Extension to finite alphabets and positive interaction weights

Theorem 1.6. *Let $\Omega = \prod_{i=1}^n X_i$ with each X_i finite and nonempty, and let $w : \Omega \to (0,\infty)$ be arbitrary. Define the same two heat-bath exchange kernels using w instead of the Ising weight. For each array $\boldsymbol{\pi} = (\pi_i)_i$ of one-site marginals, let $\mu_{\boldsymbol{\pi}}$ uniquely maximize $H(q) + q[\log w]$ subject to those marginals. There are $C(w) < \infty$ and $c(w) > 0$ such that*

$$\|T^t(p) - \mu_{(p_i)_i}\|_{\mathrm{TV}} \leq C(w)e^{-c(w)t}$$

uniformly over every initial law and every integer $t \geq 0$.

Proof. We detail the changes needed to include faces on which several alleles are absent. The fixed tangent space is now

$$E = \{v : \sum_x v(x) = 0, \sum_{x:x_i=\alpha} v(x) = 0 \text{ for all } i, \alpha \in X_i\}.$$

The entropy maximizer is positive on the product $S = \prod_i S_i$ of the marginal supports and has density proportional to $w(x)\prod_i \lambda_i(x_i)$ there, with all factors positive. The Gibbs section is continuous: to change marginal π_i to π_i', use a Markov kernel obtained from a maximal coupling of those two laws. It changes the joint law by at most $\|\pi_i - \pi_i'\|_{\mathrm{TV}}$ and preserves the other marginals. The preceding compactness argument then applies. The comparison with Q^0, the linear map L_π, and the positive-box entropy proof are unchanged.

On $S \times S$, the block kernel is conditional expectation on the classes determined by the unordered pair at each coordinate. All subsets realizing a given output pair have the same multiplicity, namely 2^r, where r is the number of agreeing input coordinates. Thus this is indeed the Gibbs heat bath on each orbit. The one-site kernels are again conditional expectations on exchange orbits. The compression identity and the spectral interval $[0, 1]$ remain valid. The equality condition says, for every $\alpha, \beta \in S_i$, that $h(\alpha, u) - h(\beta, u)$ is independent of u. It follows that h is a sum of one-site functions; the full marginal constraints annihilate all such neutral modes.

At a boundary equilibrium, classify every configuration by its mutant coordinates $B = \{i : x_i \notin S_i\}$ together with the corresponding absent-allele labels $(x_i)_{i\in B}$. A mutant-resident collision partitions this labeled set. Ordering by its cardinality makes the derivative block triangular. Every block with at least two mutant coordinates is strictly substochastic, since a one-site exchange splits its labeled set. A block with just one absent allele is strictly positive and stochastic: swapping the mutant coordinate transfers that allele to a resident with any prescribed allowed configuration on the remaining coordinates. Its unique eigenvalue-1 direction has nonzero mass in that absent allele. In an eigenvector belonging to E, the separate marginal constraint for this absent allele kills that direction after the larger mutant blocks have vanished. The resident block has only the already described additive neutral modes, also killed by E. Therefore $\rho(DT(\mu)|_E) < 1$ on every face. The compact common-iterate argument proves the result. □

1.9 A nonlinear entropy reweighting bound in continuous time

This section concerns the different evolution $\dot{p} = T(p) - p$. We keep its distinction from discrete iteration explicit. Let μ be the Gibbs law with the same one-site marginals as p, and define its entropy production by

$$\mathcal{I}_J(p) = -\sum_x (T(p)(x) - p(x)) \log \frac{p(x)}{\mu(x)}.$$

For positive p and μ, put $f = p/\mu$. Reversibility gives the standard symmetrized formula

$$\mathcal{I}_J(p) = \frac{1}{4} \sum_{x,y,u,v} \mu(x)\mu(y)Q_J(x, y; u, v) \cdot [f(x)f(y) - f(u)f(v)] \log \frac{f(x)f(y)}{f(u)f(v)}.$$

Let κ_0 be any constant for which the zero-interaction kernel satisfies

$$\mathcal{I}_0(q) \geq \kappa_0 D(q \| \bigotimes_i q_i).$$

For $n \geq 2$, the actual single-site heat bath used here admits $\kappa_0 = 1/[2(n-1)]$; the factor $1/2$ is its holding normalization. For the uniform block kernel, one may take $\kappa_0 = (1 - 2^{1-n})/(n-1)$.

These established recombination bounds hold for arbitrary finite coordinate alphabets [1.2]. The conservative alternatives $1/(2n)$ and $1/n$, respectively, remain valid when some coordinates are pinned; when at most one coordinate remains nontrivial, the desired entropy is identically zero.

Theorem 1.7. *For $R = \max w / \min w$ and positive laws on the allowed product support,*

$$\mathcal{I}_J(p) \geq \begin{cases} \kappa_0 R^{-2} D(p\|\mu), & \textit{single-site heat bath,} \\ \kappa_0 R^{-4} D(p\|\mu), & \textit{uniform block heat bath.} \end{cases}$$

Thus the continuous-time evolution has the corresponding global exponential relative-entropy decay, with prefactor one, uniformly over the equilibrium fields. Boundary initial laws follow by approximation on their allowed product support.

Proof. Put $m = \min w$, $M = \max w$, and define the probability q by

$$q(x) = \frac{p(x)/w(x)}{\sum_z p(z)/w(z)}, \qquad Z = \sum_x q(x)w(x), \qquad p = wq/Z.$$

Let $\pi = \bigotimes_i q_i$ and $\mu' = w\pi/\pi[w]$. Both μ and μ' belong to the same exponential family, so $\log(\mu/\mu')$ is additive. Since p and μ have the same one-site marginals,

$$D(p\|\mu') = D(p\|\mu) + D(\mu\|\mu') \geq D(p\|\mu).$$

Apply to q and π the same accept/reject channel, accepting an input x with probability $w(x)/M$ and recording the accepted value. The accepted laws are p and μ', and the acceptance probability under q is Z/M. Data processing and the chain rule for relative entropy give

$$D(p\|\mu) \leq D(p\|\mu') \leq (M/Z)D(q\|\pi). \tag{1.9.1}$$

All discarded terms in the chain rule are nonnegative.

Write $\mu = w\gamma/\gamma[w]$ with product γ. On every admissible swap, $\gamma(x)\gamma(y) = \gamma(u)\gamma(v)$. Substitution in the production formula therefore gives

$$\mathcal{I}_J(p) = \frac{1}{4Z^2} \sum_{x,y,u,v} w(x)w(y)Q_J(x,y;u,v)[q(x)q(y) - q(u)q(v)] \log \frac{q(x)q(y)}{q(u)q(v)}.$$

Every bracket-log product is nonnegative.

For a single-site exchange set $a = w(x)w(y)$ and $b = w(u)w(v)$. Its heat-bath conductance is $ab/(a+b) \geq m^2/2$, while the zero-interaction exchange probability is $1/2$. The site-selection factors agree. Thus

$$\mathcal{I}_J(p) \geq (m^2/Z^2)\mathcal{I}_0(q) \geq \kappa_0 \frac{m^2}{MZ} D(p\|\mu) \geq \kappa_0 R^{-2} D(p\|\mu).$$

For the full block heat bath, write $\mathcal{O}$ for the pair orbit and $a_z = w(z_1)w(z_2)$. The ratio of the weighted edge coefficient to its zero-interaction counterpart is

$$\frac{a_{(x,y)}a_{(u,v)}}{|\mathcal{O}|^{-1}\sum_{z\in\mathcal{O}} a_z} \geq \frac{m^4}{M^2}.$$

Consequently

$$\mathcal{I}_J(p) \geq \frac{m^4}{M^2Z^2}\mathcal{I}_0(q) \geq \kappa_0 \frac{m^4}{M^3Z} D(p\|\mu) \geq \kappa_0 R^{-4} D(p\|\mu).$$

The entropy differential identity and Gronwall's inequality prove continuous-time decay. For initial laws with zeros, restrict to the product of their marginal supports and approximate the law by its mixture with the Gibbs equilibrium. The finite-dimensional flow is continuous in the initial condition, and relative entropy to the fixed positive equilibrium is continuous, so the result passes to the limit. □

For completeness, uniformly over initial laws,

$$D(p\|\mu) = \mathcal{F}(\mu) - \mathcal{F}(p) \le n \log 2 + \log R$$

in the binary case (replace $n \log 2$ by $\sum_i \log |X_i|$ for finite alphabets). Pinsker therefore supplies an explicit continuous-time total-variation prefactor as well. None of this identifies the sharp nonlinear constants, and the discrete-time entropy loss is not identified with its continuous-time derivative.

1.10 Limitations

- The discrete-time rate has the explicit lower bounds proved above, while its prefactor remains nonquantitative because of the uniform entrance-time argument. The continuous-time entropy estimate has prefactor one and a fully explicit rate.
- The total-variation estimate in discrete time and the entropy estimate in continuous time are distinct. Neither computes a sharp fixed-reference recombination constant.
- The qualitative convergence mechanism is established prior art and is rederived here only for completeness. The 2026 covariance manuscript is discussed for context and is not an input to this proof.

Chapter 2

Uniform Exponential Convergence for Nonlinear Independent Set Exchange

Chapter overview

For a fixed finite graph of maximum degree Δ and every $\beta < 1/(\Delta+1)$, we prove uniform exponential convergence of the discrete-time nonlinear single-site and mean-field independent-set exchange maps. The initial vertex marginals, respectively mean density, may range throughout $[0, \beta]$, including zero values. The asymptotic exponents are explicit; the common prefactors are not. An empty-set deletion estimate controls the linearization uniformly as fugacities vanish. Spectral continuity and fixed-coordinate adapted norms then give a uniform nonlinear contraction neighborhood and entrance time. The result concerns exponential ergodicity, not the stronger nearly-linear-rate conjecture.

2.1 Background, definitions, and statement

The source [2.1], revised 20 October 2026, proves qualitative convergence in Proposition 1.1 and asks in Section 6(1) whether it is exponential throughout the same region, allowing a very small dimension-dependent rate. We consider precisely its discrete-time single-site and mean-field collision rules, not heat-bath variants. The stronger rate and algorithmic questions in Section 6(2) are not addressed here.

Classical mass-action theory already gives local exponential stability at positive equilibria [2.2, Theorems 5.4–5.6]. Here $T(p) - p$ is a reversible quadratic mass-action field on the probability simplex. Together with conserved-mode removal, aperiodicity, and qualitative convergence, this supplies a fixed-positive-profile route. The estimates below instead make the rate explicit and uniform as conserved quantities vanish; local stability itself is not new.

Let $G = (V, \mathcal{E})$ be a finite simple graph, $n = |V| \geq 1$, and let Ω be its independent sets. Write $N[v]$ for the closed neighborhood of v and Δ for the maximum degree. For an ordered pair $(I, J) \in \Omega^2$, select vertices v, u and exchange the occupancy bit at v in I with that at u in J, accepting the exchange exactly when both resulting sets are independent. Otherwise keep (I, J) unchanged. The single-site rule selects $v = u$ uniformly in V; the mean-field rule selects v, u independently uniformly in V. Denote the pair transition matrix by Q, its first-output marginal by Q_1, and define

$$T(p)(K) = \sum_{I,J\in\Omega} p(I)p(J)Q_1(K \mid I, J).$$

Under the input law $p \otimes p$, both output marginals agree. The single-site map conserves all $m_v(p) = p(v \in I)$; the mean-field map conserves $\alpha(p) = \mathbb{E}_p|I|/n$.

For a vector $\lambda = (\lambda_v)_{v\in V} \geq 0$, put

$$\mu_\lambda(I) = \frac{\prod_{v\in I} \lambda_v}{Z(\lambda)}, \qquad Z(\lambda) = \sum_{I\in\Omega} \prod_{v\in I} \lambda_v.$$

A zero fugacity excludes that vertex. The single-site Gibbs equilibrium $\mu_{m(p)}$ is the unique entropy maximizer with marginals $m(p)$. The mean-field equilibrium $\mu_{\alpha(p)}$ has a common fugacity and uniquely maximizes entropy at that density.

Theorem 2.1. *Fix $0 < \beta < 1/(\Delta+1)$ and write*

$$d = 1 - (\Delta+1)\beta, \qquad L = \beta/d, \qquad Z_* = (1+L)^n.$$

There exist finite constants $C_{\mathrm{ss}}(G,\beta), C_{\mathrm{mf}}(G,\beta)$ such that, for every integer $t \geq 0$,

$$\|T^t_{\mathrm{ss}}(p) - \mu_{m(p)}\|_{\mathrm{TV}} \leq C_{\mathrm{ss}}(G,\beta)\exp\{-t/(16n^2 Z_*^2)\}, \qquad \max_v m_v(p) \leq \beta,$$
$$\|T^t_{\mathrm{mf}}(p) - \mu_{\alpha(p)}\|_{\mathrm{TV}} \leq C_{\mathrm{mf}}(G,\beta)\exp\{-t/(48n^3 Z_*^2)\}, \qquad \alpha(p) \leq \beta.$$

The constants are uniform as positive conserved quantities approach zero, and include exactly zero values. The prefactors are not asserted to be explicit.

Every initial law in the strict region of [2.1, Proposition 1.1] belongs to one of these closed subregions by choosing a larger β still below the threshold. No uniformity as $\beta \uparrow 1/(\Delta+1)$ is asserted.

2.2 Stationarity, entropy, and compactness

Each specified vertex exchange, with rejection, is an involution on ordered pair states. Thus Q is symmetric and stochastic. It is reversible for $\mu_\lambda \otimes \mu_\lambda$: for the single-site kernel all vertex counts are preserved, while for the mean-field kernel the common-fugacity weight is preserved.

Lemma 2.2. *If G has an edge, every diagonal entry of either pair kernel is at least $\theta = 1/n^2$. Hence $Q \succeq (-1+2\theta)I$ in every reversible weighted inner product.*

Proof. If I and J agree at some vertex v, selecting $v = u$ leaves the pair unchanged. If they disagree at every vertex, then they partition V. For any edge, its endpoints must lie in different sets. Transferring either endpoint to the other set is rejected. In either case a choice of probability at least $1/n^2$ holds the pair. Write $Q = \theta I + (1-\theta)\widetilde{Q}$, where $\widetilde{Q}$ is a reversible Markov matrix with spectrum in $[-1,1]$. □

Lemma 2.3. *On each parameter region in Theorem 2.1, the Gibbs equilibrium exists, is unique, and $T^t(p)$ converges to it. The equilibrium section is continuous on the entire closed region, including zero marginals.*

Proof. First assume G has an edge. Shannon entropy satisfies

$$2H(Tp) \geq H((p\otimes p)Q) \geq H(p\otimes p) = 2H(p).$$

The first inequality is subadditivity; the second follows because Q is doubly stochastic. Equality $H(Tp) = H(p)$ forces equality in the second inequality. Strict concavity of entropy, together with the positive diagonal from Lemma 2.2, implies that the density $p(I)p(J)$ is constant across every positive transition of Q. This assertion includes zero entries. In particular $(p\otimes p)Q = p\otimes p$, so $Tp = p$, and the support of p is closed under admissible exchanges.

We classify such equality laws in the specified region. Choose a supported set I of minimum cardinality. For the single-site rule, if $v \in I$, then

$$p(J \cap N[v] = \varnothing) \geq 1 - \sum_{u\in N[v]} m_u(p) \geq d > 0.$$

A supported J avoiding $N[v]$ permits the exchange $(I,J) \mapsto (I\setminus\{v\}, J\cup\{v\})$. Support closure contradicts minimality. Hence $p(\varnothing) > 0$. In the mean-field case,

$$\frac{1}{n}\sum_u p(J\cap N[u] = \varnothing) \geq 1 - \frac{1}{n}\sum_u \sum_{z\in N[u]} m_z(p) \geq 1 - (\Delta+1)\alpha(p) \geq d.$$

Thus some u and supported J have $J \cap N[u] = \varnothing$; transferring any $v \in I$ to this u gives the same contradiction. Again $p(\varnothing) > 0$.

The admissible same-site exchange with an empty set yields

$$p(I)p(\varnothing) = p(I \setminus \{v\})p(\{v\}) \qquad (v \in I).$$

Induction implies $p(I) = p(\varnothing) \prod_{v \in I} \lambda_v$, where $\lambda_v = p(\{v\})/p(\varnothing) \geq 0$. For the mean-field kernel, exchanging a singleton with the empty set at any target vertex gives $p(\{v\}) = p(\{u\})$; its fugacity is common. Such a Gibbs law uniquely maximizes entropy on its conserved fiber: on its allowed support, $\log \mu$ is an affine combination of the conserved observables, so for any feasible q,

$$D(q\|\mu) = H(\mu) - H(q).$$

For the single-site law, vertices with $\lambda_v = 0$ have zero conserved marginal and are excluded from every law on the same fiber. The case of zero mean density is simply the point mass at the empty set.

Entropy is continuous and nondecreasing along every trajectory in the compact simplex. Any subsequential limit q satisfies $H(Tq) = H(q)$, because successive entropies have the same limit. The preceding classification and entropy uniqueness show that every subsequential limit is the same Gibbs law. This proves existence and convergence without needing a separate feasible-profile construction.

For a Gibbs equilibrium in the single-site region,

$$m_v = \lambda_v \mu(I \cap N[v] = \varnothing), \qquad 0 \leq \lambda_v \leq \beta/d = L.$$

For a common fugacity, summing this identity gives

$$\alpha = \frac{\lambda}{n} \sum_v \mu(I \cap N[v] = \varnothing) \geq \lambda[1 - (\Delta + 1)\alpha],$$

so again $\lambda \leq L$. All equilibria therefore have fugacities in a compact cube $[0, L]^n$. If conserved profiles converge, take a convergent subsequence of their fugacities. The Gibbs formula is continuous on that cube because $Z \geq 1$. Its limit has the desired conserved profile and hence is its unique entropy maximizer. This proves continuity of the equilibrium section.

If G is edgeless, the maps on each conserved fiber are the linear refresh maps

$$T_{\mathrm{ss}} p = \frac{1}{n} \sum_v p_v \otimes p_{-v}, \qquad T_{\mathrm{mf}} p = \frac{1}{n} \sum_v \mathrm{Bern}(\alpha) \otimes p_{-v}.$$

Their convergence to the corresponding product equilibrium follows, for example, by conditioning on the event that all coordinates have been refreshed. The probability of its complement is at most $n(1 - 1/n)^t$ (with the one-coordinate case interpreted directly). The equilibrium sections are then explicitly continuous. □

2.3 The derivative and an empty-set comparison

Let E_{ss} be the fixed vector space of signed measures annihilating mass and every vertex occupancy, and E_{mf} the space annihilating mass and $|I|$. These spaces are independent of the conserved profile and invariant under the corresponding derivative $A_\mu = DT(\mu)$. For a full-support equilibrium write $v = \mu h$ and put $Mh(I, J) = h(I) + h(J)$. Reversibility and replica symmetry give

$$\langle h, A_\mu(\mu k)/\mu \rangle_\mu = \tfrac{1}{2} \langle Mh, QMk \rangle_{\mu \otimes \mu}.$$

If $\mu[h] = 0$, then $\|Mh\|_{\mu^2}^2 = 2\|h\|_\mu^2$. Thus the derivative is self-adjoint on centered densities, and its deficit form is

$$\begin{aligned}\mathcal{D}_\mu(h) &= \|h\|_\mu^2 - \langle h, A_\mu(\mu h)/\mu\rangle_\mu \\ &= \frac{1}{4}\sum_{I,J,U,W} \mu(I)\mu(J)Q(I,J;U,W)[h(I)+h(J)-h(U)-h(W)]^2.\end{aligned}$$

The general compression viewpoint is classical in reversible quadratic dynamics; see [2.3, Section 2.3].

Suppose now $0 < \lambda_v \le L$. Two elementary estimates are

$$\mu(\varnothing) \ge Z_*^{-1}, \qquad \sum_{I \supseteq J} \mu(I) \le Z_* \mu(J). \tag{2.3.1}$$

Indeed $Z(\lambda) \le \prod_v (1+\lambda_v) \le Z_*$, and after factoring the weight of J, the weights of its admissible extensions are at most the sum over all subsets of $V \setminus J$.

Fix an ordering of vertices. For $v \in J$, set

$$e(J,v) = h(J) + h(\varnothing) - h(J \setminus \{v\}) - h(\{v\}).$$

Remove the vertices of I in the fixed order. With $a_v = h(\{v\}) - h(\varnothing)$, the residual

$$R(I) = h(I) - h(\varnothing) - \sum_{v \in I} a_v$$

is the sum of the corresponding removal defects. Cauchy–Schwarz and (2.3.1) show

$$\sum_I \mu(I) R(I)^2 \le n Z_* \sum_J \sum_{v \in J} \mu(J) e(J,v)^2. \tag{2.3.2}$$

To justify the summation, a fixed stage (J, v) can arise only from initial sets $I \supseteq J$; its total initial weight is therefore at most $Z_*\mu(J)$. Overcounting all stages only enlarges the bound.

Proposition 2.4. *On the fixed conserved tangent spaces, every equilibrium with fugacities bounded by L, including zero fugacities, satisfies*

$$\rho(A_\mu|_{E_{\mathrm{ss}}}) \le 1 - \varepsilon_{\mathrm{ss}}, \quad \varepsilon_{\mathrm{ss}} = \frac{1}{4n^2 Z_*^2}, \qquad \rho(A_\mu|_{E_{\mathrm{mf}}}) \le 1 - \varepsilon_{\mathrm{mf}}, \quad \varepsilon_{\mathrm{mf}} = \frac{1}{12n^3 Z_*^2}.$$

In the mean-field statement the fugacity is common. A zero-dimensional tangent space is understood to have spectral radius zero.

Proof. First take positive fugacities. For the single-site rule, h corresponding to a vector in E_{ss} is orthogonal to all additive occupancy functions. Consequently $\|h\|_\mu^2 \le \|R\|_\mu^2$. The transition

$$(J, \varnothing) \longmapsto (J \setminus \{v\}, \{v\})$$

has probability at least $1/n$. Keeping just these distinct directed edges in the nonnegative deficit sum gives

$$\mathcal{D}_\mu(h) \ge \frac{\mu(\varnothing)}{4n} \sum_J \sum_{v \in J} \mu(J) e(J,v)^2.$$

Using (2.3.2) and (2.3.1) yields $\|h\|_\mu^2 \le 4n^2 Z_*^2 \mathcal{D}_\mu(h)$.

For the mean-field rule, h is orthogonal to 1 and $|I|$. Set $b = n^{-1}\sum_v a_v$ and $d_v = a_v - b$. Then

$$h(I) - h(\varnothing) - b|I| = R(I) + B(I), \qquad B(I) = \sum_{v \in I} d_v.$$

The same deletion transitions now have probability at least $1/n^2$, so

$$\|R\|_\mu^2 \le \frac{4n^3 Z_*}{\mu(\varnothing)} \mathcal{D}_\mu(h).$$

Since $m_v \le \lambda$ for common fugacity λ,

$$\|B\|_\mu^2 \le n \sum_v m_v d_v^2 \le n\lambda \sum_v d_v^2.$$

The singleton-transfer edges $(\{v\}, \varnothing) \mapsto (\varnothing, \{u\})$, each with probability at least $1/n^2$, give

$$\mathcal{D}_\mu(h) \ge \frac{\lambda \mu(\varnothing)^2}{4n^2} \sum_{v,u} (d_v - d_u)^2 = \frac{\lambda \mu(\varnothing)^2}{2n} \sum_v d_v^2.$$

Hence $\|B\|_\mu^2 \le 2n^2 \mu(\varnothing)^{-2} \mathcal{D}_\mu(h)$. Orthogonality and $(R+B)^2 \le 2R^2 + 2B^2$ imply

$$\|h\|_\mu^2 \le (8n^3 + 4n^2) Z_*^2 \mathcal{D}_\mu(h) \le 12 n^3 Z_*^2 \mathcal{D}_\mu(h).$$

The deletion and singleton lower bounds are used separately; no disjointness between these two collections of edges is required.

These bounds put the largest restricted derivative eigenvalue below $1 - \varepsilon$. If G has an edge, Lemma 2.2 and compression put its smallest centered eigenvalue above $-1 + 2/n^2$. Both stated ε are at most $2/n^2$, so the same estimate controls the absolute spectral radius. If G is edgeless, the explicit refresh operators have nonnegative restricted spectrum with gap at least $1/n$, which is more than sufficient.

Finally, A_μ depends continuously on μ in the fixed coordinates of signed measures, and the tangent spaces E do not change with μ. Replace any zero fugacity by a positive number tending to zero, still bounded by L. The positive-fugacity estimates proved above require only that upper bound, not any marginal restriction. Continuity of the Gibbs formula and of finite-dimensional spectral radius proves the same estimate at every boundary equilibrium. □

2.4 A uniform nonlinear neighborhood and global entrance

We give the upgrade explicitly to avoid relying on density norms that degenerate at zero fugacity. Let K be either of the compact convex sets of laws in Theorem 2.1; it is invariant under the map. Write $r(p)$ for its continuous equilibrium section and E for the appropriate fixed tangent space. Set ε to the corresponding constant of Proposition 2.4 and $a = 1 - \varepsilon/2$.

The matrices $A_\mu|_E$, $\mu \in r(K)$, form a compact family with spectral radii at most $1 - \varepsilon$. A circle of radius strictly between $1 - \varepsilon$ and a has uniformly bounded resolvents over this family. The matrix Cauchy integral formula therefore bounds their powers uniformly at that radius. Consequently

$$N_\mu(v) = \sum_{k=0}^{\infty} a^{-k} \|A_\mu^k v\|_1$$

is a norm on E and there is a finite B, independent of μ, such that

$$\|v\|_1 \le N_\mu(v) \le B\|v\|_1, \qquad N_\mu(A_\mu v) \le a N_\mu(v).$$

No positive lower bound on the cells of μ is used.

Quadraticity and stochasticity give

$$T(\mu + v) - \mu = A_\mu v + T(v), \qquad \|T(v)\|_1 \le \|v\|_1^2.$$

For a same-fiber displacement v, the remainder lies in E. Therefore, whenever $N_\mu(v) \le \eta = \varepsilon/(4B)$,

$$N_\mu(T(\mu+v)-\mu) \le (1-\varepsilon/4)N_\mu(v).$$

By Lemma 2.3, every trajectory eventually has ordinary distance to its equilibrium less than $\eta/(2B)$. At its chosen entrance time, continuity of T^t and r gives an open neighborhood of initial laws with distance less than η/B. Every law in that neighborhood thus enters the invariant adapted ball at that same time. A finite subcover of the compact K gives a common finite upper bound on these entrance times. Subsequent adapted contractions and $\|v\|_1 \le N_\mu(v)$ imply

$$\|T^t(p) - r(p)\|_{\mathrm{TV}} \le C\exp(-\varepsilon t/4)$$

with finite C independent of $p \in K$. This proves Theorem 2.1.

2.5 Why the strict threshold matters

Uniformity up to the endpoint cannot in general be added to the single-site result. Let G be two disjoint edges $\{1,2\}$ and $\{3,4\}$, so $\Delta = 1$. Every law supported on the four sets choosing exactly one vertex from each edge is frozen: transferring a vertex absent from the other replica would conflict with the occupied opposite endpoint. The law

$$p_0 = \tfrac{1}{2}\delta_{\{1,3\}} + \tfrac{1}{2}\delta_{\{2,4\}}$$

has all marginals $1/2$, but differs from the maximum-entropy law, which is uniform on the four sets. Set $p_\eta = (1-\eta)p_0 + \eta\delta_\varnothing$. Its marginals are strictly below $1/2$. Its entropy-maximizing equilibrium is the product over the two edges of a law assigning probabilities $\eta, (1-\eta)/2, (1-\eta)/2$ to the empty and two singleton states. As $\eta \downarrow 0$, these equilibria converge to the uniform law on the four maximal sets, whereas $T^t(p_\eta) \to p_0$ for every fixed t. Therefore no single finite prefactor and positive exponent can hold uniformly over the entire open region with no distance from its threshold.

2.6 Scope and qualifications

The argument proves an explicit asymptotic exponent with a nonexplicit common prefactor, not an explicit mixing-time estimate and not a rate of order $1/n$ near the full threshold. Its entropy monotonicity and support-closure mechanisms are prior ingredients of the qualitative theory. Its compression and local spectral-stability framework also have classical precedents. The proof's quantitative step is the empty-set deletion comparison uniform as fugacities vanish; no historical novelty claim for that step is made here.

Chapter 3

Fast Entropy Contraction for Nonlinear Independent Set Exchange

Chapter overview

We prove global relative entropy contraction for nonlinear independent-set exchange. For single-site exchange, maximum marginal $\beta \le 1/(2\Delta+1)$ gives entropy decay at rate $(1-2\Delta\beta)/n$ and an explicit total-variation bound. A heterogeneous local condition is also sufficient. The mean-field map admits the same discrete-time bound with its conserved density replacing β. Each nonlinear update is represented by a reversible linear kernel with a varying Gibbs reference. A contractive coupling and the curvature theorem of Caputo–Salez give a discrete entropy inequality, which transfers to the fixed equilibrium because the relevant moments are conserved. The result enlarges known fast-convergence regions but does not resolve the full single-site conjecture.

3.1 Statement and relation to the original question

Let $G=(V,\mathcal{E})$ be a finite simple graph, $n=|V|\ge 1$, and Ω_G its independent sets. The nonlinear single-site map T samples I,J independently from p, chooses $v\in V$ uniformly, and exchanges the two occupancy bits at v if both resulting sets are independent; otherwise the pair is unchanged. The law of its first output is $T(p)$. This is exactly the single-site rule of [3.1], not a heat-bath modification. It preserves every marginal $m_v = p(v\in I)$.

Coordinates with marginal zero are absent, and coordinates with marginal one are deterministically occupied. Remove these deterministic coordinates and restrict to the resulting allowed independent-set support. Write $F=\{v : 0<m_v<1\}$ for the remaining free coordinates; no vertex in F neighbors a deterministically occupied vertex. The original normalization $1/n$ is retained, so updates at removed coordinates are holding steps. Put

$$s_v = \sum_{u\in N(v)} m_u \qquad (v\in F).$$

On the fixed-marginal fiber, let μ_m be the maximum-entropy law. Under the assumptions below it has full support on the allowed support and the hard-core Gibbs form there.

Theorem 3.1. *Suppose $F\neq\varnothing$ and*

$$m_v + 2s_v \le 1 \qquad (v\in F). \tag{3.1.1}$$

Set $\kappa = n^{-1}\min_{v\in F}(1-2s_v)>0$. Let $h(r)=-r\log r-(1-r)\log(1-r)$ denote binary entropy, with $0\log 0=0$. For every law p with these marginals,

$$D(Tp\|\mu_m)\le(1-\kappa)D(p\|\mu_m).$$

Consequently, for every integer $t\ge 0$,

$$D(T^tp\|\mu_m)\le(1-\kappa)^tD(p\|\mu_m),\qquad \|T^tp-\mu_m\|_{\mathrm{TV}}\le\sqrt{\frac12\sum_{v\in V}h(m_v)}\,e^{-\kappa t/2}.$$

If $F=\varnothing$, the fiber consists of one law and all errors are zero.

Theorem 3.2. *Let $\Delta \geq 1$ be the maximum degree and $0 \leq \beta \leq 1/(2\Delta+1)$. Uniformly over all initial laws with $\max_v m_v \leq \beta$, Theorem 3.1 holds with*

$$\kappa = \frac{1-2\Delta\beta}{n}.$$

The endpoint $\beta = 1/(2\Delta+1)$ is included, as are zero marginals. The total-variation prefactor is at most $\sqrt{nh(\beta)/2}$. For edgeless graphs, the same conclusion holds with $\kappa = 1/n$ after removing deterministic coordinates.

Theorem 3.2 extends the allowed marginal range in [3.1, Theorem 1.2], which gives an explicit total-variation bound when $m_v \leq (1-\delta)/[3(\Delta+1)]$. We do not claim our total-variation exponent is larger at every common parameter value. The conjecture in [3.1, Section 6(2)] asks for a nearly-linear rate throughout the larger region $m_v < 1/(\Delta+1)$; that remaining interval is not settled here. The curvature theorem used below is an established general result, not a new principle claimed in this note.

3.2 The frozen reversible kernel

For a law p on the allowed support, define

$$a_v(p) = p(I \cap N[v] = \varnothing), \qquad b_v(p) = m_v + a_v(p) = p(I \cap N(v) = \varnothing), \quad v \in F.$$

Thus $b_v \leq 1$ and

$$a_v(p) \geq 1 - m_v - s_v \geq s_v. \tag{3.2.1}$$

Moreover $a_v(p) > 0$: if $s_v > 0$ this follows from (3.2.1); if $s_v = 0$, then $a_v(p) = 1 - m_v > 0$.

Lemma 3.3. *There is a Markov kernel K_p such that $Tp = pK_p$. It selects a coordinate with probability $1/n$, inserts a vacant vertex v with probability m_v if all its neighbors are vacant, and removes an occupied vertex v with probability $a_v(p)$. Its invariant Gibbs law ν_p has fugacities*

$$\tau_v = \frac{m_v}{a_v(p)} \quad (v \in F).$$

On the allowed support K_p is reversible and irreducible.

Proof. For an insertion in the first parent, the donor must be occupied at v, an event of probability m_v; removal from the donor is always admissible. For a deletion from the first parent, the donor must have its entire closed neighborhood of v vacant, an event of probability $a_v(p)$. These are precisely the stated transition probabilities. Detailed balance on each insertion/deletion edge holds because the Gibbs weight ratio is $\tau_v = m_v/a_v(p)$. All free-site insertion and deletion rates are positive, and the empty free configuration connects the state space. □

For completeness, the equilibrium μ_m exists and is positive on the allowed support under (3.1.1). Entropy has a unique maximizer on the nonempty compact marginal fiber. Its support contains the support of every other feasible law: otherwise mixing with that law would give an infinite positive entropy derivative at a zero cell. Since applying T preserves the fiber, this support is closed under admissible pair exchanges. A supported independent set of minimum size must be empty: if it contains v, a supported donor avoiding $N[v]$ exists by $1 - m_v - s_v > 0$, and the exchange deletes v. Positive free marginals then supply every singleton, and support closure builds every independent set. Lagrange multipliers now give the positive Gibbs form. This argument is on the free induced graph after removing deterministic coordinates.

3.3 A nonexpanding contractive coupling

Use Hamming distance $d(I,J) = |I \triangle J|$ on independent sets of the free induced graph. This is a geodesic metric: delete $I \setminus J$ and then insert $J \setminus I$ to obtain a path of length $d(I,J)$.

Lemma 3.4. *For each fixed p satisfying* (3.1.1) *and all allowed I, J, the rows $K_p(I,\cdot)$ and $K_p(J,\cdot)$ admit a coupling (X,Y) with*

$$d(X,Y) \le d(I,J) \quad \textit{almost surely}, \qquad \mathbb{E}d(X,Y) \le (1-\kappa)d(I,J).$$

Proof. First let $J = I \cup \{v\}$ be adjacent. Couple the chosen update coordinates synchronously. At v, use the coupling with masses a_v/n on (I,I), m_v/n on (J,J), and $(1-a_v-m_v)/n$ on (I,J). This is valid because $a_v + m_v \le 1$. Away from v, couple identical births or deaths whenever possible. The only pairs at distance two are

$$(I \cup \{u\}, J), \qquad u \in N(v), \quad I \cap N[u] = \varnothing,$$

each of mass m_u/n. Write B for their total mass, so $B \le s_v/n \le a_v/n$.

For each such u, remove mass m_u/n both from its distance-two pair and from the good pair (I,I), replacing it with the same mass on each of

$$(I \cup \{u\}, I), \qquad (I,J).$$

Both coupling marginals are unchanged, sufficient good mass is available, and all resulting distances are at most one. The expected distance is unchanged: the replaced distances were two and zero, while the new distances are one and one. Hence it is

$$1 - \frac{a_v + m_v}{n} + B \le 1 - \frac{a_v + m_v - s_v}{n} \le 1 - \frac{1 - 2s_v}{n} \le 1 - \kappa.$$

The penultimate step uses the union bound $a_v + m_v \ge 1 - s_v$. Glue these couplings along a Hamming geodesic between arbitrary I, J. The triangle inequality gives both the almost-sure and expected bounds. □

We invoke the following precise finite-state case of [3.2, Theorem 2, $M = 1$ and $\ell = 1$]. If K is reversible for a positive probability ν, is nonexpanding in W_∞, and contracts W_1 by a factor $1 - \kappa$, then

$$D(qK\|\nu) \le (1-\kappa)D(q\|\nu) \qquad \text{for every probability } q. \tag{3.3.1}$$

Its regularity hypothesis is automatic on the finite metric space. This is a full discrete-step entropy inequality; no conversion from a continuous-time modified logarithmic Sobolev inequality is used. Lemma 3.4 verifies its hypotheses for each frozen K_p.

3.4 Transfer through the conserved exponential family

Both μ_m and ν_p are Gibbs measures on the same allowed support, so $\log(\mu_m/\nu_p)$ is an affine combination of free-site occupancies. If q has the same marginals as p, then

$$D(q\|\nu_p) = D(q\|\mu_m) + D(\mu_m\|\nu_p). \tag{3.4.1}$$

Applying (3.3.1) to $q = p$, and then (3.4.1) to p and Tp, gives the slightly stronger inequality

$$D(Tp\|\mu_m) \le (1-\kappa)D(p\|\mu_m) - \kappa D(\mu_m\|\nu_p).$$

Discard its last nonpositive term. The marginal profile and κ are conserved, so iteration proves the entropy assertion. Finally,

$$D(p\|\mu_m) = H(\mu_m) - H(p) \le \sum_{v\in V} h(m_v) \le n\log 2.$$

Pinsker's inequality and $(1-\kappa)^t \le e^{-\kappa t}$ prove the stated total-variation bound. In the uniform regime $\beta \le 1/(2\Delta+1) \le 1/3$, binary entropy is increasing on $[0,\beta]$, giving $\sum_v h(m_v) \le nh(\beta)$. The uniform condition in Theorem 3.2 implies (3.1.1), because $m_v + 2s_v \le (2\Delta+1)\beta \le 1$, and gives $1-2s_v \ge 1-2\Delta\beta$. On an edgeless graph all $s_v = 0$ and the kernel is ordinary coordinate refresh at rate $1/n$. This completes both proofs.

3.5 The discrete-time mean-field extension

The mean-field map T_{mf} chooses the updated vertex v and donor vertex u independently uniformly from V, rather than setting $u = v$. It conserves the mean density $\alpha = \mathbb{E}_p|I|/n$. Denote its maximum-entropy Gibbs equilibrium by μ_α.

Theorem 3.5. *Suppose $\Delta \ge 1$ and $0 \le \alpha \le 1/(2\Delta+1)$. Put $\kappa_\alpha = (1-2\Delta\alpha)/n$. Then*

$$D(T_{\mathrm{mf}}^t p\|\mu_\alpha) \le (1-\kappa_\alpha)^t D(p\|\mu_\alpha), \qquad \|T_{\mathrm{mf}}^t p - \mu_\alpha\|_{\mathrm{TV}} \le \sqrt{\frac{nh(\alpha)}{2}}\, e^{-\kappa_\alpha t/2}.$$

The zero-density case is trivial. On edgeless graphs the same statements hold with $\kappa_\alpha = 1/n$, including the deterministic endpoint densities.

Proof. Assume $\alpha > 0$, and set $A(p) = n^{-1}\sum_u p(I \cap N[u] = \varnothing)$. The frozen kernel inserts at every available vertex with probability α/n and removes each occupied vertex with probability $A(p)/n$. Indeed the donor's randomly chosen bit is occupied with probability α, while a deletion requires its randomly chosen closed neighborhood to be vacant. Thus its reversible Gibbs reference ν_p has common fugacity $\alpha/A(p)$. The identities and union bounds give

$$\alpha + A(p) \le 1, \qquad A(p) \ge 1 - (\Delta+1)\alpha \ge \Delta\alpha > 0.$$

For adjacent configurations the coupling proof above applies with m_v replaced by α and a_v by $A(p)$. Bad mass is at most $\Delta\alpha/n$, and the diagonal mass $A(p)/n$ is sufficient for the same repair. The expected distance is at most

$$1 - \frac{A(p)+\alpha-\Delta\alpha}{n} \le 1 - \frac{1-2\Delta\alpha}{n}.$$

Caputo–Salez therefore gives full-step entropy contraction for this frozen kernel.

The equilibrium exists and has full support: the preceding maximum-entropy support argument applies, since the positive average vacancy $A(p)$ allows deletion of any occupied vertex by transfer to some vacant closed neighborhood. Once the empty set belongs to the support, cross-site transfers supply every singleton and then every independent set. At fixed density, entropy uniqueness gives a Gibbs law with common fugacity. Now $\log(\mu_\alpha/\nu_p)$ is affine in $|I|$, whose expectation is conserved. The same Pythagorean calculation proves the one-step inequality and its iteration. Finally,

$$D(p\|\mu_\alpha) = H(\mu_\alpha) - H(p) \le \sum_v h(\mu_\alpha(v \in I)) \le nh(\alpha)$$

by concavity of binary entropy. Pinsker completes the proof. In the edgeless case the frozen kernel is ordinary Bernoulli-α coordinate refresh, and the deterministic endpoints have zero error. □

This is a discrete-time statement. The broader critical-density entropy estimate in [3.1, Theorem 1.3] concerns the continuous-time equation $\dot{p} = T_{\mathrm{mf}}(p) - p$; it is not being transferred to discrete time here. Theorem 3.5 instead extends the discrete-time low-density regime of [3.1, Theorem 1.2] by the direct coupling just proved.

3.6 Remaining limitation

The threshold $1/(2\Delta + 1)$ arises from the sufficient supply of diagonal coupling mass $a_v \geq \sum_{u \in N(v)} m_u$ (or $A \geq \Delta\alpha$ in the mean-field case). This need not hold throughout $m_v < 1/(\Delta + 1)$. No extension beyond this coupling condition, and no claim to have resolved the full near-linear-rate conjecture, is made. The result and its proof are also separate from the earlier qualitative-to-exponential compactness argument: the bound here is global, has an explicit prefactor, and is obtained in one discrete step.

Chapter 4

A Tail Obstruction to an Exponential Convergence Conjecture

Chapter overview

The constant exchange kernel satisfies the monotonicity and growth conditions in the concluding conjecture of Esenturk and Velazquez, but permits initial laws for which convergence to equilibrium cannot be exponential, even in total variation. A Poisson-clock coupling gives a necessary exponential-tail condition and an explicit counterexample with finite second moment, as well as a variant with every finite polynomial moment. This is a direct application of a known slow-tail mechanism, discussed by Graham for the same mean-field zero-range dynamics. It does not contradict the stronger hypotheses of the proved exponential-convergence theorem.

4.1 Scope and the constant-kernel dynamics

The concluding conjecture in [4.1] asks for exponential convergence under the hypotheses of its Theorem 1 and monotonicity of the exchange kernel: nondecreasing in the donor coordinate and nonincreasing in the receiver coordinate. The relevant growth and data assumptions are $K(j,k) \le Cjk$ for large j,k and $M_p(0) < \infty$ for some $p > 1$, where $M_p(t) := \sum_{k\ge1} k^p c_k(t)$. Consider

$$K(j,k) = 1 \quad (j \ge 1,\ k \ge 0), \qquad K(0,k) = 0. \tag{4.1.1}$$

This kernel has all these properties. Normalize the total cluster concentration to one and write $c_k(t)$ for the concentration at size $k \in \mathbb{N}_0$. The exchange equations reduce to

$$\dot c_0 = c_1 - b(t)c_0, \tag{4.1.2}$$

$$\dot c_k = c_{k+1} + b(t)c_{k-1} - (1+b(t))c_k, \qquad k \ge 1, \tag{4.1.3}$$

$$b(t) = \sum_{k\ge1} c_k(t) = 1 - c_0(t) \in [0,1].$$

Thus $c(t)$ is the law of a reflected birth–death process with unit death rate away from zero and birth rate $b(t)$.

Here and below, for probability laws u, v on $\mathbb{N}_0$,

$$d_{\mathrm{TV}}(u,v) = \sup_{A\subseteq\mathbb{N}_0} |u(A) - v(A)| = \frac{1}{2}\sum_{k\ge0} |u_k - v_k|, \qquad \|u - v\|_{1,+} = \sum_{k\ge1} |u_k - v_k|. \tag{4.1.4}$$

The second expression is the positive-coordinate norm in the exponential estimate under discussion. In particular, for $A \subseteq \{1,2,\dots\}$, both quantities in (4.1.4) are at least $u(A) - v(A)$. More generally $d_{\mathrm{TV}}(u,v) \le \|u - v\|_{1,+}$, since $|u_0 - v_0| \le \|u - v\|_{1,+}$.

4.2 A probability representation and a necessary tail condition

For completeness, (4.1.2)–(4.1.3) define a unique global probability-valued solution for every initial probability law. Indeed, the death and birth operators are bounded on ℓ^1, and $c \mapsto 1 - c_0$ is Lipschitz. The resulting vector field is locally Lipschitz, so the Banach-space ordinary differential equation has a unique local solution. Nonnegativity and total mass one are preserved (equivalently, use the positive, conservative propagator for its bounded time-dependent generator). The ℓ^1 norm therefore stays one, which gives global existence.

Once this deterministic function $b(t)$ is fixed, construct the process from independent rate-one Poisson clocks: a death-attempt clock D_t, whose rings remove a particle only when the state is positive, and a birth-attempt clock B_t, thinned with acceptance probability $b(t)$. Take both clocks, and the thinning marks, independent of X_0. Uniqueness for the bounded linear forward equation identifies its law with $c(t)$. Pathwise,

$$X_0 - D_t \le X_t \le X_0 + B_t, \qquad D_t, B_t \sim \mathrm{Pois}(t). \tag{4.2.1}$$

If $\mathbb{E}X_0^q < \infty$ for some $q > 0$, the upper bound and the finite moments of B_t propagate that moment on every finite time interval. In particular, when $\mathbb{E}X_0 < \infty$, compensation of the two jump counts gives

$$\mathbb{E}X_t - \mathbb{E}X_0 = \int_0^t \big[b(s) - \mathbb{P}(X_s > 0)\big]\, ds = 0.$$

Thus the mean ρ is conserved, without any formal interchange of an infinite weighted series.

The stationary probability laws are precisely

$$q_k = (1-r)r^k, \qquad 0 \le r < 1. \tag{4.2.2}$$

To see this, the stationary flux $J_k = bq_k - q_{k+1}$ satisfies $J_0 = 0$ by (4.1.2) and $J_k = J_{k-1}$ by (4.1.3). Hence $q_{k+1} = bq_k$. Normalization gives (4.2.2) with $r = b$; $b = 1$ cannot give a probability law. The mean is $r/(1-r)$, so the equilibrium of mean ρ has $r = \rho/(1+\rho)$.

Lemma 4.1 (An exponential initial tail is necessary)**.** *Suppose a probability curve $\mu_t = \mathcal{L}(X_t)$ admits the lower coupling in* (4.2.1)*, with D_t independent of X_0. Let q satisfy*

$$Q(m) := q(\{m, m+1, \ldots\}) \le Ae^{-am} \quad (m \ge 1)$$

for some $A, a > 0$. If $d_{\mathrm{TV}}(\mu_t, q) \le Ce^{-ct}$ for all $t \ge 0$, with $C, c > 0$, then, for every integer $N \ge 1$,

$$\mathbb{P}(X_0 \ge N) \le 2Ce^{-cN/4} + 2Ae^{-aN/2}. \tag{4.2.3}$$

Proof. Set $t = N/4$ and $L = \lceil N/2 \rceil$. Markov's inequality gives $\mathbb{P}(D_t > N/2) \le \mathbb{E}D_t/(N/2) = 1/2$. On $\{X_0 \ge N,\ D_t \le N/2\}$, the integer-valued lower bound in (4.2.1) implies $X_t \ge L$. Independence therefore yields

$$\tfrac{1}{2}\mathbb{P}(X_0 \ge N) \le \mathbb{P}(X_t \ge L) \le d_{\mathrm{TV}}(\mu_t, q) + Q(L).$$

Substitute the assumed bounds and use $L \ge N/2$. □

The same conclusion, with adjusted constants, holds if the assumed exponential convergence starts only at a finite time. This is a necessary condition, not a sufficiency statement. In particular, finitely many polynomial moment assumptions cannot replace it for these dynamics.

4.3 Explicit initial laws

Proposition 4.2 (A finite-second-moment counterexample)**.** *For* (4.1.2)–(4.1.3)*, take*

$$c_0(0) = \frac{4}{7}, \qquad c_{2^j}(0) = \frac{3}{8^j} \quad (j \geq 1), \qquad c_k(0) = 0 \quad \textit{otherwise.} \tag{4.3.1}$$

This is a probability law of mean one and second moment three. With $\pi_k = 2^{-k-1}$ *and* $t_j = 2^{j-2}$*, for every* $j \geq 6$*,*

$$d_{\mathrm{TV}}(c(t_j), \pi) \geq \frac{3}{256\, t_j^3}, \qquad \|c(t_j) - \pi\|_{1,+} \geq \frac{3}{256\, t_j^3}. \tag{4.3.2}$$

Consequently neither distance admits a bound Ce^{-ct} *with* $C, c > 0$*, even when the constants may depend on this initial law.*

Proof. The geometric-series identities are

$$\frac{4}{7} + 3\sum_{j\geq 1} 8^{-j} = 1, \qquad 3\sum_{j\geq 1} 4^{-j} = 1, \qquad 3\sum_{j\geq 1} 2^{-j} = 3.$$

More generally, $M_p(0) = 3\sum_{j\geq 1} 2^{(p-3)j} < \infty$ for every $0 < p < 3$. By the preceding construction, the solution is global and retains mean one, whose stationary law is π.

Put $L_j = 2^{j-1} = 2t_j$. By Markov's inequality, $\mathbb{P}(D_{t_j} \leq 2t_j) \geq 1/2$. A particle starting at 2^j is then still at or above L_j. Independence and (4.2.1) give

$$c(t_j)(\{L_j, L_j + 1, \ldots\}) \geq \frac{3}{2 \cdot 8^j}, \qquad \pi(\{L_j, L_j + 1, \ldots\}) = 2^{-2^{j-1}}. \tag{4.3.3}$$

For $j \geq 6$, $2^{-2^{j-1}} \leq 3/(4 \cdot 8^j)$. For example, the inequality holds at $j = 6$, and $2^{j-1} - 3j$ is increasing thereafter. Applying (4.1.4) to this positive tail set proves

$$d_{\mathrm{TV}}(c(t_j), \pi),\ \|c(t_j) - \pi\|_{1,+}\ \geq \frac{3}{4 \cdot 8^j} = \frac{3}{256\, t_j^3}.$$

Since $t_j \to \infty$, this lower bound is incompatible with any Ce^{-ct}. □

The obstruction persists even if every polynomial moment is finite. For a second example, set

$$c_{2^j}(0) = 2^{-j^2-j} \quad (j \geq 1), \qquad c_0(0) = 1 - \sum_{j\geq 1} 2^{-j^2-j}, \tag{4.3.4}$$

with all other coordinates zero. The sum is less than one, and

$$\rho = \sum_{j\geq 1} 2^{-j^2}, \qquad M_q(0) = \sum_{j\geq 1} 2^{-j^2+(q-1)j} < \infty \quad \text{for every } q > 0.$$

For its mean-matched equilibrium $q_k = (1 - r)r^k$, $r = \rho/(1 + \rho) \in (0, 1)$, the same tail event at $t_j = 2^{j-2}$ gives

$$d_{\mathrm{TV}}(c(t_j), q),\ \|c(t_j) - q\|_{1,+}\ \geq 2^{-j^2-j-1} - r^{2^{j-1}}\ \geq 2^{-j^2-j-2}$$

for all sufficiently large j. The last inequality follows because 2^j grows faster than j^2. The resulting lower bound again dominates every Ce^{-ct_j} eventually. Neither example has an exponentially bounded initial tail. The argument also excludes exponential approximation to any other fixed stationary law, since every such law has a geometric tail.

4.4 What the counterexample does and does not establish

Theorem 10 of [4.1] imposes more than the concluding conjecture. In particular, its structural hypothesis H3 includes

$$K(j,k) = ja_k + b_j + \varepsilon\beta_j\alpha_k, \qquad a_k \geq \widetilde{a} > 0,$$

with b_j nondecreasing and a bounded perturbation. For fixed k, the right side is at least $j\widetilde{a} + b_1 - O(1)$ and diverges as $j \to \infty$, unlike (4.1.1). The accepted Theorem 10 also requires $p > 2$, which (4.3.1) can meet. The failure identified here concerns the broader conjecture, not the proved theorem. No convergence theorem, or optimal decay-rate assertion, is needed for the counterexample.

The mechanism is not new. Graham [4.2], Section 1 (journal p. 500; arXiv PDF p. 4), explicitly explains that the initial tail affects relaxation and obstructs exponential relative-entropy decay when it is not geometrically bounded. Section 5 describes the same time-inhomogeneous birth–death equation used above. His relative-entropy observation alone does not imply a lower bound in total variation: the usual entropy-to-total-variation inequality has the opposite logical direction. The coupling proof here supplies that direct lower bound and makes the obstruction explicit for the later conjecture. This note makes no claim of priority for the mechanism or for identifying its relevance to that conjecture.

Part II

Partition Products and Coefficient Inequalities

These three chapters concern coefficient inequalities for partition-related generating functions. Each retains its complete analytic argument, finite coverage analysis and executable certificates. Their claims are separate: denominator log concavity, shifted-polynomial unimodality and regular-partition log concavity. Unimodality in Chapter 6 must not be read as global coefficient log concavity.

Chapter 5

Log Concavity of Reciprocal Subsum Denominators

Chapter overview

We prove that the coefficients of the reciprocal-subsum denominator $D_n(x)$ of Ballantine–Beck–Feigon–Maurischat are log-concave exactly when $n \notin \{3, 5, 6, 7\}$. The proof groups its known dyadic factorization into balanced four-row blocks. A saddle-point determinant identity, with explicit phase and entire-frequency tail control, proves strict log-concavity away from a bounded edge. An effective eta–Bessel estimate for four-colored distinct partitions and exact truncation bounds settle all block parameters at least 28000. Integer-only directed dyadic certificates establish the remaining 1 119 898 742 edge inequalities for parameters 257 through 27999; smaller bases are checked exactly. All rational analytic checkers and the finite certificate engine are embedded as reproducible source listings. The finite and analytic regions are explicitly matched, including floors, midpoints and boundary indices.

5.1 The denominator and the theorem

For a finite sequence of nonnegative numbers, log-concavity means $a_k^2 \geq a_{k-1}a_{k+1}$ at every internal index. Strict log-concavity means that all these inequalities are strict. A polynomial is called log-concave when its coefficient sequence is log-concave. All products considered below have positive coefficients throughout their degree intervals.

For a partition λ put $\mathrm{sp}(\lambda, x) = \prod_i (1 + x^{\lambda_i})$. Ballantine, Beck, Feigon and Maurischat define the denominator $D_n(x) = \mathrm{den}(n, x)$ associated to the sum of the reciprocals of these subsum polynomials. More explicitly, put $D_n^*(x) = \prod_{i=1}^{n}(1 + x^i)^{\lfloor n/i \rfloor}$, $h_\lambda = D_n^*/\mathrm{sp}(\lambda, x)$ and $G_n = \gcd_{\lambda \vdash n} h_\lambda$, with monic normalization; then $D_n = D_n^*/G_n$. Their Proposition 3.8(a) gives the identity

$$D_n(x) = \prod_{i=1}^{n}(1 + x^i)^{1+\lfloor \log_2(n/i) \rfloor} = \prod_{r \geq 0} Q_{\lfloor n/2^r \rfloor}(x), \qquad Q_m(x) = \prod_{j=1}^{m}(1 + x^j), \quad Q_0 = 1. \tag{5.1.1}$$

We use this known factorization, and do not claim it as a result of this paper. In particular the object here is exactly the ordinary denominator in BBFM Conjecture 4, not a binary numerator or a modified conjecture [5.1]. The corresponding entry of Table 1 of Chen–Ono–Zhang is marked open [5.2].

Theorem 5.1.1 (Denominator log-concavity)**.** *For every positive integer n, the coefficients of D_n are log-concave if and only if $n \notin \{3, 5, 6, 7\}$.*

The proof is computer-assisted in a precise, finite sense. Its analytic part treats continuously varying saddle parameters using rational interval envelopes, and treats all sufficiently large block sizes. Its remaining finite coefficient inequalities are certified by integer-only directed dyadic arithmetic. The complete source listings needed to reproduce the calculations are included in this standalone manuscript. Ordinary machine integer arithmetic, a C++17 compiler supporting 128-bit integers, and Python arbitrary-precision integers and rational fractions are the computational trust base; this is not a proof-assistant formalization.

Define the balanced four-row block

$$B_n(x) = Q_n(x)Q_{\lfloor n/2 \rfloor}(x)Q_{\lfloor n/4 \rfloor}(x)Q_{\lfloor n/8 \rfloor}(x) = \sum_{k=0}^{M_n} b_{n,k}x^k, \quad M_n = \sum_{a=0}^{3} \frac{\lfloor n/2^a \rfloor(\lfloor n/2^a \rfloor + 1)}{2}. \tag{5.1.2}$$

Theorem 5.1.2 (Balanced blocks)**.** *For every integer $n \geq 8$, B_n is strictly log-concave at every internal coefficient index.*

Here is the coverage map for its proof. Palindromicity always permits restriction to $1 \leq k \leq \lfloor M_n/2 \rfloor$.

Block parameter	Index region	Proof component
$8 \leq n \leq 256$	Entire half-range	Exact finite base
$257 \leq n \leq 27999$	$k \leq 40399$	Directed dyadic certificate
$n \geq 100$	Saddle $0 \leq t \leq 1/100$	Fourier determinant estimate
$n \geq 28000$	$1 \leq k \leq 40400$	Four-color truncation transfer

The Fourier estimate implies that any uncovered lower-half index for $n \geq 100$ is strictly smaller than 40400. Thus these regions overlap and exhaust the domain. No numerical sampling of an unbounded parameter is used to infer a universal inequality.

5.1.1 Relation to prior work

The factorization (5.1.1) and its increment recurrence are from BBFM. Convolution preservation, eta transformation, Bessel identities, Fourier inversion and elementary interval arithmetic are classical tools. Agarwal–Garg–Maji, Theorem 2.6, already imply eventual log-concavity of the four-colored distinct-part sequence by specializing their colored regular partition theorem to $(r, \ell, d) = (4, 2, 2)$ [5.3]. We need an explicit error bound and a verified all-index version, which we supply below. Chern's work treats Fourier coefficients of eta quotients including weight zero [5.4]; a direct contour argument here avoids any reliance on a negative-weight-only Rademacher formula. The contribution asserted by this manuscript is the explicit balanced-block proof and its assembled finite certificates, followed by the denominator conclusion. No priority claim for the auxiliary infinite sequence, or for a new general principle of log-concavity, is needed.

5.2 Elementary reduction to balanced blocks

Lemma 5.2.1 (Convolution)**.** *The convolution of two nonnegative log-concave sequences with interval support is log-concave and has interval support.*

Proof. Extend each sequence by zero to the integers. For such a sequence a, its Toeplitz matrix $T(a)_{ij} = a_{j-i}$ has nonnegative minors of size two. Indeed, on positive entries this follows by multiplying inequalities of the decreasing ratios a_{r+1}/a_r: among pairs with equal sum, the pair with indices closer together has the larger product. If a diagonal entry is zero while both off-diagonal entries are positive, interval support rules this out; all other zero cases are immediate. Conversely, adjacent minors give log-concavity. Since $T(a * b) = T(a)T(b)$, the size-two Cauchy–Binet formula expresses every minor of the product as a sum of products of nonnegative minors. Each relevant sum is finite for finite supports. The convolution support is the sum of the two support intervals. □

The support of Q_m is its entire interval: by induction the subset sums of $1, \dots, m$ fill $[0, m(m+1)/2]$. Products inherit this property. For $n \geq 8$, choose the unique $j \geq 0$ with $8 \cdot 16^j \leq n < 128 \cdot 16^j$ and set $m_s = \lfloor n/16^s \rfloor$. Nested floors in (5.1.1) give

$$D_n = B_{m_0} B_{m_1} \cdots B_{m_{j-1}} D_{m_j}, \qquad m_s \geq 128 \ (s < j), \quad 8 \leq m_j \leq 127. \tag{5.2.1}$$

Consequently Theorem 5.1.2 and the finite base $D_8, \dots, D_{127}$ imply Theorem 5.1.1 for $n \geq 8$ by Lemma 5.2.1. The small cases and the finite bases are treated explicitly in the final verification section.

5.3 Fourier inversion and the saddle criterion

Consider first any polynomial $P(x) = \prod_j (1 + x^j)^{m_j}$ of degree M, where the multiplicities are nonnegative integers and $m_1 > 0$. For an integer $0 < k \leq M/2$, the strictly decreasing continuous function

$$\mu(t) = \sum_j \frac{m_j j}{1 + e^{tj}}$$

has a unique solution $t \geq 0$ to $\mu(t) = k$; its range is $(0, M/2]$. Write $r = e^{-t}$, $p_j = r^j/(1 + r^j)$, $v_j = p_j(1 - p_j)$, $\sigma^2 = \sum_j m_j j^2 v_j$ and $M_q = \sum_j m_j j^q v_j$. The centered characteristic function is

$$\phi(\theta) = \frac{P(re^{i\theta})}{P(r)} e^{-ik\theta}.$$

Fourier inversion gives $r^{k+s}a_{k+s}/P(r) = (2\pi)^{-1}\int_{-\pi}^{\pi}\phi(\theta)e^{-is\theta}\,d\theta$ for integral s. Multiplication for $s = 0$ and for $s = -1, 1$, followed by symmetrization, therefore gives the exact determinant identity

$$\frac{r^{2k}(a_k^2 - a_{k-1}a_{k+1})}{P(r)^2} = \frac{1}{(2\pi)^2}\operatorname{Re}\iint_{[-\pi,\pi]^2}\phi(\theta)\phi(\psi)[1-\cos(\theta-\psi)]\,d\theta\,d\psi. \tag{5.3.1}$$

The use of $\phi(\theta)\phi(\psi)$, not a complex conjugate, is important. The imaginary part vanishes by simultaneous reflection of both variables.

On a symmetric core $C = [-\delta, \delta]$ choose the continuous phase $\phi = Ae^{i\alpha}$, with A even and α odd. Put

$$J_q = \int_C \theta^q A(\theta)\,d\theta, \quad H_q = \int_C \theta^q\alpha(\theta)^2A(\theta)\,d\theta, \quad \Lambda = \int_C \theta\alpha(\theta)A(\theta)\,d\theta.$$

The inequalities $1 - \cos u \le u^2/2$ and $\cos v \ge 1 - v^2/2$ bound the phase loss on the core by one quarter of

$$\iint_{C^2}(\theta-\psi)^2(\alpha(\theta)+\alpha(\psi))^2A(\theta)A(\psi)\,d\theta\,d\psi = 2(H_2J_0 + H_0J_2) - 4\Lambda^2. \tag{5.3.2}$$

This identity follows simply by expansion and parity. Discarding the negative square is safe; replacing absolute cubic phase bounds by a signed pointwise cubic expression would not be safe. If $|\alpha(\theta)| \le c|\theta|^3$ and $A(\theta) \le e^{-a\sigma^2\theta^2/2}$ on C, full Gaussian moments bound the loss by

$$\frac{c^2}{2}(J_8J_0 + J_6J_2) \le \frac{120\pi c^2}{a^5\sigma^{10}}. \tag{5.3.3}$$

Here the Gaussian double factorials are 105 and 15.

For $T_q = \int_{[-\pi,\pi]\setminus C}\theta^qA(\theta)\,d\theta$, parity and the quadratic kernel bound show that the absolute contribution outside C^2 is at most

$$T_2J_0 + T_0J_2 + T_0T_2. \tag{5.3.4}$$

All subsequent bounds on the double integral omit the common positive factor $(2\pi)^{-2}$ and are normalized by multiplication by σ^4.

5.4 Balanced moment and tail certificates

5.4.1 Moment bounds with real scale and floored row lengths

Lemma 5.4.1. *Let $n \ge h \ge 100$, with n integral, and set $v(x) = e^x/(1+e^x)^2$, $S_q = \sum_{a=0}^{3}\sum_{j=1}^{\lfloor n/2^a\rfloor} j^qv(j/h)$, $L = \min(n, 5h)$. With $\sigma^2 = S_2$, $M_3 = S_3$, $M_4 = S_4$,*

$$M_3^2/\sigma^6 < 15/(2h), \qquad \sigma^2/L^2 > 79h/1000, \qquad M_4/\sigma^2 < 14h^2, \qquad \sigma^2 < 7h^3.$$

Proof.

5.4.1.1 A single error bound, including the floor

For a nonnegative differentiable unimodal function f with $f(0) = 0$, let $M_x = \max_{0\le u\le x} f(u)$. For any real endpoint $x \ge 0$ and mesh $\Delta > 0$,

$$\left|\Delta\sum_{j=1}^{\lfloor x/\Delta\rfloor} f(j\Delta) - \int_0^x f(u)\,du\right| \le \Delta M_x. \tag{5.4.1}$$

In particular, a separate floor error need not be added to the usual Riemann-sum error. To prove (5.4.1), put $r(u) = u - \Delta\lfloor u/\Delta \rfloor$. Integration by parts against the step function gives the exact error

$$E = \int_0^x r(u) f'(u)\, du - r(x) f(x).$$

Its upper bound is at most $\Delta \int_0^x (f')_+ \le \Delta M_x$. If f has not reached its maximum by x, the lower bound is $-\Delta f(x) = -\Delta M_x$. Otherwise, the negative variation is $M_x - f(x)$, so the lower bound is again $-\Delta(M_x - f(x)) - \Delta f(x) = -\Delta M_x$. This also handles an endpoint on the mesh, where $r(x) = 0$.

Write $f_q(x) = x^q v(x)$, $I_q(z) = \int_0^z f_q(x)\, dx$. The logarithmic derivative $q/x - \tanh(x/2)$ decreases strictly, so f_q is unimodal. Since $v(x) \le 1/4$ and $v(x) \le e^{-x}$,

$$\max_{0 \le u \le x} f_2(u) \le A_2(x) := \min(1, x^2/4), \qquad \max_{0 \le u \le x} f_3(u) \le A_3(x) := \min(2, x^3/4).$$

The global constants follow from $4/e^2 < 1$ and $27/e^3 < 2$. With $z = n/h$, (5.4.1) applied to each row gives

$$\left| \frac{S_q}{h^{q+1}} - F_q(z) \right| \le \frac{E_q(z)}{h}, \quad F_q(z) := \sum_{a=0}^{3} I_q(z/2^a), \quad E_q(z) := \sum_{a=0}^{3} A_q(z/2^a). \tag{5.4.2}$$

This is exactly where the actual balanced multiplicities are retained.

5.4.1.2 Rational grid certificate

The grid has denominator $G = 256$, with integral endpoints i/G for $0 \le i \le 16G$. The exponential is enclosed by its positive Taylor sum through degree 150 and its geometric remainder

$$T_{150}(x) \le e^x \le T_{150}(x) + \frac{x^{151}/151!}{1 - x/152}, \qquad 0 \le x \le 16.$$

Bounds are rounded outward to denominator 10^{18}. Because $u/(1+u)^2$ decreases for $u \ge 1$, substituting the exponential bounds in reverse order encloses $v(x)$. On a grid cell $[l, u]$, monotonicity of v then gives rational lower and upper integrals by multiplying the appropriate endpoint bound by $(u^{q+1} - l^{q+1})/(q+1)$. Their cumulative sums are denoted by L_q, U_q.

For a parameter cell $z \in [i/G, (i+1)/G]$, the exact checker in Appendix 5.A forms

$$s_i = \sum_{a=0}^{3} L_2\left(\frac{\lfloor i/2^a \rfloor}{G} \right) - \frac{E_2((i+1)/G)}{100},$$

$$t_i = \sum_{a=0}^{3} U_3\left(\frac{\lceil (i+1)/2^a \rceil}{G} \right) + \frac{E_3((i+1)/G)}{100}.$$

By (5.4.2) and monotonicity, $S_2/h^3 \ge s_i$ and $S_3/h^4 \le t_i$. The exact checks establish

$$s_i > 0, \qquad t_i^2/s_i^3 < 15/2 \quad (G \le i < 16G),$$

and

$$\frac{s_i}{((i+1)/G)^2} > 79/1000 \quad (G \le i < 5G).$$

For $z \ge 16$, use

$$S_2/h^3 \ge F_2(16) - 4/100, \qquad S_3/h^4 \le 4 I_3(\infty) + 8/100.$$

Here $I_3(\infty) = \frac{9}{2}\zeta(3)$ and $\zeta(3) < 1203/1000$, the latter following from the bound $\sum_{j=1}^{10} j^{-3} + 1/200 < 1203/1000$. These bounds prove the same ratio throughout the infinite remaining parameter range.

For $z \geq 5$, monotonicity of the row sums in their endpoints gives

$$\frac{\sigma^2}{25h^3} \geq \frac{F_2(5) - E_2(5)/100}{25} > 79/1000.$$

Together with the finite grid this proves the core-variance inequality.

5.4.1.3 Fourth moment and upper variance

To justify the fourth-moment reduction explicitly, put $w_j = j^2 v(j/h)$. For any finite upper endpoint, the identity

$$\Big(\sum w_j\Big)\Big(\sum w_j m_j j^2\Big) - \Big(\sum w_j m_j\Big)\Big(\sum w_j j^2\Big) = \frac{1}{2}\sum_{i,j} w_i w_j (m_i - m_j)(i^2 - j^2) \leq 0$$

holds because the balanced multiplicities m_j decrease. It bounds M_4/σ^2 by the single-row truncated ratio. Extending that row only increases the weighted average of j^2. Since $\max f_4 < 5$, the integral identities $I_4(\infty) = 7\pi^4/30$ and $I_2(\infty) = \pi^2/6$ give

$$\frac{M_4}{h^2\sigma^2} \leq \frac{7(22/7)^4/30 + 5/100}{(3.14)^2/6 - 1/100} < 14.$$

Finally each of the four rows is bounded above by its infinite sum; (5.4.1) and $\max f_2 < 1$ imply

$$\frac{\sigma^2}{h^3} \leq 4\left(\frac{(22/7)^2}{6} + \frac{1}{100}\right) < 7.$$

For completeness, the integrals used here follow by integration by parts from $v(x) = -\frac{d}{dx}(1 + e^x)^{-1}$ and expansion into exponentials:

$$I_q(\infty) = q!\,(1 - 2^{1-q})\zeta(q) \quad (q > 1).$$

All finite comparisons in this proof are reproduced by the moment checker in Appendix 5.A (`verify_balanced_moments.py`). Its loops cover all 3840 parameter cells, while the separate monotone bounds cover $z \geq 16$ and the variance range $z \geq 5$. □

5.4.2 Absolute Fourier tails

Lemma 5.4.2. *Under the hypotheses of Lemma 5.4.1, let* $A(\theta) = |B_n(e^{-1/h+i\theta})/B_n(e^{-1/h})|$ *and* $L = \min(n, 5h)$. *Then*

$$\sigma \int_{2/L \leq |\theta| \leq \pi} A(\theta)\, d\theta < 1/3000, \qquad \sigma^3 \int_{2/L \leq |\theta| \leq \pi} \theta^2 A(\theta)\, d\theta < 1/50.$$

The same bounds hold with $L = n$ *and the actual tilted variance when* $n \geq 100$ *and* $0 \leq t \leq 1/n$.

Proof. The following finite rational envelope construction proves the inequalities uniformly on the indicated continuous domain. Its full implementation is `verify_balanced_tail_h100.py` in Appendix 5.A.

5.4.2.1 Global modulus and Abel summation

Write $m_j = \#\{a : j \le \lfloor n/2^a \rfloor\}$ and $w_j = m_j v(j/h)$. The sequence w_j decreases, and $w_1 \le 1$. The Bernoulli product gives

$$\log A(\theta) \le -2\sum_j w_j \sin^2(j\theta/2) = -W_0 + \sum_j w_j \cos(j\theta), \quad W_0 = \sum_j w_j.$$

For $0 < |\theta| \le \pi$, every partial cosine sum is at most

$$\sum_{j=1}^{N} \cos(j\theta) = \frac{\sin((N+1/2)|\theta|)}{2\sin(|\theta|/2)} - \frac{1}{2} \le C(\theta) := \frac{1}{2\sin(|\theta|/2)} - \frac{1}{2}.$$

Abel summation, and $C \ge 0$, therefore give

$$\sum_j w_j \cos(j\theta) \le w_1 C(\theta) \le C(\theta).$$

This is the global gain over a bound using only the first $\lfloor h \rfloor$ weights.

Put $z = n/h$ and

$$J(z) = \sum_{a=0}^{3} \int_0^{z/2^a} v(x)\,dx = \sum_{a=0}^{3} \left(\frac{1}{2} - \frac{1}{1+e^{z/2^a}} \right).$$

Since v decreases, for $N = \lfloor n/2^a \rfloor$,

$$\sum_{j=1}^{N} v(j/h) \ge h \int_{1/h}^{(N+1)/h} v(x)\,dx \ge h \int_0^{z/2^a} v(x)\,dx - \frac{1}{4}.$$

Consequently $W_0 \ge hJ(z) - 1$.

For $s = h|\theta| \le 8$, one has $|\theta| \le 0.08 < 1$, so $\sin(|\theta|/2) \ge 23|\theta|/48$. Hence, writing $H = 100$,

$$A(s/h) \le \exp\{-hF_A(s,z)\}, \quad F_A(s,z) = J(z) - \frac{1}{2H} - \frac{24}{23s}.$$

For every $8/h \le |\theta| \le \pi$, monotonicity of $\sin(|\theta|/2)$ instead gives

$$A(\theta) \le \exp\{-hF_\infty(z)\}, \quad F_\infty(z) = J(z) - \frac{1}{2H} - \frac{3}{23}.$$

5.4.2.2 A complementary prefix bound

Let a parameter cell have $z \ge z_0$, and a frequency cell be $s \in [\ell, u]$. For each of the four rows choose only the weights

$$1 \le j \le \lfloor hc_a \rfloor, \qquad c_a = \min(z_0/2^a, 2/u).$$

They exist in that row and satisfy $|j\theta| \le 2$. Therefore $\sin(j|\theta|/2) \ge 5j|\theta|/12$.

Define $I_2(x) = \int_0^x u^2 v(u)\,du$. The function $f_2(x) = x^2 v(x)$ is unimodal, has maximum less than one, and increases on $[0,2]$: its logarithmic derivative is $2/x - \tanh(x/2) > 0$ there. The standard unimodal right-Riemann error is at most $1/h$. On the increasing interval the right-Riemann sum is a lower bound for the integral without an error subtraction. Thus the selected variance in row a, divided by h^3, is at least

$$I_2(c_a - 1/H) - \frac{\mathbf{1}_{\{c_a > 2\}}}{H}.$$

All arguments $c_a - 1/H$ in the certificate are positive. The resulting modulus estimate on that frequency cell is

$$A(s/h) \le e^{-hF_P}, \qquad F_P = \frac{25\ell^2}{72} \sum_{a=0}^{3} \left[I_2(c_a - 1/H) - \frac{\mathbf{1}_{\{c_a > 2\}}}{H} \right].$$

Combining both independent upper bounds for the modulus uses the exponent $F = \max(F_P, F_A(\ell, z_0))$, not their sum.

5.4.2.3 Variance upper bounds

For a finite parameter cell $z \in [z_0, z_1] \subset [1, 5]$, let $x_a = z_1/2^a$. The same Riemann estimate gives

$$\frac{\sigma^2}{h^3} \le B(z_1) := \sum_{a=0}^{3} \left[I_2(x_a) + \frac{E(x_a)}{H} \right], \qquad E(x) = \begin{cases} x^2 v(x), & x \le 2, \\ 1, & x > 2. \end{cases}$$

For $x \le 2$, the endpoint value bounds the sum-to-integral error; for $x > 2$, the global maximum bound suffices. For $z \ge 5$, the cruder bound $\sigma^2 < 9h^3$ suffices: $v(x) \le e^{-x}$, the integral of $x^2 e^{-x}$ is 2, and its maximum is less than one, giving $\sigma^2/h^3 \le 4(2 + 1/H) < 9$.

5.4.2.4 Exact finite certificate covering continuous parameters

The tail checker uses $z_0 = i/32$, $z_1 = (i+1)/32$, for $32 \le i < 160$, and a final cell $z \ge 5$. For a finite cell the true starting frequency is $s = 2/\min(z, 5) \ge 2/\min(z_1, 5)$. For the final cell it is $s = 2/5$. Round this lower endpoint downward to the grid of denominator 40, and partition up to $s = 8$ into cells of width $1/40$. This only enlarges the integration region.

The integral enclosures for I_2 use a grid of denominator 256 on $[0, 5]$. At every endpoint the exponential is enclosed by its degree-80 Taylor polynomial and a geometric upper bound for the positive remainder. Bounds are rounded outward to denominator 10^{24}. Monotonicity of v then bounds each integral cell by the appropriate endpoint value of v times the exact integral of x^2; a final partial cell is treated the same way. The formula for J is enclosed directly from the exponential lower bounds.

For each parameter cell choose a rational $R > \sqrt{B}$. If E_F is a rational upper bound for e^{-HF}, the near-tail moment bounds are

$$x_{\text{near}} \le 2R\sqrt{H} \sum_{[\ell,u]} (u - \ell) E_F,$$

$$y_{\text{near}} \le 2R^3 H^{3/2} \sum_{[\ell,u]} \frac{u^3 - \ell^3}{3} E_F.$$

These also hold at every $h \ge H$, because the certificate checks $F > 3/(2H)$, making both relevant powers of h times e^{-hF} decreasing. The far tail satisfies

$$x_{\text{far}} \le 2\pi R H^{3/2} e^{-HF_\infty}, \qquad y_{\text{far}} \le \frac{2\pi^3}{3} R^3 H^{9/2} e^{-HF_\infty}.$$

Here $F_\infty > 9/(2H)$ is also checked, and $\pi < 22/7$ is used.

All negative exponential bounds are rational: a Taylor lower bound for $e^{1/100}$, iterated with downward rounding, certifies $e^{k/100}$ from below. Rounding the desired positive exponent down to that grid and taking the reciprocal, rounded upward, gives E_F. The tail checker proves $x_{\text{near}} + x_{\text{far}} < 1/3000$ and $y_{\text{near}} + y_{\text{far}} < 1/50$ separately in every parameter cell. Thus this is a finite certificate for a uniform analytic tail lemma, not a finite coefficient scan.

5.4.2.5 Central-tilt extension

For $0 \le t \le 1/n$, set $h = n$. Every variance weight satisfies $v(tj) \ge v(j/n)$; thus all preceding damping estimates for $z_0 = 1$ remain valid. Abel summation still applies because $m_j v(tj)$ decreases and its first term is at most one. The actual variance satisfies

$$\frac{\sigma^2}{n^3} \le \sum_{a=0}^{3} \left(\frac{2^{-3a}}{12} + \frac{2^{-2a}}{8n} + \frac{2^{-a}}{24n^2} \right) < \frac{1}{10} \quad (n \ge 100),$$

by $v \le 1/4$ and the exact sum-of-squares formula. The tail checker repeats the first parameter cell's tail bounds with $R > \sqrt{1/10}$, which still yields the two displayed bounds. Its slightly enlarged starting region contains the required $|\theta| \ge 2/n$.

Every comparison asserted by that checker is between exact rational numbers. In particular its final decimal reports are not inputs to any proof comparison. This completes the proof of the tail lemma. □

5.5 Positivity of the Fourier determinant at scale 100

Theorem 5.5.1. *Let $n \ge 100$. At every lower-half integer coefficient index of B_n whose saddle satisfies $0 \le t \le 1/100$, the Turán determinant is strictly positive. The reflected index has the same property.*

Proof. First suppose $n \ge h = 1/t \ge 100$. The case $0 \le tn \le 1$, which includes all remaining saddles in the statement, is treated at the end.

5.5.1 Normalization and moment inputs

Use the centered characteristic function $\phi = Ae^{i\alpha}$, variance σ^2, and identity (5.3.1). All contributions below omit its common positive factor $(2\pi)^{-2}$, and are normalized by multiplying by σ^4. Put

$$L = \min(n, 5h), \quad \delta = 1/L, \quad C = [-\delta, \delta], \quad a = 11/12, \quad b = 64/49.$$

Lemma 5.4.1 gives

$$M_3^2/\sigma^6 < 15/(2h), \qquad \sigma^2/L^2 > 79h/1000, \qquad M_4/\sigma^2 < 14h^2.$$

If $n \le 5h$, $M_4 \le n^2\sigma^2$; otherwise $M_4 < 14h^2\sigma^2 < L^2\sigma^2$. Thus in either case $M_4 \le L^2\sigma^2$.

5.5.2 A sharper cubic phase estimate

For a Bernoulli factor write

$$f_p(u) = \log(1 - p + pe^{iu}) - ipu, \qquad v = p(1-p).$$

For $|u| \le 1$, put $s = \sin^2(u/2) \le 1/4$. The exact derivative is

$$|f_p'''(u)|^2 = v^2 \frac{1 - 4v + 4vs}{(1 - 4vs)^3} \le v^2.$$

Indeed, the denominator minus the numerator is

$$4v(1 - 4s + 12vs^2 - 16v^2s^3) \ge 0,$$

since $1-4s\ge 0$ and $3-4vs>0$. Taylor's integral remainder therefore proves

$$|\operatorname{Im} f_p(u)|\le v|u|^3/6. \tag{5.5.1}$$

This estimate retains every phase term.

For clarity, the rare-factor series used below is

$$\operatorname{Im} f_p(u)=\sum_{\ell\ge 1}\frac{(-1)^{\ell-1}q^\ell}{\ell}(\sin(\ell u)-\ell u),\qquad \sum_{\ell\ge 1}\ell^2 q^\ell=\frac{q(1+q)}{(1-q)^3},$$

where $p=q/(1+q)$ and $0<q<1$. The absolute cubic bound follows directly by summing $|\sin(\ell u)-\ell u|\le|\ell u|^3/6$. All weights $j\le L$ are active on C. If there are weights $j>L$, then $L=5h$ and their $q=e^{-j/h}<e^{-5}<1/144$. The absolutely convergent logarithm series, using $|\sin u-u|\le|u|^3/6$, gives

$$|\operatorname{Im} f_p(j\theta)|\le\frac{vj^3|\theta|^3}{6}\left(\frac{1+q}{1-q}\right)^3<\frac{7}{40}vj^3|\theta|^3,$$

because $(145/143)^3/6<7/40$. Summing this and (5.5.1) yields

$$|\alpha(\theta)|\le(7/40)M_3|\theta|^3\qquad(\theta\in C). \tag{5.5.2}$$

The phase is continuous and odd. The active denominator bound and the rare-factor bound $((1-q)/(1+q))^2>49/64$ give $A(\theta)\ge e^{-b\sigma^2\theta^2/2}$, by $\log(1-y)\ge -y/(1-y)$ and $\sin^2 u\le u^2$.

The modulus identity is $A(\theta)^2=\prod_j(1-4v_j\sin^2(j\theta/2))^{m_j}$. For active factors, $1-4v_j\sin^2(j\theta/2)\ge\cos^2(1/2)\ge 49/64$. Together with the rare-factor denominator just bounded, this and $\log(1-y)\ge -y/(1-y)$ give the stated lower Gaussian. For the upper bound use the global inequality $\sin^2(u/2)\ge u^2/4-u^4/48$ and $\log(1-y)\le -y$. It gives

$$A(\theta)\le\exp\{-\sigma^2\theta^2/2+M_4\theta^4/24\}\le e^{-a\sigma^2\theta^2/2}\qquad(\theta\in C). \tag{5.5.3}$$

5.5.3 Positive core and phase loss

The lower-Gaussian endpoint satisfies

$$b\sigma^2\delta^2>b(79/10)>(16/5)^2.$$

The standard Gaussian tail estimates, with $e^{128/25}>160$ and $\sqrt{2\pi}>5/2$, show that both retained moments of orders 0 and 2 exceed 49/50 of their full Gaussian moments. Since $L\ge 100$, the kernel correction $1-1/(3L^2)>999/1000$. The normalized positive-modulus contribution is therefore greater than

$$\frac{999}{1000}\left(\frac{49}{50}\right)^2\frac{2\pi}{b^2}>\frac{7}{2}. \tag{5.5.4}$$

Use the exact phase parity identity (5.3.2), including its negative square term before safely discarding this nonpositive contribution to the loss. With (5.5.2), the loss is at most

$$\frac{120\pi}{a^5}\left(\frac{7}{40}\right)^2\frac{M_3^2}{\sigma^6}<\frac{120\pi}{a^5}\left(\frac{7}{40}\right)^2\frac{15}{2h}<\frac{27}{20}\qquad(h\ge 100). \tag{5.5.5}$$

Also, (5.5.3) gives the absolute core moments $J_0<3/\sigma$ and $J_2<3/\sigma^3$.

5.5.4 First tail: between one and two core widths

Set $U = \sigma^2/L^2 > 79h/1000$ and $r = L|\theta|$. The global quartic bound and $M_4 \le L^2\sigma^2$ give

$$A(\theta) \le e^{-Uf(r)/2}, \qquad f(r) = r^2 - r^4/12.$$

The function f is increasing on $[1,2]$. For $p = 1/2, 3/2$, $U^p e^{-Uf(r)/2}$ decreases in $U \ge 79/10$, since $(79/10)f(1) > 3$. Therefore the normalized moments satisfy

$$x_1 := \sigma T_{0,1} \le 2\sqrt{79/10}\int_1^2 e^{-(79/20)f(r)}dr < 1/40,$$

$$y_1 := \sigma^3 T_{2,1} \le 2(79/10)^{3/2}\int_1^2 r^2 e^{-(79/20)f(r)}dr < 1/4. \tag{5.5.6}$$

These two integral inequalities are certified by 256 rational rectangles: use the left endpoint in the increasing f, the right endpoint in r^2, $\sqrt{79/10} < 3$, and a positive rational Taylor sum to bound the exponential reciprocal above. The checker is `verify_balanced_h100_constants.py` in Appendix 5.A.

5.5.5 Remaining tail and positivity

Lemma 5.4.2 proves

$$x_2 := \sigma\int_{2/L\le|\theta|\le\pi} A(\theta)d\theta < 1/3000,$$

$$y_2 := \sigma^3\int_{2/L\le|\theta|\le\pi} \theta^2 A(\theta)d\theta < 1/50. \tag{5.5.7}$$

It retains all four balanced row lengths in low-frequency prefix bounds and uses Abel summation for the full far-frequency interval; no frequency interval is omitted. Its finite grid verifies analytic envelopes for all real $h \ge 100$ and all integer $n \ge h$, not just sampled saddles.

For total tail moments set $x = x_1 + x_2$, $y = y_1 + y_2$. Keeping the quadratic kernel, the normalized absolute contribution outside C^2 is at most $3x + 3y + xy$. By (5.5.6)–(5.5.7),

$$3x + 3y + xy < 9/10.$$

Combining (5.5.4), (5.5.5), and this bound gives a normalized real double integral strictly greater than

$$7/2 - 27/20 - 9/10 = 5/4 > 0.$$

This proves the tilted theorem. All remaining scalar comparisons are reproduced by the same core-constants checker; the moment and remaining-tail checks were proved separately.

5.5.6 Central extension down to block size 100

Suppose $n \ge 100$ and $0 \le tn \le 1$. Set $L = n$, regardless of whether $t = 0$. Since $v(tj) \ge v(j/n)$, the balanced moment lower bound applied at scale $h = n$ implies

$$\sigma^2/n^2 > (79/1000)n.$$

Every weight is active on $C = [-1/n, 1/n]$. Thus (5.5.1) gives the stronger phase bound $|\alpha(\theta)| \le M_3|\theta|^3/6$, and $M_3 \le n\sigma^2$. The modulus bounds (5.5.3) and the positive core bound (5.5.4) are unchanged, since $M_4 \le n^2\sigma^2$. The normalized phase loss is now less than

$$\frac{120\pi}{36a^5}\frac{n^2}{\sigma^2} < \frac{120\pi}{36a^5}\frac{1000}{79n} < \frac{41}{20} \qquad (n \ge 100).$$

The first-tail argument still uses only $U = \sigma^2/n^2 > 79n/1000$ and $M_4 \le n^2\sigma^2$, so (5.5.6) holds. The remaining-tail certificate includes this central case: the modulus only decreases as t decreases from $1/n$, while the normalization is bounded separately by $\sigma^2 < n^3/10$. Thus (5.5.7) also holds. The normalized determinant integral is therefore greater than

$$7/2 - 41/20 - 9/10 = 11/20 > 0.$$

This proves the central extension including $t = 0$.

□

Corollary 5.5.2. *For $n \ge 100$, any lower-half index not covered by Theorem 5.5.1 satisfies $1 \le k < 40400$.*

Proof. An uncovered saddle has $h = 1/t < 100$. Since all multiplicities are at most four,

$$k \le 4\sum_{j\ge1} je^{-j/h} \le 4(h^2 + h) < 40400.$$

For the middle inequality, for $j - 1 \le x \le j$ one has $je^{-j/h} \le (x + 1)e^{-x/h}$; integrate and sum to obtain $\sum_{j\ge1} je^{-j/h} \le \int_0^\infty (x + 1)e^{-x/h}dx = h^2 + h$. □

5.6 An effective all-index four-color theorem

Put

$$F(q) = \prod_{j\ge1}(1 + q^j)^4 = \prod_{j\ge1,\ j\ \text{odd}} (1 - q^j)^{-4} = \sum_{n\ge0} q_4(n)q^n.$$

The identity follows by cancelling the even factors in Euler's product; it is also a bijection between colored distinct-part and colored odd-part partitions.

Theorem 5.6.1. *The sequence $q_4(n)$ is strictly log-concave at every integer $n \ge 1$. For $n \ge 10000$, with $s = n + 1/6$, it moreover satisfies*

$$2\log q_4(n) - \log q_4(n - 1) - \log q_4(n + 1) > \frac{9}{10(s + 3)^2}.$$

Proof. The analytic proof treats every $n \ge 10000$; the finite base is specified at the end of this section.

5.6.1 Standard identities and the main term

Put $A = \pi^2/3$, so $3 < A < 4$. Dedekind eta inversion gives, for $\Re z > 0$,

$$F(e^{-z}) = \frac{1}{4}\exp\left(\frac{A}{z} + \frac{z}{6}\right) G(z), \qquad G(z) = \frac{1}{F(e^{-2\pi^2/z})}. \tag{5.6.1}$$

The sign is **plus** $z/6$. For example, (5.6.1) follows by taking the ratio of the eta inversion identities for $E(z) = \prod_{j\ge1}(1 - e^{-jz})$:

$$E(z) = \sqrt{2\pi/z}\, e^{-\pi^2/(6z)+z/24}E(4\pi^2/z), \qquad F(e^{-z}) = (E(2z)/E(z))^4.$$

For real $s > 0$, define

$$M(s) = \frac{1}{4}\sqrt{A/s}\ I_1(2\sqrt{As}) = \frac{A}{4}\sum_{j\ge0}\frac{(As)^j}{j!(j + 1)!}. \tag{5.6.2}$$

In particular, $4M(s)$ is the residue at zero of $\exp(sz + A/z)$.

The eta transformation and the Bessel integral and canonical product are the classical identities recorded in [5.5]. Their use here does not involve an asymptotic expansion with an uncontrolled remainder.

5.6.2 A deliberately weak lower bound on the main term

For $x \geq 1$, the standard integral is

$$I_1(x) = \frac{x}{\pi} \int_{-1}^{1} e^{xu} \sqrt{1-u^2}\, du.$$

Restrict to $1 - 1/x \leq u \leq 1 - 1/(2x)$. Then $u \geq 0$, $e^{xu} \geq e^{x-1}$, and $\sqrt{1-u^2} \geq 1/\sqrt{2x}$. Consequently

$$I_1(x) \geq \frac{e^{x-1}}{2\pi\sqrt{2x}} > \frac{e^x}{25\sqrt{x}}. \tag{5.6.3}$$

The numerical inequality is elementary: $2\pi\sqrt{2}e < 25$, using $\pi < 22/7$, $\sqrt{2} < 10/7$, and $e < 11/4$. Put $h = \sqrt{s/A}$, $t = 1/h$. From (5.6.2)–(5.6.3), for $h \geq 50$,

$$M(s) > \frac{e^{2Ah}}{300h^{3/2}}, \tag{5.6.4}$$

since $x = 2Ah \geq 1$ and $100\sqrt{2A} < 300$.

5.6.3 An elementary global minor-arc bound

For $r = e^{-t}$ and real θ, the odd-part product gives the absolutely convergent expansion

$$\log F(r) - \log |F(re^{i\theta})| = 4 \sum_{j \text{ odd}} \sum_{m \geq 1} \frac{r^{jm}}{m} (1 - \cos(jm\theta)) \geq 4S(t, \theta),$$

where summing the first harmonic yields exactly

$$S(t,\theta) = \sum_{j \text{ odd}} e^{-jt}(1 - \cos(j\theta)) = \frac{1}{2\sinh t} - \frac{\sinh t \cos\theta}{2(\sinh^2 t + \sin^2\theta)}. \tag{5.6.5}$$

Suppose $0 < t \leq 1/50$ and $t \leq |\theta| \leq \pi$. By evenness assume $\theta \geq t$. On $[t, \pi/2]$, use $\sin\theta \geq \sin t$ and $\cos\theta \leq 1$; on $[\pi/2, \pi]$, the second term in (5.6.5) is nonpositive. In both cases,

$$S(t,\theta) \geq \frac{\sin^2 t}{2\sinh t(\sinh^2 t + \sin^2 t)}.$$

The elementary bounds $\sin t \geq (99/100)t$, $\sinh t \leq (101/100)t$, and $\sin t \leq t$, valid on the stated interval, imply

$$4S(t,\theta) \geq \frac{1960200}{2040301} \frac{1}{t} > \frac{9}{10t}. \tag{5.6.6}$$

For clarity, $\sin t \geq t - t^3/6$ proves the sine lower bound. For sinh, its power series divided by t is at most $\sum_{m \geq 0} t^{2m} = 1/(1-t^2) < 101/100$.

As $G(t) < 1$ on the positive real line, (5.6.1) yields $F(e^{-t}) < (1/4)e^{A/t+t/6}$. For an integer n let $s = n + 1/6$ and choose $t = \sqrt{A/s}$. The coefficient integral outside $|\theta| \le t$ is therefore bounded in absolute value by

$$E_{\text{minor}} \le \frac{1}{4} e^{2Ah-(9/10)h}. \tag{5.6.7}$$

The factor $1/(2\pi)$ in coefficient extraction cancels the crude upper bound 2π for total minor-arc length.

5.6.4 Major-arc correction and explicit closure to the residue

On $z = t + i\theta$, $|\theta| \le t$, we have $\Re(1/z) \ge 1/(2t)$. Write $w = e^{-2\pi^2/z}$, $u = |w| \le e^{-\pi^2 h}$. From the product in (5.6.1),

$$|\log G(z)| \le 4 \sum_{j \ge 1} \frac{u^j}{1-u^j} \le \frac{4u}{(1-u)^2}.$$

For $u \le 1/100$, the last quantity is at most $5u$, and $e^{5u} - 1 \le 10u$. Thus $|G(z) - 1| \le 10e^{-\pi^2 h}$. Also $\Re(sz + A/z) \le 2Ah$ on the vertical major segment. Integrating its length $2t$ with factor $1/(8\pi)$, we get

$$E_{\text{transform}} \le \frac{5t}{2\pi} e^{2Ah-\pi^2 h} < te^{2Ah-\pi^2 h}. \tag{5.6.8}$$

Orient the vertical segment from $t - it$ to $t + it$. Close it by the **left** arc of $|z| = \sqrt{2}t$, oriented counterclockwise from angle $\pi/4$ to $7\pi/4$. This is a positively oriented contour around zero. On the arc,

$$\Re(sz + A/z) = \left(s + \frac{A}{2t^2}\right) \Re z = \frac{3s}{2} \Re z \le \frac{3}{2} Ah,$$

because $\Re z \le t$. Its length is $(3\pi/2)\sqrt{2}t$. By the residue theorem and (5.6.2), the difference between the major integral without G and $M(s)$ has absolute value at most

$$E_{\text{arc}} \le \frac{3\sqrt{2}t}{16} e^{(3/2)Ah} < \frac{t}{2} e^{(3/2)Ah}. \tag{5.6.9}$$

Only the explicit exponential $e^{sz+A/z}$, not F, is continued to the left half-plane. No invalid use of an infinite product there is involved.

5.6.5 Uniform relative error

Combining (5.6.4), (5.6.7)–(5.6.9), using $A > 3$ and $\pi^2 > 9$, gives

$$\left| \frac{q_4(n)}{M(n + 1/6)} - 1 \right| \le 75h^{3/2}e^{-9h/10} + 150h^{1/2}e^{-3h/2} + 300h^{1/2}e^{-9h}. \tag{5.6.10}$$

For $h \ge 50$, division by $h^{3/2}e^{-9h/10}$ bounds the right side by $75 + (150/h)e^{-3h/5} + (300/h)e^{-81h/10} < 84 < 300$. Therefore the relative error is at most

$$\epsilon(h) := 300h^{3/2}e^{-9h/10} < \frac{1}{100(s+3)^2}. \tag{5.6.11}$$

Here are elementary checks of the last strict inequality, to avoid relying on floating-point evaluation. Since $s = Ah^2 < 4h^2$ and $h \geq 50$, $s+3 < 5h^2$. It suffices to prove $750000h^{11/2}e^{-9h/10} < 1$. Replace $h^{11/2}$ by h^6. The function $h^6e^{-9h/10}$ decreases on $[50, \infty)$, and

$$750000 \cdot 50^6 = 11718750000000000 < 20^{15} < e^{45}.$$

The final inequality follows from $e^3 > 20$; the Taylor sum through degree 8 already exceeds 20.

5.6.6 Main-term curvature from the canonical product

The classical canonical product for J_1, equivalently for I_1, gives

$$M(s) = \frac{A}{4} \prod_{m \geq 1} \left(1 + \frac{4As}{j_{1,m}^2}\right),$$

where $j_{1,m}$ are the positive real zeros of J_1. Local uniform convergence permits twice differentiating the logarithm for $s > 0$:

$$-\frac{d^2}{ds^2} \log M(s) = \sum_{m \geq 1} \frac{1}{(s + j_{1,m}^2/(4A))^2} > \frac{1}{(s+2)^2}. \tag{5.6.12}$$

For the last inequality, $j_{1,1} < 4$ and $A > 3$ suffice. The zero bound does not require a numerical zero table: the alternating series

$$J_1(4) = \sum_{m \geq 0} \frac{(-1)^m 2^{2m+1}}{m!(m+1)!}$$

has, after its partial sum through summation index 6, a nonpositive alternating remainder; the partial sum is exactly $-934/14175 < 0$. Since $J_1(x) > 0$ for small positive x, it has a positive zero below 4. All its nonzero zeros are real, as required by the standard product for order 1.

Integrating (5.6.12) against the triangular kernel on $[-1, 1]$, for $s > 1$,

$$2 \log M(s) - \log M(s-1) - \log M(s+1) > \frac{1}{(s+3)^2}. \tag{5.6.13}$$

5.6.7 Transfer to the exact coefficients

Let $n \geq 10000$, $s = n + 1/6$. The parameters for all three indices $n-1, n, n+1$ satisfy $h \geq 50$, since $A < 10/3$ and $(9999 + 1/6)/(10/3) > 2500$. Thus (5.6.11) applies to all three. Write $q_4(m) = M(m + 1/6)(1 + \delta_m)$. On these three indices,

$$|\delta_m| < \frac{1}{100(s+2)^2} < \frac{1}{2}, \qquad |\log(1+\delta_m)| \leq 2|\delta_m|.$$

The possible loss in the log-concavity expression (5.6.13) is therefore less than $8/[100(s+2)^2]$, and

$$\frac{8}{100(s+2)^2} < \frac{1}{(s+3)^2}$$

for $s > 1$ (indeed for all $s \geq 0$). Hence $2 \log q_4(n) - \log q_4(n-1) - \log q_4(n+1) > 0$ for every $n \geq 10000$.

For the stronger displayed bound, use $(s+3)/(s+2) \le 10/9$ for $s \ge 7$ and $(8/100)(10/9)^2 = 8/81 < 1/10$ in the preceding curvature comparison.

5.6.8 The finite base

The script `verify_four_color_prefix.py` in Appendix 5.A multiplies the four-colored odd-part factors through degree 10000 using arbitrary-precision integers. Each color of an odd part j is inserted by the ascending recurrence $a[k] \leftarrow a[k] + a[k-j]$, for $j \le k \le 10000$; ascending order allows arbitrary repetitions. It checks strict determinants for every $1 \le k \le 9999$. Independently it verifies all coefficients by the formal logarithmic-derivative identity

$$nq_4(n) = 4\sum_{j=1}^{n} \sigma_{\mathrm{odd}}(j) q_4(n-j), \qquad \sigma_{\mathrm{odd}}(j) = \sum_{d \mid j,\ d \ mod 2 \neq 0} d.$$

The minimum integer determinant in this finite range is 4. The SHA-256 of the newline-separated decimal coefficient list, including the final newline, is

`b682e4acb8c8ed380987918a0512ce917e75a05c0635d009334ce44bc52da9ee.`

Thus the exact finite base and the proved analytic tail cover every index. □

5.7 Quantitative transfer to finite four-row products

We next compare any four-row product $P = \prod_{i=1}^{4} Q_{L_i}$ with F. The result is used only for the balanced blocks, whose shortest row is $\lfloor n/8 \rfloor$.

5.7.1 The coefficient deficit: a genuine union bound

Let

$$F(x) = \prod_{j \ge 1} (1+x^j)^4 = \sum_{k \ge 0} q_k x^k, \qquad P(x) = \prod_{i=1}^{4} Q_{L_i}(x) = \sum_{k} b_k x^k, \qquad d = \min_i L_i.$$

All parts carry one of four colors, and each colored size is used at most once. A partition counted by $q_k - b_k$ contains a colored part (i, j) with $j > L_i$. Removing that specified part injects the set of such partitions into the unrestricted colored partitions of $k - j$. Thus, with $q_t = 0$ for $t < 0$,

$$0 \le q_k - b_k \le \sum_{i=1}^{4} \sum_{j > L_i} q_{k-j} \le 4 \sum_{j > d} q_{k-j}. \tag{5.7.1}$$

Multiple offending parts merely produce overcounting, in the permissible direction. This argument does not divide by a polynomial and assume positivity of an alternating inverse.

The odd-part representation of F injects partitions of $k-1$ into those of k by adding a part of size 1 in a fixed color; partitions having no size-1 part of that color give a nonempty complement. Hence $q_k > q_{k-1} > 0$. Theorem 5.6.1 implies

$$R_k := q_{k-1}/q_k < 1 \ (k \ge 1), \qquad R_{k-1} < R_k \ (k \ge 2).$$

Consequently, for $1 \le j \le k$,

$$\frac{q_{k-j}}{q_k} = \prod_{u=k-j+1}^{k} R_u \le R_k^j.$$

Define $\delta_k = 1 - b_k/q_k$. Summing the geometric series in (5.7.1) gives

$$0 \le \delta_k \le \frac{4R_k^{d+1}}{1-R_k}. \tag{5.7.2}$$

5.7.2 Explicit ratio bound

The preceding proof supplies, for all the indices used below,

$$q_k = M(k+1/6)(1+\eta_k), \qquad M(s) = \tfrac{1}{4}\sqrt{A/s}\, I_1(2\sqrt{As}), \qquad A = \pi^2/3 > 3,$$

and $|\eta_k| < 1/[100(k+19/6)^2]$. For $k \ge 10000$, both k and $k-1$ satisfy the auxiliary proof's $h \ge 50$ hypothesis. Writing $s = k + 1/6$, their errors are each less than $1/[100(s+2)^2]$.

For $f = \log M$, direct differentiation gives

$$f'(s) = \sqrt{A/s}\,\frac{I_0(2\sqrt{As})}{I_1(2\sqrt{As})} - \frac{1}{s} > \sqrt{A/s} - \frac{1}{s}. \tag{5.7.3}$$

Here $I_1'(x) = I_0(x) - I_1(x)/x$ follows from the power series; the standard angular integral gives, for $x > 0$,

$$I_0(x) - I_1(x) = \frac{1}{\pi}\int_0^\pi e^{x\cos\theta}(1-\cos\theta)\,d\theta > 0.$$

Integrating (5.7.3) from $s-1$ to s, and using $|\log(1+\eta)| \le 2|\eta|$ for $|\eta| < 1/2$, yields

$$\log(q_k/q_{k-1}) > \sqrt{3/s} - \frac{1}{s-1} - \frac{1}{25(s+2)^2}. \tag{5.7.4}$$

For every $k \ge 10000$,

$$\sqrt{3k/s} > \frac{173}{100}, \qquad \frac{\sqrt{k}}{s-1} < \frac{1}{50}, \qquad \frac{\sqrt{k}}{25(s+2)^2} < \frac{1}{100}.$$

For the first, $3k/s \ge 180000/60001 > (173/100)^2$. For the second, $s - 1 > k/2$; for the third, $s + 2 > k$. Therefore (5.7.4) proves

$$\boxed{R_k < e^{-17/(10\sqrt{k})}} \qquad (k \ge 10000). \tag{5.7.5}$$

This constant is deliberately nonoptimal.

5.7.3 How much curvature must be retained?

Let $\kappa_k = 2\log q_k - \log q_{k-1} - \log q_{k+1}$. The preceding proof, with $s = k + 1/6$, gives

$$\kappa_k > \frac{1}{(s+3)^2} - \frac{8}{100(s+2)^2} > \frac{9}{10(s+3)^2} =: \kappa_k^0 \qquad (k \ge 10000). \tag{5.7.6}$$

Indeed $(s+3)/(s+2) \le 10/9$ for $s \ge 7$, and $(8/100)(10/9)^2 = 8/81 < 1/10$.

Only the deficit at the center index k is needed. Since $b_{k\pm1} \le q_{k\pm1}$,

$$\frac{b_k^2}{b_{k-1}b_{k+1}} \ge (1-\delta_k)^2 e^{\kappa_k}. \tag{5.7.7}$$

This is stated when neighboring coefficients are positive; if their product is zero and $b_k > 0$, strict log-concavity is immediate. The exact sufficient condition is $\delta_k < 1 - e^{-\kappa_k/2}$. A convenient certified sufficient condition is

$$\boxed{\delta_k \le \kappa_k^0/4 = \frac{9}{40(k+19/6)^2}.} \tag{5.7.8}$$

In fact $\kappa_k^0 < 1$, so (5.7.8) implies $\delta_k < 1/2$, and $2\log(1-\delta_k) \ge -4\delta_k \ge -\kappa_k^0$. The strict inequality $\kappa_k > \kappa_k^0$ then makes (5.7.7) exceed 1. No error bounds at the neighboring truncated coefficients are required.

Combining the deficit bound and the ratio estimate gives

$$\delta_k < 4\left(1 + \frac{10\sqrt{k}}{17}\right)\exp\left(-\frac{17(d+1)}{10\sqrt{k}}\right) \qquad (k \ge 10000). \tag{5.7.9}$$

Indeed $e^a \ge 1+a$ implies $(1-e^{-a})^{-1} \le 1 + 1/a$.

5.7.4 A uniform cutoff at block size 28000

Proposition 5.7.1. *If $d = \min_i L_i \ge 3500$, then P is strictly log-concave at every index $1 \le k \le 40400$ in its support. In particular, every B_n with $n \ge 28000$ is strictly log-concave at every internal index.*

Proof. There are two index ranges; both are needed. For $1 \le k \le 9999$, the exact small-transfer checker and its included independent-prefix dependency in Appendix 5.A prove

$$R_{9999} < 99/100, \qquad (q_k^2 - q_{k-1}q_{k+1})/q_k^2 > 10^{-9}.$$

The dependency generates coefficients through 10001 by the displayed logarithmic-derivative recurrence; every division is checked exact. Thus it is independent of the colored-factor multiplication above. The checker also verifies the single integer inequality

$$400 \cdot 10^{12} \cdot 99^{3501} < 100^{3501}.$$

Theorem 5.6.1 makes R_k increase, so $4R_k^{d+1}/(1-R_k) < 10^{-12}$ throughout this range. Consequently

$$b_k^2 = (1-\delta_k)^2 q_k^2 > (1 - 2\cdot 10^{-12})q_k^2 > (1-10^{-9})q_k^2 > q_{k-1}q_{k+1} \ge b_{k-1}b_{k+1}.$$

This treats the entire interval, in particular the transition indices $3501, \dots, 9999$, not merely the indices where the finite and infinite products agree exactly.

For $10000 \le k \le 40400$, we have $\sqrt{k} < 201$, $1 + 10\sqrt{k}/17 < 120$, and $17(d+1)/(10\sqrt{k}) > 17 \cdot 3501/(10 \cdot 201) > 59/2$. Positive Taylor sums through degrees 8 and 10 give $e^3 > 20$ and $e^{5/2} > 12$. Hence (5.7.9) yields

$$\delta_k < \frac{480}{12 \cdot 20^9} < \frac{9}{40 \cdot 40404^2} < \frac{9}{40(k+19/6)^2}.$$

These comparisons too are exact rational assertions in the small-transfer checker. The sufficient curvature condition already proved therefore applies. Finally for $n \ge 28000$ the shortest row of B_n is at least 3500. Corollary 5.5.2 and this edge argument cover the whole lower half, and palindromicity covers the other half. □

5.8 Finite certificates and completion of the proof

5.8.1 Exact finite bases and the exceptions

For the denominator family the recurrence

$$\frac{D_n}{D_{n-1}} = \prod_{a=0}^{v_2(n)} (1 + x^{n/2^a})$$

is the increment identity of BBFM, Proposition 3.10. Starting with $D_0 = 1$, the included `check.py` multiplies these factors by descending integer-array updates, checks palindromicity, and tests every lower-half Tur'an determinant for $1 \le n \le 256$. The only failing parameters are $3, 5, 6, 7$, and in particular $D_8, \dots, D_{127}$ pass. The following explicit witnesses also verify all four exceptions:

n	k	(a_{k-1}, a_k, a_{k+1})	$a_k^2 - a_{k-1}a_{k+1}$
3	2	$(2, 2, 3)$	-2
5	5	$(11, 14, 18)$	-2
6	2	$(3, 5, 9)$	-2
7	2	$(3, 5, 9)$	-2

The same computation verifies D_1, D_2, D_4. If $n = 0$ is allowed, $D_0 = 1$ is vacuously log-concave.

For balanced blocks the quotient is instead

$$\frac{B_n}{B_{n-1}} = \prod_{\substack{0 \le a \le 3 \\ 2^a \mid n}} (1 + x^{n/2^a}). \tag{5.8.1}$$

The fourth-row stopping condition is essential. The included `check_balanced_blocks.py` verifies ordinary log-concavity for $8 \le n \le 256$ with arbitrary-precision integers. The directed certificate described next certifies strict inequalities throughout those complete half-ranges as well.

5.8.2 Directed dyadic arithmetic: why the finite predicate is a proof

A nonzero dyadic endpoint is represented as $m2^e$, with $2^{63} \le m \le 2^{64} - 1$ and e a signed 64-bit integer; zero is $(0, 0)$. Every coefficient is stored between two such nonnegative endpoints. All operations in `dyadic_certificate.cpp` are integer-only. Here are the invariants sufficient to verify its soundness.

For an exact unsigned 128-bit numerator, normalization discards low bits for downward rounding and increments the retained mantissa exactly when the discarded remainder is nonzero for upward rounding. A carry to 2^{64} is normalized exactly to 2^{63} at the next exponent. For addition order the exponents $e_a \ge e_b$ and put $d = e_a - e_b$. If $d \le 64$, the exact numerator $m_a 2^d + m_b$ is at most $(2^{64} - 1)2^{64} + (2^{64} - 1) = 2^{128} - 1$ and so is formed without overflow. If $d > 64$, the smaller operand is strictly less than one unit at exponent e_a; the larger operand and its next representable successor enclose the sum. Zero operands are handled exactly. Exponent calculations use signed 128-bit intermediates and reject any out-of-range conversion. Therefore downward addition of lower endpoints and upward addition of upper endpoints preserve the interval invariant.

To compare products, each pair of 64-bit mantissas is multiplied exactly into 128 bits. A product with mantissa bit-length b and exponent e has top-bit position $e + b$. Unequal positions

decide the comparison. If positions agree, the shorter mantissa is shifted to the larger bit-length, at most 128, which aligns the exponents exactly without discarding a bit. Zero products are treated separately. Thus the certificate predicate

$$\mathrm{lo}(b_{n,k})^2 > \mathrm{hi}(b_{n,k-1})\,\mathrm{hi}(b_{n,k+1}) \tag{5.8.2}$$

is an exact sufficient condition for the true strict inequality. A failed comparison is reported as inconclusive, never as a counterexample.

Starting at $B_0 = 1$, the descending update $p[j] \leftarrow p[j]+p[j-w]$ for a new weight w implements multiplication by $1 + x^w$. Both right-hand entries are old values. The loop in (5.8.1) stops at the first failed divisibility test, since failure for 2^a implies failure for higher powers. Entries above a chosen cap may be discarded: no positive weight can feed a discarded high index back into a smaller one. The stored cap is $K + 1$, and the checked inclusive endpoint is $\min(K, \lfloor M_n/2 \rfloor)$; both neighbors are available. Choosing a first reported row does not skip its earlier construction. Each polynomial is palindromic, so the checked prefix also certifies its reflected prefix. The even-degree midpoint is included.

The requested construction has at most 52495 factors; both exact coefficients and upper endpoints are bounded inductively by the exactly representable 2^{52495}. All exponent and degree storage bounds are therefore amply within range. In particular $M_{27999} = 520598750$. These facts verify the arithmetic, recurrence, truncation, and index invariants independently of performance tests of the implementation.

5.8.3 Completed finite ranges and integrity checks

The exact source SHA-256 is

`b3154e0381004bb878a33af90f513d5e20b3ab157db6080bf1fc34c2ccb9c7fe.`

The completed scans have the following scope. The parameter K is inclusive; "strict'' counts only the proved predicates (5.8.2), not estimated determinant signs.

Parameter range	K	Rows	Strict inequalities	Inconclusive
8–256	40400	249	1 878 532	0
257–27999	40399	27 743	1 119 898 742	0

The first scan covers complete half-ranges. The second covers complete half-ranges only through $n = 348$, and otherwise the lower-half prefix through 40399 and its reflection. It is not represented as a complete coefficient scan for every polynomial. The missing middle indices are covered analytically by Corollary 5.5.2.

For the two newline-delimited JSON outputs, respectively, the SHA-256 identifiers are

`fc6f65c4a2650f67b7cc97656460da2412ba4887d4cb99dfb13c2ede565be57a`
`93e20573a804fc4e213286519b3ff26b725fa584709253d95de7766c72d4e14f.`

Each output was checked for its metadata and source identity, consecutive complete parameter rows, degree formula, retained cap, checked and reflected index endpoints, empty inconclusive lists, strict counts and terminal summary. The exact expected total is independently computed as $\sum_n \min(K, \lfloor M_n/2 \rfloor)$. Process exit was successful; completeness checks are additionally necessary because a stream-write failure alone need not produce a C++ exception. The supplemental validator in the appendix specifies and reproduces these checks. Hashes identify artifacts; neither a hash nor a self-reported source macro substitutes for the mathematical arithmetic invariants or for actually executing the scan.

5.8.4 Completion

For $8 \le n \le 256$, the complete finite certificate proves Theorem 5.1.2. For $257 \le n \le 27999$, every index that is not covered by the Fourier theorem has distance at most 40399 from an edge, and is covered by the second certificate. For $n \ge 28000$, Proposition 5.7.1 proves the whole claim. This establishes Theorem 5.1.2 for every integer in its stated range. Finally (5.2.1), Lemma 5.2.1, and the finite D_n base prove Theorem 5.1.1, with the four displayed witnesses showing that the exceptions are genuine. □

5.9 Reproduction protocol

The document contains all executable source text; no repository notes, external data files, external bibliography, or TeX input files are required. The historical explanatory comments inside the verbatim listings describe the local scope of each checker; the full assembled proof is the preceding text. Source names in those comments are not runtime dependencies unless an import is present. The only sibling-file dependency among the analytic checkers is the small-transfer script's use of the included independent q_4 prefix script.

Save the appendix listings under their indicated names in `research/bbfm_logconcavity/`, starting from an otherwise empty working directory. The extraction script in the appendix automates this from this single TeX file. Use Python without the `-O` flag, since assertions are part of the certificate. Run the moment, tail and core scripts, the four-color prefix and small-transfer scripts, and both finite base scripts. Compile the C++ source with its actual SHA-256 macro, then run the two finite ranges. The commands and validator are included below. The C++ runs require neither floating-point arithmetic nor mathematical libraries. The Python programs use only the standard library.

A successful rerun must reproduce the exact strict counts and zero inconclusives, not merely return quickly or print a partial prefix. A nonzero exit, assertion failure, incomplete output, or different source hash is not a completed reproduction. No computation in this proof relies on TeX shell escape.

5.A Complete executable sources

The following listings are the full source files, not excerpts. Their numeric comparisons use integers or rational fractions; floating-point conversions in reporting fields do not affect certification. In the TeX source, each listing has a machine-readable source marker. The extraction utility after the listings preserves the source bytes, including final newlines. Python requires version 3.8 or later; C++ requires a compiler with the GNU-compatible `__int128` integer extension.

5.A.1 verify_balanced_moments.py

SHA-256: `ee5953f5b7037b6f84fdc48593dd01ee17777cbabb0f129421497d1b7e930bf5`.

```
"""Exact all-parameter moment certificate for four balanced rows, h >= 100.

No coefficient scan. Every inequality is rational; displayed decimals are reports.
See balanced_moments.md for the sum/integral and floor arguments.
"""
from fractions import Fraction as F
from pathlib import Path
import json

H = 100
GRID = 256
LAST = 16 * GRID
DEN = 10**18

def down(x):
    return F(x.numerator * DEN // x.denominator, DEN)

def up(x):
    return F(-((-x.numerator * DEN) // x.denominator), DEN)

def exp_interval(x):
    term = total = F(1)
    for j in range(1, 151):
        term *= x / j
        total += term
    nxt = term * x / 151
    return down(total), up(total + nxt / (1 - x / 152))

vl, vu = [], []
for i in range(LAST + 1):
    el, eu = exp_interval(F(i, GRID))
    vl.append(down(eu / (1 + eu)**2))
    vu.append(up(el / (1 + el)**2))
lo, hi = {2: [F(0)], 3: [F(0)]}, {2: [F(0)], 3: [F(0)]}
for i in range(LAST):
    l, u = F(i, GRID), F(i + 1, GRID)
    for q in (2, 3):
        area = (u**(q + 1) - l**(q + 1)) / (q + 1)
        lo[q].append(lo[q][-1] + vl[i + 1] * area)
        hi[q].append(hi[q][-1] + vu[i] * area)

def max_bound(q, x):
    return min(F(1 if q == 2 else 2), x**q / 4)

def balanced_interval(q, i, upper=False):
    # z >= i/GRID for lower, z <= i/GRID for upper.
    result = F(0)
    for a in range(4):
        d = 2**a
```

```
        j = (i + d - 1)//d if upper else i//d
        result += (hi if upper else lo)[q][j]
    return result

def error(q, z):
    return sum((max_bound(q, z / 2**a) for a in range(4)), F(0)) / H

worst = (F(0), None)
variance_worst = (F(100), None)
for i in range(GRID, LAST):
    z_upper = F(i + 1, GRID)
    s_lower = balanced_interval(2, i) - error(2, z_upper)
    t_upper = balanced_interval(3, i + 1, upper=True) + error(3, z_upper)
    assert s_lower > 0
    ratio = t_upper**2 / s_lower**3
    if ratio > worst[0]:
        worst = ratio, i
    assert ratio < F(15,2), (i, float(ratio))
    if i < 5 * GRID:
        vratio = s_lower / z_upper**2
        if vratio < variance_worst[0]:
            variance_worst = vratio, i
        assert vratio > F(79,1000), (i, float(vratio))

zeta_upper = sum((F(1,j**3) for j in range(1,11)), F(0)) + F(1,200)
assert zeta_upper < F(1203,1000)
s_tail = balanced_interval(2, LAST) - F(4,H)
t_tail = 4 * F(9,2) * F(1203,1000) + F(8,H)
tail_ratio = t_tail**2 / s_tail**3
assert tail_ratio < F(15,2)
v_tail = (balanced_interval(2, 5*GRID) - error(2,F(5))) / 25
assert v_tail > F(79,1000)

# Fourth moment covariance reduces to the single infinite row.
fourth_ratio = (F(7,30)*F(22,7)**4 + F(5,H))/(F(314,100)**2/6 - F(1,H))
assert fourth_ratio < 14
# Four rows; each right sum differs by at most max(f_2)/h.
variance_upper = 4 * (F(22,7)**2/6 + F(1,H))
assert variance_upper < 7
# Constants used to bound max f_q, from e > 27/10.
assert F(4)/F(27,10)**2 < 1
assert F(27)/F(27,10)**3 < 2
assert F(256)/F(27,10)**4 < 5

result = {
    'status': 'PASS', 'h_min': H,
    'third_moment_ratio_constant': '15/2',
    'core_variance_constant': '79/1000',
    'fourth_moment_constant': 14,
    'variance_upper_constant': 7,
    'grid_denominator': GRID, 'z_interval': [1,16],
    'exp_taylor_degree': 150, 'rounding_denominator': DEN,
    'third_moment_worst_cell': [f'{worst[1]}/{GRID}',f'{worst[1]+1}/{GRID}'],
    'third_moment_upper_decimal': float(worst[0]),
    'third_moment_upper_exact': [str(worst[0].numerator),str(worst[0].denominator)],
    'third_moment_tail_upper_decimal': float(tail_ratio),
    'core_variance_worst_cell': [f'{variance_worst[1]}/{GRID}',f'{variance_worst[1]+1}/{GRID
    ↪ }'],
    'core_variance_lower_decimal': float(variance_worst[0]),
    'core_variance_lower_exact': [str(variance_worst[0].numerator),str(variance_worst[0].
    ↪ denominator)],
    'core_variance_tail_lower_decimal': float(v_tail),
    'fourth_moment_upper_decimal': float(fourth_ratio),
```

```
    'variance_upper_decimal': float(variance_upper),
}
Path(__file__).with_suffix('.json').write_text(json.dumps(result,indent=2)+'\n')
print(json.dumps(result,indent=2))
```

5.A.2 verify_balanced_tail_h100.py

SHA-256: `16448f7811ca7c2ee926cf9ad46979572a44fe570ce9d1ea52c6f765356de95a`.

```
"""Exact rational Fourier absolute-moment tail certificate for balanced blocks.

For h>=100,n>=h,B_n=Q_n Q_floor(n/2) Q_floor(n/4) Q_floor(n/8),
x=sigma*integral_{2/min(n,5h)<=|theta|<=pi}|phi(theta)|dtheta < 1/3000,
y=sigma^3*integral_{2/min(n,5h)<=|theta|<=pi}theta^2|phi(theta)|dtheta < 1/50.
See balanced_tail_h100.md for the proof and definitions.
All certificate comparisons use fractions; decimal output is descriptive only.
"""
from fractions import Fraction as F
from math import isqrt
from pathlib import Path
import json

H=100
GRID=256
DEN=10**24
TAIL_DEN=10**32

def down(x,den=DEN):
    return F(x.numerator*den//x.denominator,den)
def up(x,den=DEN):
    return F(-((-x.numerator*den)//x.denominator),den)
def exp_interval(x):
    term=total=F(1)
    for j in range(1,81):
        term*=x/j
        total+=term
    nxt=term*x/81
    return down(total),up(total+nxt/(1-x/82))

elo=[]; vlow=[]; vhigh=[]
for i in range(5*GRID+1):
    lo,hi=exp_interval(F(i,GRID))
    elo.append(lo)
    vlow.append(down(hi/(1+hi)**2))
    vhigh.append(up(lo/(1+lo)**2))
Ilo=[F(0)]; Ihi=[F(0)]
for i in range(5*GRID):
    l,u=F(i,GRID),F(i+1,GRID)
    area=(u**3-l**3)/3
    Ilo.append(Ilo[-1]+vlow[i+1]*area)
    Ihi.append(Ihi[-1]+vhigh[i]*area)

def I2lo(x):
    assert 0<=x<=5
    i=x.numerator*GRID//x.denominator
    if i==5*GRID:return Ilo[i]
    return down(Ilo[i]+vlow[i+1]*(x**3-F(i,GRID)**3)/3)

def I2hi(x):
    assert 0<=x<=5
    i=x.numerator*GRID//x.denominator
```

```
    if i==5*GRID:return Ihi[i]
    return up(Ihi[i]+vhigh[i]*(x**3-F(i,GRID)**3)/3)

def Jlo(z):
    ans=F(0)
    for a in range(4):
        x=z/2**a
        assert (x*GRID).denominator==1
        el=elo[int(x*GRID)]
        ans+=F(1,2)-1/(1+el)
    return down(ans)

# e^(k/100) lower bounds by one exact Taylor lower bound and rounded
# multiplication. Induction certifies every table entry independently
# of floating point. Tail reciprocals are rounded upward.
r=down(sum((F(1,100)**j / __import__('math').factorial(j) for j in range(21)),F(0)),TAIL_DEN)
exp_table=[F(1)]
for k in range(1,20001):
    exp_table.append(down(exp_table[-1]*r,TAIL_DEN))
def neg_exp_up(x):
    assert 0<=x<=200
    k=x.numerator*100//x.denominator
    return up(1/exp_table[k],TAIL_DEN)

def sqrt_up(x):
    den=10**6
    t=isqrt(x.numerator*den*den//x.denominator)+1
    ans=F(t,den)
    assert ans*ans>x
    return ans

results=[]
first_tail=None
for zi in range(32,161):
    zlo=F(zi,32)
    zhi=F(zi+1,32) if zi<160 else None
    mass=Jlo(zlo)
    if zhi is None:
        # f2(x)<=x^2 e^-x is unimodal with maximum<1; its integral is2.
        # Four rows give sigma^2/h^3 <=4(2+1/H)<9.
        assert 4*(2+F(1,H))<9
        B=F(9)
    else:
        # f2 increases on[0,2]: 2/x-tanh(x/2)>0.
        B=sum((I2hi(zhi/2**a)+((zhi/2**a)**2*vhigh[int(zhi/2**a*GRID)] if zhi/2**a<=2 else F
    ↪ (1))/H for a in range(4)),F(0))
    R=sqrt_up(B)
    s0=s2=F(0)
    minimum=F(100)
    start=2/min(zhi or F(5),F(5))
    first=start.numerator*40//start.denominator
    for si in range(first,320):
        l,u=F(si,40),F(si+1,40)
        V=F(0)
        for a in range(4):
            c=min(zlo/2**a,2/u)
            V+=I2lo(c-F(1,H))-(F(1,H) if c>2 else F(0))
        prefix=F(25,72)*l*l*V
        abel=mass-F(1,2*H)-F(24,23)/l
        rate=max(prefix,abel)
        # h^(1/2)e^-rate*h and h^(3/2)e^-rate*h decrease for h>=H.
        assert rate>F(3,2*H)
        minimum=min(minimum,rate)
```

```
        E=neg_exp_up(H*rate)
        s0+=(u-l)*E
        s2+=(u**3-l**3)/3*E
    xn=2*R*10*s0
    yn=2*R**3*1000*s2
    ratefar=mass-F(1,2*H)-F(3,23)
    assert ratefar>F(9,2*H)
    E=neg_exp_up(H*ratefar)
    xf=2*F(22,7)*R*1000*E
    yf=F(2,3)*F(22,7)**3*R**3*10**9*E
    x,y=xn+xf,yn+yf
    if zi==32:first_tail=(x/R,y/R**3)
    assert x<F(1,3000),(zi,float(x))
    assert y<F(1,50),(zi,float(y))
    results.append({"z_lower":str(zlo),"z_upper":str(zhi) if zhi is not None else "infinity
    ↪ ","sigma_squared_upper":float(B),"minimum_near_rate":float(minimum),"far_rate":float(
    ↪ ratefar),"x_upper":float(x),"y_upper":float(y)})
central_variance=sum((F(1,12*2**(3*a))+F(1,8*H*2**(2*a))+F(1,24*H**2*2**a) for a in range(4))
    ↪ ,F(0))
assert central_variance<F(1,10)
Rc=sqrt_up(F(1,10))
xc,yc=first_tail[0]*Rc,first_tail[1]*Rc**3
assert xc<F(1,3000) and yc<F(1,50)
report={"central_tn_le_1":{"variance_bound":"1/10","variance_certificate_upper":float(
    ↪ central_variance),"x_upper":float(xc),"y_upper":float(yc)},"status":"PASS","h_min":H
    ↪ ,"frequency_start":"2/min(n,5h)","x_bound":"1/3000","y_bound":"1/50","
    ↪ integral_grid_denominator":GRID,"z_grid_denominator":32,"s_grid_denominator":40,"
    ↪ exp_grid_denominator":100,"rounding_denominator":DEN,"tail_rounding_denominator":
    ↪ TAIL_DEN,"max_x_upper":max(r['x_upper']for r in results),"max_y_upper":max(r['y_upper
    ↪ ']for r in results),"parameter_cells":results}
Path(__file__).with_suffix('.json').write_text(json.dumps(report,indent=2)+'\n')
print(json.dumps({k:v for k,v in report.items()if k!='parameter_cells'},indent=2))
```

5.A.3 verify_balanced_h100_constants.py

SHA-256: `3f06b305f1da42a90c036b2f6ce5603ec031a5f3ac70fb8e9c427cc9247e7d35`.

```
"""Exact scalar/core/first-tail checks for balanced_fourier_h100.md.

Moment and remaining-tail rational certificates are separate. No coefficient scan.
"""
from fractions import Fraction as F
from pathlib import Path
import json


def exp_lower(x, terms=100):
    term=total=F(1)
    for j in range(1,terms+1):
        term *= x/j
        total += term
    return total

H=100
a,b=F(11,12),F(64,49)
C=F(15,2)
phase_coefficient=F(7,40)
assert exp_lower(F(5))>144
assert F(145,143)**3/6 < phase_coefficient
assert F(143,145)**2>F(49,64)
assert F(14,25)<1
```

```
assert 2*F(22,7)/a**3<9

# Lower Gaussian endpoint is > 16/5. Each retained moment >49/50.
U=F(79,10)
assert b*U>F(16,5)**2
assert exp_lower(F(128,25))>160
assert F(5,2)**2 < 2*F(314,100)
tail_fraction_0=F(4,5)*F(5,16)/160
tail_fraction_2=F(4,5)*(F(16,5)+F(5,16))/160
assert tail_fraction_0<F(1,50) and tail_fraction_2<F(1,50)
assert 1-F(1,3*H**2)>F(999,1000)
positive=F(999,1000)*F(49,50)**2*2*F(314,100)/b**2
assert positive>F(7,2)
phase=120*F(22,7)*phase_coefficient**2*C/(a**5*H)
assert phase<F(27,20)

# delta < |theta| <= 2 delta. Put r=L |theta| and U=(sigma/L)^2.
# f(r)=r^2-r^4/12 increases on [1,2]. For p=1/2,3/2,
# U^p exp[-Uf(r)/2] decreases when U>=79/10.
# Integrate via exact positive rectangles, with sqrt(79/10)<3.
N=256
assert U<9
assert U*F(11,12)>3
x=y=F(0)
for i in range(N):
    l=1+F(i,N)
    u=1+F(i+1,N)
    f=l*l-l**4/12
    eminus=1/exp_lower(U*f/2)
    x+=F(6,N)*eminus
    y+=F(6,N)*U*u*u*eminus
assert x<F(1,40),float(x)
assert y<F(1,4),float(y)
central_phase=120*F(22,7)/(36*a**5*F(79,1000)*H)
assert central_phase<F(41,20)
xt=F(1,40)+F(1,3000)
yt=F(1,4)+F(1,50)
assert 3*xt+3*yt+xt*yt<F(9,10)
assert F(7,2)-F(27,20)-F(9,10)==F(5,4)
assert F(7,2)-F(41,20)-F(9,10)==F(11,20)
result={
  'status':'PASS', 'h_min':H,
  'positive_core_lower_decimal':float(positive),
  'phase_loss_upper_decimal':float(phase),
  'central_phase_loss_upper_decimal':float(central_phase),
  'total_tail_bound':'9/10',
  'first_tail_x_upper_decimal':float(x),
  'first_tail_y_upper_decimal':float(y),
  'first_tail_x_bound':'1/40', 'first_tail_y_bound':'1/4',
  'first_tail_grid':N,
}
Path(__file__).with_suffix('.json').write_text(json.dumps(result,indent=2)+'\n')
print(json.dumps(result,indent=2))
```

5.A.4 verify_four_color_prefix.py

SHA-256: `fa5dffc5c3eb42abf182560c7f1e173bef4431ae9c725398cf70d5d985b34556`.

```
"""Exact finite part of the all-index q4 proof; see four_color_partition_tail.md.

No finite list alone is claimed to prove an infinite assertion.
```

```
The analytic argument covers indices n >= 10000.
"""
import hashlib, json, time
from pathlib import Path

N = 10000
start = time.time()
# F(x) = product_{j odd}(1-x^j)^(-4).  Four independent colors.
a = [1] + [0] * N
for j in range(1, N + 1, 2):
    for color in range(4):
        for k in range(j, N + 1):
            a[k] += a[k-j]
assert a[:10] == [1, 4, 10, 24, 51, 100, 190, 344, 601, 1024]
T = [a[k]*a[k] - a[k-1]*a[k+1] for k in range(1, N)]
assert all(v > 0 for v in T)
# Independently recover each coefficient using n a_n = 4 sum sigma_odd(j) a_(n-j).
sigma_odd = [0] * (N + 1)
for d in range(1, N + 1, 2):
    for k in range(d, N + 1, d):
        sigma_odd[k] += d
for n in range(1, N + 1):
    assert n*a[n] == 4*sum(sigma_odd[j]*a[n-j] for j in range(1,n+1))
coeff_bytes = ("\n".join(map(str,a)) + "\n").encode()
result = {
    "coefficient_range": [0,N],
    "strict_log_concavity_range": [1,N-1],
    "failed_indices": [],
    "minimum_integer_turan": min(T),
    "coefficient_sha256": hashlib.sha256(coeff_bytes).hexdigest(),
    "independent_recurrence_verified": True,
    "elapsed_seconds": time.time()-start,
}
path = Path(__file__).with_name('four_color_prefix.json')
path.write_text(json.dumps(result,indent=2)+"\n")
print(json.dumps(result,indent=2))
```

5.A.5 check_q4_prefix_independent.py

SHA-256: 2e2326abcd2fab7cc863bd70a13f7e9b1aec7c8baab0f84760376ab95aa8e375.

```
"""Exact q4 prefix via log derivative; not an infinite-tail proof."""
import json,hashlib
from pathlib import Path
N=10001
b=[0]*(N+1)
for d in range(1,N+1,2):
 for j in range(d,N+1,d):b[j]+=4*d
a=[0]*(N+1);a[0]=1
for n in range(1,N+1):
 s=sum(b[j]*a[n-j] for j in range(1,n+1))
 assert s%n==0
 a[n]=s//n
bad=[n for n in range(1,N) if a[n]*a[n]<a[n-1]*a[n+1]]
result={'checked_internal_indices':[1,N-1],'failures':bad,'coefficient_sha256':hashlib.sha256
    ↪ (('\n'.join(map(str,a))+'\n').encode()).hexdigest(),'method':'n*a_n=4 sum_(j=1)^n
    ↪ sigma_odd(j)*a_(n-j), exact integer arithmetic','scope':'finite verification only'}
Path(__file__).with_suffix('.json').write_text(json.dumps(result,indent=2)+'\n')
print(json.dumps(result))
```

5.A.6 verify_small_q4_transfer.py

SHA-256: 1c2d8594fb17d8e4f88c7ccb58d787845814e28d618126d40201c059d89c96c2.

```
"""Exact finite bridge for q4 indices below 10000, not a BBFM proof alone."""
from fractions import Fraction as F
from pathlib import Path
import runpy,contextlib,io,json
with contextlib.redirect_stdout(io.StringIO()):d=runpy.run_path(str(Path(__file__).with_name
    ↪ ('check_q4_prefix_independent.py')))
a=d['a']
assert 100*a[9998]<99*a[9999]
worst=F(1)
for k in range(1,10000):
 T=a[k]*a[k]-a[k-1]*a[k+1]
 assert 10**9*T>a[k]*a[k],k
 worst=min(worst,F(T,a[k]*a[k]))
assert 400*10**12*99**3501<100**3501
# Large-index bridge with the already-proved R_k<exp(-1.7/sqrt(k)).
assert 40400<201**2
assert F(17*3501,10*201)>F(59,2)
def lower_exp(x,N):
 t=s=F(1)
 for j in range(1,N+1):
  t*=x/j;s+=t
 return s
assert lower_exp(F(3),8)>20
assert lower_exp(F(5,2),10)>12
assert F(4*120,12*20**9)<F(9,40*40404**2)
result={'status':'PASS','small_indices':[1,9999],'R_9999_upper':'99/100','
    ↪ relative_Turan_lower':'1/1000000000','minimum_relative_Turan_exact':[str(worst.
    ↪ numerator),str(worst.denominator)],'minimum_relative_Turan_decimal':float(worst),'
    ↪ d_min':3500,'deficit_upper':'1/1000000000000','large_indices':[10000,40400],'
    ↪ balanced_parameter_cutoff':28000,'scope':'conditional on the independently audited H
    ↪ =100 Fourier theorem; not a finite residual scan'}
Path(__file__).with_suffix('.json').write_text(json.dumps(result,indent=2)+'\n')
print(json.dumps(result))
```

5.A.7 check.py

SHA-256: 0a657b45ba725cabb4ebb7e42af74804927ac15d12cf2714720b00342b37564c.

```
import json, time
from fractions import Fraction
from pathlib import Path
N=256
p=[1]
rows=[]
t0=time.time()
for n in range(1,N+1):
    m=n
    while True:
        p += [0]*m
        for k in range(len(p)-1,m-1,-1): p[k]+=p[k-m]
        if m%2:break
        m//=2
    bad=[]
    # palindromicity halves work, exact integer certificate.
    assert p==p[::-1]
    for k in range(1,(len(p)+1)//2):
        d=p[k]*p[k]-p[k-1]*p[k+1]
```

```
        if d<0:bad.append((k,p[k-1],p[k],p[k+1],d))
    rows.append({'n':n,'degree':len(p)-1,'bad_half_indices':bad})
    if bad:print('failure',n,bad[:8],flush=True)
    if n%32==0:print('checked',n,'degree',len(p)-1,'sec',time.time()-t0,flush=True)
Path('research/bbfm_logconcavity/exact_check.json').write_text(json.dumps(rows))
print('DONE',time.time()-t0)
```

5.A.8 check_balanced_blocks.py

SHA-256: `8f0fbd1cd3a12ad569e6d84cdbc2dc9524af65d87b0fdccd2febb10b02de2c2f`.

```
import json
p=[1];bad=[]
for n in range(1,257):
 for s in range(4):
  if n%(1<<s):break
  k=n//(1<<s);q=p+[0]*k
  for j,a in enumerate(p):q[j+k]+=a
  p=q
 if n>=8:
  hits=[j for j in range(1,(len(p)+1)//2) if p[j]*p[j]<p[j-1]*p[j+1]]
  if hits:bad.append([n,hits])
print(json.dumps({'range':[8,256],'failures':bad,'last_degree':len(p)-1,'status':'finite
    ↪ exact verification only'}))
```

5.A.9 dyadic_certificate.cpp

SHA-256: `b3154e0381004bb878a33af90f513d5e20b3ab157db6080bf1fc34c2ccb9c7fe`.

```
// Finite certificate engine for B_n = Q_n Q_floor(n/2) Q_floor(n/4) Q_floor(n/8).
// All arithmetic is integer-only. Build via test_dyadic_certificate.py so that
// SOURCE_SHA256 records this exact source. No assertion is made beyond the
// explicitly reported finite (n,k) ranges. See dyadic_certificate.md.
#include <algorithm>
#include <cstdint>
#include <iostream>
#include <limits>
#include <stdexcept>
#include <string>
#include <vector>
#ifndef SOURCE_SHA256
#error "Define SOURCE_SHA256 as the SHA256 of this source file."
#endif
using U128 = unsigned __int128;
using I128 = __int128;
constexpr uint64_t HIGH = uint64_t(1) << 63;
struct Dyadic { uint64_t m = 0; int64_t e = 0; };
struct Interval { Dyadic lo, hi; };

static int bits(U128 x) {
    uint64_t hi = uint64_t(x >> 64);
    return hi ? 128 - __builtin_clzll(hi) :
        (uint64_t(x) ? 64 - __builtin_clzll(uint64_t(x)) : 0);
}
static int64_t exponent(I128 e) {
    if (e < std::numeric_limits<int64_t>::min() ||
        e > std::numeric_limits<int64_t>::max())
        throw std::overflow_error("dyadic exponent overflow");
    return int64_t(e);
```

```
}
static bool valid(Dyadic a) { return a.m ? a.m >= HIGH : a.e == 0; }
// Exact numerator n, then directed normalization to a 64-bit mantissa.
static Dyadic normalize(U128 n, int64_t e, bool upward) {
    if (!n) return {};
    int shift = bits(n) - 64;
    if (shift <= 0)
        return {uint64_t(n << -shift), exponent(I128(e) + shift)};
    U128 q = n >> shift;
    if (upward && (n - (q << shift))) ++q;
    I128 out_e = I128(e) + shift;
    if (q == (U128(1) << 64)) { q >>= 1; ++out_e; }
    return {uint64_t(q), exponent(out_e)};
}
static Dyadic add(Dyadic a, Dyadic b, bool upward) {
    if (!a.m) return b;
    if (!b.m) return a;
    if (a.e < b.e) std::swap(a,b);
    I128 d = I128(a.e) - b.e;
    if (d <= 64) {
        // Maximum is (2^64-1)*2^64 + (2^64-1) = 2^128-1.
        U128 numerator = (U128(a.m) << unsigned(d)) + b.m;
        return normalize(numerator, b.e, upward);
    }
    // 0 < b < 2^a.e: lower rounds to a, upper adds exactly one ulp.
    if (!upward) return a;
    if (a.m == std::numeric_limits<uint64_t>::max())
        return {HIGH, exponent(I128(a.e) + 1)};
    return {a.m + 1, a.e};
}
static Interval add(Interval a, Interval b) {
    return {add(a.lo,b.lo,false), add(a.hi,b.hi,true)};
}
// Compare a*b with c*d exactly; exponent sums use signed 128-bit integers.
static int compare_products(Dyadic a, Dyadic b, Dyadic c, Dyadic d) {
    U128 p = U128(a.m)*b.m, q = U128(c.m)*d.m;
    if (!p || !q) return (p>q)-(p<q);
    int bp=bits(p), bq=bits(q);
    I128 ep=I128(a.e)+b.e, eq=I128(c.e)+d.e;
    I128 tp=ep+bp, tq=eq+bq;
    if (tp != tq) return tp > tq ? 1 : -1;
    // Equal top-bit positions: align to the larger mantissa bit length.
    // The shifted operand has at most max(bp,bq)<=128 significant bits.
    if (bp < bq) p <<= bq-bp;
    if (bq < bp) q <<= bp-bq;
    return (p>q)-(p<q);
}
static uint64_t natural(const std::string& s) {
    if (s.empty() || s.find_first_not_of("0123456789") != std::string::npos)
        throw std::invalid_argument("expected nonnegative decimal integer");
    size_t end=0;
    uint64_t n=std::stoull(s,&end);
    if (end != s.size()) throw std::invalid_argument("invalid integer");
    return n;
}
static Dyadic read_dyadic() {
    std::string m,e;
    if (!(std::cin>>m>>e)) throw std::invalid_argument("missing dyadic endpoint");
    size_t end=0; int64_t ee=std::stoll(e,&end);
    if (end != e.size()) throw std::invalid_argument("invalid exponent");
    Dyadic x{natural(m),ee};
    if (!valid(x)) throw std::invalid_argument("endpoint is not normalized");
    return x;
```

```
}
static void arithmetic() {
    std::string op;
    while (std::cin>>op) {
        if (op == "ADD") {
            Dyadic a=read_dyadic(),b=read_dyadic();
            Dyadic lo=add(a,b,false),hi=add(a,b,true);
            std::cout<<lo.m<<' '<<lo.e<<' '<<hi.m<<' '<<hi.e<<'\n';
        } else if (op == "CMP") {
            Dyadic a=read_dyadic(),b=read_dyadic(),c=read_dyadic(),d=read_dyadic();
            std::cout<<compare_products(a,b,c,d)<<'\n';
        } else throw std::invalid_argument("unknown arithmetic operation");
    }
}
static void emit_endpoint(Dyadic a) { std::cout<<'['<<a.m<<','<<a.e<<']'; }
static void emit_interval(Interval a) {
    std::cout<<'[';emit_endpoint(a.lo);std::cout<<',';emit_endpoint(a.hi);std::cout<<']';
}
static std::string decimal(U128 n) {
    std::string s;
    do { s.push_back(char('0'+n%10)); n/=10; } while(n);
    std::reverse(s.begin(),s.end());return s;
}
int main(int argc,char**argv) {
    try {
        if (argc==2 && std::string(argv[1])=="--arithmetic") {arithmetic();return 0;}
        if (argc<3) throw std::invalid_argument(
            "usage: dyadic_certificate NMAX KMAX [--nmin N] [--dump-intervals]");
        uint64_t nmax=natural(argv[1]),kmax=natural(argv[2]),nmin=8;
        bool dump=false;
        for (int i=3;i<argc;++i) {
            std::string opt=argv[i];
            if (opt=="--nmin" && i+1<argc) nmin=natural(argv[++i]);
            else if(opt=="--dump-intervals") dump=true;
            else throw std::invalid_argument("unknown or incomplete option: "+opt);
        }
        if (!nmin || nmin>nmax || nmax>1000000000ULL)
            throw std::invalid_argument("require 1 <= nmin <= nmax <= 1000000000");
        if (!kmax || kmax>std::numeric_limits<size_t>::max()/sizeof(Interval)-2)
            throw std::invalid_argument("kmax must be positive and safely allocatable");
        const uint64_t cap=kmax+1;
        const Dyadic one{HIGH,-63};
        std::vector<Interval> p(1,Interval{one,one});
        uint64_t degree=0;
        U128 total_checked=0,total_inconclusive=0,total_strict=0;
        std::cout<<"{\"type\":\"metadata\",\"schema\":1,\"source_sha256\":\""<<SOURCE_SHA256
          <<"\",\"family\":\"B_n=Q_n*Q_floor(n/2)*Q_floor(n/4)*Q_floor(n/8)\","
          <<"\"scope\":\"finite only; no analytic skips\",\"nmin\":"<<nmin<<",\"nmax\":"<<
    ↪ nmax
          <<",\"kmax\":"<<kmax<<",\"stored_cap\":"<<cap
          <<",\"built_but_not_checked_n\":[1,"<<nmin-1<<"]}\n";
        for (uint64_t n=1;n<=nmax;++n) {
            for (unsigned s=0;s<4;++s) {
                uint64_t divisor=uint64_t(1)<<s;
                if (n%divisor) break;
                uint64_t w=n/divisor;
                degree+=w; // n<=10^9 bounds degree < 10^18.
                uint64_t end=std::min(cap,degree);
                p.resize(size_t(end)+1);
                if (w<=end)
                    for (uint64_t j=end;j>=w;--j) p[j]=add(p[j],p[j-w]);
            }
            if (n<nmin) continue;
```

```
            uint64_t end=std::min(kmax,degree/2),bad=0;
            std::cout<<"{\"type\":\"row\",\"n\":"<<n<<",\"degree\":"<<degree
              <<",\"stored_through\":"<<p.size()-1<<",\"checked_k\":[1,"<<end
              <<"],\"reflected_k\":["<<degree-end<<','<<degree-1
              <<"],\"complete_half_range\":"<<(end==degree/2?"true":"false")
              <<",\"inconclusive_k\":[";
            for(uint64_t k=1;k<=end;++k) {
                if(compare_products(p[k].lo,p[k].lo,p[k-1].hi,p[k+1].hi)<=0) {
                    if(bad++) std::cout<<',';
                    std::cout<<k;
                }
            }
            total_checked+=end;total_inconclusive+=bad;total_strict+=end-bad;
            std::cout<<"],\"strict_certified\":"<<end-bad;
            if(dump) {
                std::cout<<",\"intervals\":[";
                for(size_t k=0;k<p.size();++k) {if(k)std::cout<<',';emit_interval(p[k]);}
                std::cout<<']';
            }
            std::cout<<"}\n";
        }
        std::cout<<"{\"type\":\"summary\",\"checked_inequalities\":"<<decimal(total_checked)
          <<",\"strict_certified\":"<<decimal(total_strict)<<",\"inconclusive\":"<<decimal(
    ↪ total_inconclusive)
          <<",\"all_requested_strict_certified\":"<<(total_inconclusive?"false":"true")
          <<",\"status\":\"finite verification only; inconclusive is not a counterexample\"}\
    ↪ n";
        return total_inconclusive ? 2 : 0;
    } catch(const std::exception& e) {
        std::cerr<<"ERROR: "<<e.what()<<'\n';return 1;
    }
}
```

5.B Extraction, execution and output validation

The driver below extracts all ten source listings (including itself) from this TeX file, runs the rational and exact checks, compiles the directed engine, executes both finite ranges, and validates every output row. Save this listing as `reproduce.py` and invoke it in a fresh directory:

```
python3 reproduce.py /absolute/path/to/KEIO_BBFM_Denominator_Log_Concavity_2026.tex
```

The driver is supplemental reproducibility code; the source files in Appendix 5.A are reproduced byte for byte from the versions used for the mathematical certificates. The single TeX file itself can be compiled independently with two ordinary `pdflatex` runs; no source extraction or certificate execution occurs during compilation.

```
"""Extract and reproduce the manuscript certificates. Python 3, no -O.
Usage: python reproduce.py /absolute/path/to/manuscript.tex
Run in an empty writable directory with g++ available.
"""
from pathlib import Path
import hashlib, json, re, subprocess, sys
assert __debug__, 'Assertions must be enabled'
tex = Path(sys.argv[1]).resolve().read_text()
root = Path.cwd()
out = root / 'research' / 'bbfm_logconcavity'
out.mkdir(parents=True, exist_ok=True)
pattern = (r'^%<source:([^>]+)>\n'
           r'\\begin\{lstlisting\}\n(.*?)'
           r'\\end\{lstlisting\}\n%</source>')
sources = re.findall(pattern, tex, re.M | re.S)
assert len(sources) == 10
for name, code in sources:
    assert Path(name).name == name
    (out / name).write_text(code)

def run_python(name):
    result = subprocess.run([sys.executable, str(out / name)],
                            check=True, text=True, capture_output=True)
    (out / (name + '.stdout')).write_text(result.stdout)
    print('PASS', name, flush=True)
    return result.stdout

for name in ('verify_balanced_moments.py',
             'verify_balanced_tail_h100.py',
             'verify_balanced_h100_constants.py',
             'verify_four_color_prefix.py',
             'verify_small_q4_transfer.py'):
    run_python(name)
run_python('check.py')
rows = json.loads((out / 'exact_check.json').read_text())
assert [r['n'] for r in rows] == list(range(1,257))
assert [r['n'] for r in rows if r['bad_half_indices']] == [3,5,6,7]
b = json.loads(run_python('check_balanced_blocks.py'))
assert b['range'] == [8,256] and b['failures'] == []
source = out / 'dyadic_certificate.cpp'
source_hash = hashlib.sha256(source.read_bytes()).hexdigest()
assert source_hash == ('b3154e0381004bb878a33af90f513d5e20'
                       'b3ab157db6080bf1fc34c2ccb9c7fe')
exe = out / 'dyadic_certificate'
subprocess.run(['g++','-std=c++17','-O3','-Wall','-Wextra',
                '-DSOURCE_SHA256="' + source_hash + '"',
                str(source),'-o',str(exe)],check=True)

```

```
def degree(n):
    return sum((n//(1<<a))*(n//(1<<a)+1)//2 for a in range(4))

def certify(first, last, cap, expected_count, expected_hash, filename):
    path = out / filename
    with path.open('w') as stream:
        subprocess.run([str(exe),str(last),str(cap),'--nmin',str(first)],
                       stdout=stream,check=True)
    assert hashlib.sha256(path.read_bytes()).hexdigest() == expected_hash
    records = [json.loads(line) for line in path.read_text().splitlines()]
    assert len(records) == last-first+3
    meta, summary = records[0], records[-1]
    assert meta['type'] == 'metadata' and meta['schema'] == 1
    assert meta['source_sha256'] == source_hash
    assert (meta['nmin'],meta['nmax'],meta['kmax']) == (first,last,cap)
    assert meta['stored_cap'] == cap+1
    assert meta['built_but_not_checked_n'] == [1,first-1]
    count = 0
    for n, row in zip(range(first,last+1),records[1:-1]):
        d = degree(n)
        end = min(cap,d//2)
        assert row['type'] == 'row' and row['n'] == n
        assert row['degree'] == d
        assert row['stored_through'] == min(cap+1,d)
        assert row['checked_k'] == [1,end]
        assert row['reflected_k'] == [d-end,d-1]
        assert row['complete_half_range'] == (end == d//2)
        assert row['inconclusive_k'] == []
        assert row['strict_certified'] == end
        count += end
    assert count == expected_count
    assert summary['type'] == 'summary'
    assert summary['checked_inequalities'] == count
    assert summary['strict_certified'] == count
    assert summary['inconclusive'] == 0
    assert summary['all_requested_strict_certified'] is True
    print('PASS',filename,'rows',last-first+1,'strict',count,flush=True)

certify(8,256,40400,1878532,
        'fc6f65c4a2650f67b7cc97656460da2412ba4887d4cb99dfb13c2ede565be57a',
        'dyadic_B8_B256.jsonl')
certify(257,27999,40399,1119898742,
        '93e20573a804fc4e213286519b3ff26b725fa584709253d95de7766c72d4e14f',
        'dyadic_B257_B27999.jsonl')
print('ALL CERTIFICATES PASS')
```

Chapter 6

Unimodality of the Shifted Nekrasov Okounkov Polynomials

Chapter overview

We prove that the original shifted Nekrasov–Okounkov polynomials $Q_n(z) = [q^n] \prod_{m\geq 1}(1-q^m)^{-z-1}$ are unimodal for every integer $n \geq 1$. A summandwise Bernoulli mean–mode bound gives a nonincreasing coefficient tail beginning at $\lceil (142/125)\sqrt{n} \rceil$ for every n. An exact two-dimensional coefficient determinant, an explicit modular transformation with complex derivative bounds, and complete Fourier-tail estimates prove strict log-concavity through $\lceil \sqrt{13n/10} \rceil$ for $n \geq 50000$. An integer-only fixed-point GMP certificate verifies the remaining 8 492 344 prefix inequalities through $n = 50000$, with exact coverage of all required indices. The complete analytic proof, executable source listings, and extraction and reproduction driver are included in one self-contained TEX file. The theorem is unimodality, not global coefficient log-concavity.

6.1 The original polynomial and the proof architecture

Put $\sigma_1(j) = \sum_{d|j} d$ and $\sigma_{-1}(j) = \sigma_1(j)/j$. Throughout,

$$F(q) = \sum_{j\geq 1} \sigma_{-1}(j) q^j, \qquad Q_n(z) = [q^n] \exp((1+z)F(q)) = \sum_{k=0}^{n} A_{n,k} z^k. \tag{6.1.1}$$

Taking the formal logarithm of the Euler product gives $e^{F(q)} = \prod_{m\geq 1}(1-q^m)^{-1}$. In particular

$$\sum_{n\geq 0} Q_n(z) q^n = \prod_{m\geq 1} (1-q^m)^{-z-1}, \qquad Q_0(z) = 1. \tag{6.1.2}$$

The Nekrasov–Okounkov identity identifies this same polynomial with

$$Q_n(z) = \sum_{\lambda \vdash n} \prod_{h\in\mathcal{H}(\lambda)} \left(1 + \frac{z}{h^2}\right),$$

where $\mathcal{H}(\lambda)$ is the multiset of hook lengths. This is the polynomial in Amdeberhan's Conjecture 2.2(b) [6.1] and Heim–Neuhauser's Conjecture 3 [6.2]; no change of the conjectured object is made here. If $P_n(x) = [q^n]e^{xF(q)}$ denotes the unshifted D'Arcais polynomial, then $Q_n(z) = P_n(z+1)$, not $P_n(z)$.

A finite real sequence $(a_0, \ldots, a_n)$ is *unimodal* if, for some m, it is nondecreasing up to m and nonincreasing thereafter. It is *log-concave* if $a_k^2 \geq a_{k-1}a_{k+1}$ at each internal index. All coefficients $A_{n,k}$ for $0 \leq k \leq n$ are positive.

Theorem 6.1.1 (Shifted Nekrasov–Okounkov unimodality)**.** *For every integer $n \geq 1$, the coefficient sequence $(A_{n,0}, A_{n,1}, \ldots, A_{n,n})$ of $Q_n(z)$ is unimodal.*

The stronger assertion of global log-concavity of the shifted polynomial is not proved. Neither is any corresponding all-n assertion for the unshifted polynomial asserted. Our exact finite and infinite regions are as follows, with

$$K_*(n) = \left\lceil \frac{142}{125}\sqrt{n} \right\rceil, \qquad K(n) = \left\lceil \sqrt{\frac{13n}{10}} \right\rceil. \tag{6.1.3}$$

Rows	Index region	Proof ingredient
Every $n \geq 1$	$k \geq K_*(n)$	Nonincreasing tail
$n \geq 50000$	$1 \leq k \leq K(n)$	Strict analytic log-concavity
$1 \leq n \leq 50000$	$1 \leq k \leq \min(n-1, K_*(n))$	Exact finite certificate
$n = 1$	Entire row	$Q_1(z) = 1 + z$

There is an intentional overlap at $n = 50000$. Computation at a log-concavity index k requires coefficient columns $k-1, k, k+1$; $K_*(50000) = 255$, so columns through 256 are computed.

Heim and Neuhauser verified an even stronger coefficient property through degree 1000 [6.2, Theorem 2]. Hong and Zhang established small-index log-concavity, a decreasing far tail, and a conditional eventual unimodality result [6.3, Theorems 1.1 and 1.3]. Zhang proved unimodality for all sufficiently large n [6.4, Theorem 1.6]. We do not present eventual unimodality as new; the role of the analysis below is to supply a completely explicit cutoff that can be joined to a finite certificate. A focused review of these sources and subsequent accessible work did not identify an all-n resolution. This is not an exhaustive priority determination.

The proof uses the classical eta transformation and Darroch's mean–mode theorem, whose precise uses are given below. All other analytic estimates needed here are derived in the text.

The appendix contains the exact rational scalar checkers, the complete C++ finite engine, and a Python extraction and reproduction driver. No external TEX inputs or bibliography files are required. The arithmetic trust base consists of Python integers and fractions, a C++17 compiler supporting unsigned 128-bit integers, and GMP on the explicitly checked LP64 platform. This is a computer-assisted mathematical proof, not a proof-assistant formalization or a claim of external peer review.

6.2 A common nonincreasing tail for every row

Expanding each factor of (6.1.2) by the binomial theorem gives

$$Q_n(z) = \sum_{\substack{m_1,m_2,\ldots\geq 0 \\ \sum j m_j = n}} R_{\boldsymbol{m}}(z), \qquad R_{\boldsymbol{m}}(z) = \prod_{j\geq 1} \binom{m_j + z}{m_j}. \tag{6.2.1}$$

Only finitely many m_j are nonzero. For a fixed summand,

$$\frac{R_{\boldsymbol{m}}(z)}{R_{\boldsymbol{m}}(1)} = \prod_j \prod_{\ell=1}^{m_j} \frac{\ell + z}{\ell + 1} \tag{6.2.2}$$

is the probability generating function of a sum of independent Bernoulli variables with success probabilities $1/(\ell+1)$. Its coefficient sequence is log-concave and thus unimodal: for example, this follows from Newton's inequalities, since all zeros of the product are negative.

We use Darroch's theorem in the following standard form: every mode m of a finite sum of independent Bernoulli variables, with mean μ, satisfies $|m - \mu| < 1$ [6.6]. A modern proof and an extension are in Gnedin [6.7, Section 5.1 and Theorem 2]. In particular, the last mode is at most $\lceil \mu \rceil$, and a log-concave coefficient sequence is nonincreasing beyond that index.

Lemma 6.2.1 (Uniform tail)**.** *For every $n \geq 1$, $A_{n,k} \geq A_{n,k+1}$ whenever $K_*(n) \leq k < n$. The same assertion holds with $K(n)$ in place of $K_*(n)$.*

Proof. Write $d_\ell = \#\{j : m_j \geq \ell\}$ for a fixed multiplicity vector in (6.2.1). Distinct part sizes are distinct positive integers, so

$$n = \sum_{\ell\geq 1} \sum_{j : m_j \geq \ell} j \geq \frac{1}{2} \sum_{\ell \geq 1} d_\ell (d_\ell + 1) \geq \frac{1}{2} \sum_{\ell \geq 1} d_\ell^2.$$

The mean μ_R of (6.2.2) satisfies, by Cauchy–Schwarz,

$$\mu_R^2 = \left(\sum_{\ell \geq 1} \frac{d_\ell}{\ell + 1} \right)^2 \leq 2n \sum_{\ell \geq 1} \frac{1}{(\ell+1)^2} = 2(\zeta(2) - 1)n. \tag{6.2.3}$$

The rational enclosure $\pi < 355/113$ implies

$$\left(\frac{142}{125}\right)^2 - \left(\frac{(355/113)^2}{3} - 2 \right) = \frac{375473}{598546875} > 0, \qquad \left(\frac{142}{125}\right)^2 < \frac{13}{10}.$$

Thus $\lceil \mu_R \rceil \leq K_*(n) \leq K(n)$. Each summand has a nonincreasing tail from this common index. Summing those inequalities proves the assertion. We do not use the false general principle that mixtures of unimodal sequences are unimodal. □

Lemma 6.2.2 (Joining a prefix and a tail)**.** *Let $a_0, \ldots, a_n$ be positive, and let T be a nonnegative integer. If $a_k^2 \geq a_{k-1} a_{k+1}$ for $1 \leq k \leq \min(T, n-1)$ and $a_k \geq a_{k+1}$ for $T \leq k < n$, then the whole sequence is unimodal.*

Proof. The first condition makes a_k/a_{k-1} nonincreasing through the joining point, so the prefix has at most one transition from a ratio at least one to a ratio below one. All remaining ratios are at most one by the second condition. If $T \geq n-1$ the first condition alone suffices. □

6.3 An exact determinant criterion

Fix integers $n > k \geq 1$ and abbreviate

$$B_k = k!A_{n,k} = [q^n]e^{F(q)}F(q)^k, \qquad D = q\frac{d}{dq}, \qquad M_j(t) = D^j F(e^{-t}).$$

Put $L = M_0(t)$, $\mu = M_1/L$, and $V = M_2/L - \mu^2$. The distribution on the positive integers with probabilities $\sigma_{-1}(j)e^{-tj}/L$ has mean μ and variance $V > 0$. The saddle equation is

$$n = M_1(t) + k\mu(t). \tag{6.3.1}$$

Its right side is strictly decreasing in t, with derivative $-(M_2 + kV) < 0$. It tends to k as $t \to \infty$ and to infinity as $t \downarrow 0$. Hence the saddle exists and is unique.

For $r = e^{-t}$, write $F_\theta = F(re^{i\theta})$ and define

$$\phi_k(\theta) = e^{F_\theta - L}(F_\theta/L)^k e^{-in\theta}, \qquad I_k = \int_{-\pi}^{\pi} \phi_k(\theta)\, d\theta, \tag{6.3.2}$$

$$a_l(\theta) = |e^{F_\theta - L}|\,|F_\theta/L|^l, \qquad J_j^{(l)} = \int_{-\pi}^{\pi} |\theta|^j a_l(\theta)\, d\theta, \tag{6.3.3}$$

$$\sigma_l^2 = M_2 + lV \qquad (l \geq 0). \tag{6.3.4}$$

The functions a_l are even. Fourier inversion gives the real number $I_k = 2\pi r^n B_k/(e^L L^k)$.

Lemma 6.3.1 (Determinant identity)**.** *At any common radius $0 < r < 1$,*

$$\frac{(k!)^2 r^{2n}}{e^{2L}L^{2k}}(A_{n,k}^2 - A_{n,k-1}A_{n,k+1}) = \frac{1}{(2\pi)^2(k+1)}\left[I_k^2 - \frac{k}{2}\int_{-\pi}^{\pi}\int_{-\pi}^{\pi}\phi_k(\theta)\phi_k(\psi)\frac{(F_\theta - F_\psi)^2}{F_\theta F_\psi}\, d\theta\, d\psi\right]. \tag{6.3.5}$$

At zeros of F_θ or F_ψ, the integrand means the expression obtained by cancellation first. In particular it is continuous for $k = 1$. A sufficient condition for strict log-concavity is

$$I_k^2 > k\mu^2 J_0^{(k-1)} J_2^{(k-1)}. \tag{6.3.6}$$

Proof. Apply Fourier inversion to B_{k-1}, B_k, B_{k+1} at the same radius. The factorial conversion contributes $k/(k+1)$ to the neighboring product. Symmetrizing that product and using

$$\frac{1}{2}\left(\frac{x}{y} + \frac{y}{x}\right) = 1 + \frac{(x-y)^2}{2xy}$$

gives (6.3.5). This calculation can be done entirely after multiplication by $x^k y^k$; thus it requires no global zero-free assumption. Since the coefficients of F are positive,

$$|F_\theta - F_\psi| \leq \sum_{j \geq 1} \sigma_{-1}(j) r^j j|\theta - \psi| = M_1|\theta - \psi|.$$

After cancellation the absolute value of the subtracted term in square brackets in (6.3.5) is bounded by

$$\frac{k\mu^2}{2}\iint a_{k-1}(\theta)a_{k-1}(\psi)(\theta-\psi)^2\,d\theta\,d\psi = k\mu^2 J_0^{(k-1)}J_2^{(k-1)}.$$

The mixed first moments vanish by evenness. This proves the sufficient condition without replacing the determinant by an asymptotic expansion. □

6.4 Explicit eta bounds, including complex derivatives

We give the details of the analytic input; the checkers in the appendix verify its scalar rational inequalities, not its logical derivation. Put

$$G(w) = F(e^{-w}), \qquad C = \frac{\pi^2}{6}, \qquad \operatorname{Re} w > 0.$$

The exact eta transformation in the partition-product normalization is

$$G(w) = \frac{C}{w} + \frac{1}{2}\log\frac{w}{2\pi} - \frac{w}{24} + R(w), \qquad R(w) = F(e^{-4\pi^2/w}). \tag{6.4.1}$$

For example, it follows from Hardy–Ramanujan [6.5, Lemma 4.31] by specializing the denominator parameter to 1 and its variable to $w/(2\pi)$. It also follows directly from $\eta(-1/\tau) = \sqrt{-i\tau}\,\eta(\tau)$ with $\tau = iw/(2\pi)$. Analytic logarithms are fixed by their values on the positive real axis; hence there is no omitted multiple of $2\pi i$. This is an exact identity with a retained remainder.

Lemma 6.4.1 (Remainder derivatives)**.** *If $0 < t \le 1/20$, $w = t - i\theta$, and $|\theta| \le t/2$, then*

$$t^j|R^{(j)}(w)| < t\,10^{-100} \qquad (0 \le j \le 4). \tag{6.4.2}$$

Proof. On $|z-w| \le t/4$ we have $\operatorname{Re} z \ge 3t/4$, $|\operatorname{Im} z| \le 3t/4$, and $|z|^2 \le 17t^2/8$. Thus $\operatorname{Re}(1/z) \ge 6/(17t)$ and $|e^{-4\pi^2/z}| \le e^{-13/t}$, since $24\pi^2/17 > 13$. The elementary bound $\sigma_{-1}(j) \le j$ gives, for $|q| \le x$ with $0 < x \le 1/4$,

$$|F(q)| \le \frac{x}{(1-x)^2} < 2x.$$

Cauchy's formula on the disk therefore gives $t^j|R^{(j)}(w)| \le 2j!4^j e^{-13/t}$. The function $e^{-13/t}/t$ increases on $(0, 1/20]$; consequently the largest needed ratio is at most $2\cdot 4!\cdot 4^4\cdot 20e^{-260} < 10^{-100}$. The last inequality follows from $e > 27/10$ by an exact rational power comparison, included in the checker. □

Factor

$$G(w) = \frac{C}{w}(1+E(w)), \qquad E(w) = \frac{w}{2C}\log\frac{w}{2\pi} - \frac{w^2}{24C} + \frac{wR(w)}{C}. \tag{6.4.3}$$

Let $\mathcal{L}(t) = \log(2\pi/t)$. For $|\theta| \le t/2$, $|w| \le 9t/8$, $|w| \ge t$, $|\arg w| \le 1/2$, and $|\log(w/(2\pi))| \le \mathcal{L}(t) + 1/2$. The last inequality uses $|w| < 2\pi$, so the absolute real logarithm is no larger than $\mathcal{L}(t)$. Direct differentiation gives

$$\begin{aligned}
E' &= \frac{\log(w/(2\pi))+1}{2C} - \frac{w}{12C} + \frac{R + wR'}{C},\\
E'' &= \frac{1}{2Cw} - \frac{1}{12C} + \frac{2R' + wR''}{C},\\
E''' &= -\frac{1}{2Cw^2} + \frac{3R'' + wR'''}{C},\\
E'''' &= \frac{1}{Cw^3} + \frac{4R''' + wR''''}{C}.
\end{aligned} \tag{6.4.4}$$

The respective non-remainder upper bounds for $|E|, t|E'|, t^2|E''|, t^3|E'''|, t^4|E''''|$ are

$$\frac{9t}{16C}(\mathcal{L}+\tfrac{1}{2})+\frac{81t^2}{1536C},\quad \frac{t}{2C}(\mathcal{L}+\tfrac{3}{2})+\frac{9t^2}{96C},\quad \frac{t}{2C}+\frac{t^2}{12C},\quad \frac{t}{2C},\quad \frac{t}{C}. \tag{6.4.5}$$

Each increases with t on the range considered. Each scaled remainder contribution is less than 10^{-100}, by Lemma 6.4.1. Using $C > 411/250$ and $\mathcal{L}(1/100) < 129/20$ in (6.4.5) proves, throughout $0 < t \le 1/100$, $|\theta| \le t/2$,

$$(|E|, t|E'|, t^2|E''|, t^3|E'''|, t^4|E''''|) < \left(\frac{1}{40}, \frac{1}{40}, \frac{1}{300}, \frac{1}{300}, \frac{1}{150}\right). \tag{6.4.6}$$

In particular G is zero-free on this core. The logarithm $H = \log(1+E)$ is defined there by its convergent power series. For the five bounds $a_0, \dots, a_4$ in (6.4.6), set $m = 1 - a_0$. Differentiation of H yields

$$\begin{aligned} t^2|H''| &\le b_2 := \frac{a_2}{m} + \frac{a_1^2}{m^2}, \\ t^3|H'''| &\le b_3 := \frac{a_3}{m} + \frac{3a_1a_2}{m^2} + \frac{2a_1^3}{m^3}, \\ t^4|H''''| &\le b_4 := \frac{a_4}{m} + \frac{4a_1a_3+3a_2^2}{m^2} + \frac{12a_1^2a_2}{m^3} + \frac{6a_1^4}{m^4}. \end{aligned} \tag{6.4.7}$$

Since $\log G = \log C - \log w + H$, exact rational substitution gives

$$t^3|\partial_\theta^3 \log F(e^{-t+i\theta})| < \frac{401}{200}, \qquad t^4|\partial_\theta^4 \log F(e^{-t+i\theta})| < \frac{601}{100}. \tag{6.4.8}$$

The conversion $\partial_\theta = -i\partial_w$ preserves absolute values. These are complex derivative bounds over the whole core, not merely real-axis cumulant estimates.

Lemma 6.4.2 (Real-axis moments)**.** *For every $0 < t \le 1/100$, independently of k,*

$$tM_0 > \frac{161}{100}, \qquad t\mu < \frac{51}{50}, \qquad \frac{199}{200} < t^2V < \frac{201}{200}, \tag{6.4.9}$$

$$\frac{82}{25} < t^3M_2 < \frac{329}{100}, \qquad M_1 < \frac{329}{200t^2}, \qquad M_3 < \frac{10}{t^4}, \quad M_4 < \frac{40}{t^5}. \tag{6.4.10}$$

Proof. Let $f_0 = C - (t/2)\mathcal{L}(t) - t^2/24$ and $a_0^* = C - t/2 + t^2/24$. Formula (6.4.1) gives $tM_0 = f_0 + tR(t) \ge f_0$. Termwise differentiation of the positive exponential series for $R(t)$ gives $R'(t) > 0$. Hence

$$t^2M_1 = a_0^* - t^2R'(t) \le a_0^* < C, \qquad t\mu \le \frac{a_0^*}{f_0} = 1 + \frac{(t/2)(\mathcal{L}(t)-1) + t^2/12}{f_0}.$$

The numerator after the 1 increases up to $1/100$; replace the denominator by $411/250 - (1/200)(129/20) - 1/240000$. This proves the stated M_0 and μ bounds by rational arithmetic. Next $V = (\log G)''(t)$, so (6.4.7) gives $1 - b_2 \le t^2V \le 1 + b_2$, which implies (6.4.9). Finally, differentiating the exact identity gives

$$\begin{aligned} t^3M_2 &= 2C - t/2 + t^3R''(t), \\ t^4M_3 &= 6C - t - t^4R'''(t), \\ t^5M_4 &= 24C - 3t + t^5R''''(t). \end{aligned}$$

Lemma 6.4.1, $411/250 < C < 329/200$, and the rational π enclosure in the checker give (6.4.10). □

For completeness, the rational checks use Machin's formula $\pi = 16\arctan(1/5) - 4\arctan(1/239)$, enclosing each arctangent between the even partial sum of its alternating series and that sum plus the next term. Positive partial sums of e^x prove $e^{129/20} > 200\pi$, as required for the logarithm bound. Thus no floating-point approximation to π, a logarithm, or an exponential enters a proof comparison. The appendix checker also verifies stronger ranges up to $t = 1/20$ and a wider angular strip; those additional assertions are not needed in the present proof.

6.5 Core and complete Fourier-tail estimates

We now assume

$$0 < t \le \frac{1}{100}, \qquad u = kt \le 3. \tag{6.5.1}$$

The earlier real-axis bounds did not require the second assumption. For $0 \le l \le k$, Lemma 6.4.2 and (6.4.8) imply the convenient weaker bounds

$$t\mu \le \frac{103}{100}, \qquad \frac{82/25 + (19/20)lt}{t^3} \le \sigma_l^2 \le \frac{33/10 + (11/10)lt}{t^3}, \tag{6.5.2}$$

$$M_1 \le \frac{33}{20t^2}, \qquad M_3 \le \frac{10}{t^4}, \quad M_4 \le \frac{40}{t^5},$$

$$|\partial_\theta^3 \log F_\theta| \le \frac{5}{2t^3}, \qquad |\partial_\theta^4 \log F_\theta| \le \frac{8}{t^4} \quad (|\theta| \le t/2). \tag{6.5.3}$$

6.5.1 Amplitude and the full centered phase

Set $\delta = t/3$, $a = 7/8$, and $b = 9/8$. On the zero-free core let $h_l = \log a_l$. It is even, $h_l(0) = h_l'(0) = 0$, and $h_l''(0) = -\sigma_l^2$. Positivity bounds each complex derivative of F by its corresponding real moment. Thus

$$|h_l^{(4)}(\theta)| \le \frac{40 + 8lt}{t^5}, \qquad \left| h_l(\theta) + \frac{\sigma_l^2\theta^2}{2} \right| \le \frac{(40 + 8lt)\theta^4}{24t^5}.$$

For $v \ge 0$, the function $(40 + 8v)/(82/25 + (19/20)v)$ decreases, with value $500/41$ at zero. The relative quadratic error is therefore at most $125/1107 < 1/8$ for $|\theta| \le \delta$, and at most $125/492 < 13/50$ for $|\theta| \le t/2$. Consequently

$$e^{-b\sigma_l^2\theta^2/2} \le a_l(\theta) \le e^{-a\sigma_l^2\theta^2/2} \qquad (|\theta| \le \delta), \tag{6.5.4}$$

$$a_l(\theta) \le e^{-(37/50)\sigma_l^2\theta^2/2} \qquad (|\theta| \le t/2). \tag{6.5.5}$$

Choose the continuous phase $\phi_k = a_k e^{i\alpha_k}$ on the core, with $\alpha_k(0) = 0$. The saddle equation makes $\alpha_k'(0) = 0$, and oddness gives $\alpha_k''(0) = 0$. The third-derivative bound in (6.5.3) and M_3 imply

$$|\alpha_k(\theta)| \le c|\theta|^3, \qquad c = \frac{10 + (5/2)u}{6t^4}. \tag{6.5.6}$$

This bounds the entire phase by Taylor's integral remainder; it does not discard any higher odd terms. Moreover

$$\frac{c^2}{\sigma_k^6} \le \frac{(10 + (5/2)u)^2 t}{36(82/25 + (19/20)u)^3} < \frac{2}{25}t \le \frac{1}{1250}. \tag{6.5.7}$$

The rational function before the factor t decreases for $u \ge 0$: after removal of positive factors its derivative has constant term $5(82/25) - 3(19/20)10 < 0$ and linear coefficient $5(19/20) - 3(19/20)(5/2) < 0$.

6.5.2 The entire Fourier complement

For every real θ, positivity and $\sigma_{-1}(j) \geq 1$ give

$$a_l(\theta) \leq e^{-S(t,\theta)}, \quad S(t,\theta) = \sum_{j\geq 1} e^{-tj}(1-\cos j\theta) = \frac{1}{2}\coth(t/2)\frac{\sin^2(\theta/2)}{\sinh^2(t/2)+\sin^2(\theta/2)}. \quad (6.5.8)$$

The identity follows by summing the geometric series. In particular it holds at all frequencies, including neighborhoods of roots of unity. Let $T_j^{(l)}$ denote the part of $J_j^{(l)}$ with $|\theta| > \delta$.

Lemma 6.5.1 (Tail moments)**.** *Under* (6.5.1)*, for every* $0 \leq l \leq k$*,*

$$\sigma_l T_0^{(l)} < 10^{-5}, \qquad \sigma_l^3 T_2^{(l)} < \frac{1}{5000}. \quad (6.5.9)$$

Proof. Split the complement into the three regions

$$\delta < |\theta| < t/2, \qquad t/2 \leq |\theta| < t, \qquad t \leq |\theta| \leq \pi.$$

For the first region put $x = \sigma_l\delta > 6$, which follows from $\sigma_l^2\delta^2 \geq (82/25)/(9t) > 36$. Using (6.5.5) with $a_2 = 37/50$, extending the integral to infinity, and integrating by parts gives

$$\sigma_l T_{0,1}^{(l)} \leq \frac{2}{a_2 x}e^{-a_2x^2/2} < \frac{1}{500000},$$
$$\sigma_l^3 T_{2,1}^{(l)} \leq 2\left(\frac{x}{a_2}+\frac{1}{a_2^2 x}\right)e^{-a_2x^2/2} < \frac{1}{20000}.$$

The expressions decrease for $x \geq 6$. Substitute $x = 6$ and use $e^{13} > 400000$; here $a_2x^2/2 = 333/25 > 13$. It is important to retain this Gaussian estimate on the first region, rather than starting the cruder geometric estimate at δ.

For $0 \leq x \leq 1/200$, elementary power series give $\sin x \geq (99/100)x$ and $\sinh x \leq (101/100)x$. Since $\sin(\theta/2)$ increases for $0 \leq \theta \leq \pi$ and $\coth(t/2) \geq 2/t$, (6.5.8) gives

$$S(t,\theta) \geq \frac{19}{100t} \quad (|\theta| \geq t/2), \qquad S(t,\theta) \geq \frac{49}{100t} \quad (|\theta| \geq t). \quad (6.5.10)$$

Indeed, the two relevant ratios are at least $9801/(40804+9801) > 19/100$ and $9801/(10201+9801) > 49/100$, respectively. From (6.5.2) and $lt \leq 3$, $\sigma_l < (13/5)t^{-3/2}$ and $\sigma_l^3 < 18t^{-9/2}$. The length of the second region is t and $\theta^2 \leq t^2$ there, so

$$\sigma_l T_{0,2}^{(l)} < \frac{13}{5}t^{-1/2}e^{-19/(100t)} \leq 26e^{-19},$$
$$\sigma_l^3 T_{2,2}^{(l)} < 18t^{-3/2}e^{-19/(100t)} \leq 18000e^{-19}.$$

For the third region use the full interval integrals 2π and $2\pi^3/3$ to obtain

$$\sigma_l T_{0,3}^{(l)} < 2\pi\frac{13}{5}t^{-3/2}e^{-49/(100t)} \leq \frac{44}{7}\frac{13}{5}1000e^{-49},$$
$$\sigma_l^3 T_{2,3}^{(l)} < \frac{2\pi^3}{3}18t^{-9/2}e^{-49/(100t)} \leq \frac{2}{3}\left(\frac{22}{7}\right)^3 18\cdot 10^9 e^{-49}.$$

Each function $t^{-p}e^{-c/t}$ used here increases for $t \leq 1/100$, since $t < c/p$; thus the endpoint bounds are valid. Positive rational Taylor sums give $e^{19} > 150000000$ and $e^{49} > 10^{21}$. Adding the three bounds proves (6.5.9). More explicitly, valid rational upper bounds are

$$\sigma_l T_0^{(l)} < \frac{2}{6a_2\cdot 400000}+\frac{26}{150000000}+\frac{(44/7)(13/5)1000}{10^{21}} < 10^{-5},$$
$$\sigma_l^3 T_2^{(l)} < \frac{2(6/a_2+1/(6a_2^2))}{400000}+\frac{18000}{150000000}+\frac{(2/3)(22/7)^3 18\cdot 10^9}{10^{21}} < \frac{1}{5000}.$$

No part of the Fourier interval is omitted or checked by sampling. □

6.5.3 Integral bounds and the positive determinant

The upper Gaussian in (6.5.4), extended to the real line, and Lemma 6.5.1 imply

$$J_0^{(l)} < \frac{27}{10\sigma_l}, \qquad J_2^{(l)} < \frac{31}{10\sigma_l^3}. \tag{6.5.11}$$

For example the scaled bounds before rounding are $\sqrt{2\pi/a} + 10^{-5}$ and $\sqrt{2\pi}/a^{3/2} + 1/5000$; squaring the rational comparisons with $\pi < 22/7$ verifies (6.5.11).

For the lower bound, use $a_k \cos\alpha_k \geq a_k - \frac{1}{2}a_k\alpha_k^2$. Bound the first term below by the lower Gaussian, and the subtracted term above by the upper Gaussian and (6.5.6). This avoids multiplication of a possibly negative cosine bound by an inappropriate amplitude bound. The missing lower-Gaussian mass, after multiplication by σ_k, is at most $2e^{-81/4}/(6b) < 10^{-8}$, using $\sigma_k\delta > 6$. The real part of the Fourier complement is bounded below by $-T_0^{(k)}$. The sixth Gaussian moment now gives

$$\sigma_k I_k \geq \frac{\sqrt{2\pi}}{\sqrt{b}} - 10^{-8} - \frac{15\sqrt{2\pi}}{2a^{7/2}}\frac{c^2}{\sigma_k^6} - 10^{-5} > \frac{23}{10}. \tag{6.5.12}$$

Here the exact square-root comparisons

$$\sqrt{2\pi} > \frac{5}{2}, \quad b^{-1/2} > \frac{47}{50}, \quad \sqrt{2\pi} < \frac{51}{20}, \quad a^{-7/2} < \frac{8}{5}$$

and (6.5.7) give the explicit lower bound $2.35 - 0.02448 - 0.00001001 = 2.32550999 > 2.3$. These terminating decimals represent exact rationals.

The $k-1$ variance loss is retained in the final comparison:

$$\sigma_{k-1}^2 \geq \frac{82/25 + (19/20)(u-t)}{t^3} \geq \frac{6541/2000 + (19/20)u}{t^3}.$$

Consequently

$$\frac{k\mu^2\sigma_k^2}{\sigma_{k-1}^4} \leq R(u) := \frac{u(103/100)^2(33/10 + (11/10)u)}{(6541/2000 + (19/20)u)^2} \leq R(3) = \frac{84023280}{149842081} < \frac{9}{16}. \tag{6.5.13}$$

To check monotonicity, write the variable part as $u(A + Bu)/(C_0 + Du)^2$. Its derivative has numerator $AC_0 + (2BC_0 - AD)u$, with both coefficients positive for $A = 33/10$, $B = 11/10$, $C_0 = 6541/2000$, $D = 19/20$. Combining (6.5.11), (6.5.12), and (6.5.13),

$$\sigma_k^2 I_k^2 > \frac{529}{100}, \qquad \sigma_k^2 k\mu^2 J_0^{(k-1)} J_2^{(k-1)} < \frac{837}{100}\frac{9}{16} < \frac{529}{100}.$$

The rounded separation is $931/1600 > 0$. Lemma 6.3.1 proves strict log-concavity at every index whose saddle satisfies (6.5.1).

6.6 An explicit cutoff and every saddle in the required range

Theorem 6.6.1 (Effective analytic prefix)**.** *For every integer $n \geq 50000$ and every integer $1 \leq k \leq K(n)$,*

$$A_{n,k}^2 > A_{n,k-1}A_{n,k+1}.$$

Consequently Q_n is unimodal for every $n \geq 50000$.

Proof. The ceiling is retained:

$$k \le \sqrt{13n/10} + 1 \le \sqrt{7n/5} \qquad (n \ge 50000). \tag{6.6.1}$$

For the second inequality it suffices that $n/10 - 1 \ge 2\sqrt{13n/10}$. Both sides are positive at 50000, their squared comparison holds there, and $(n/10 - 1)/\sqrt{n}$ increases thereafter. In particular $k < n$.

Let $t_0 = 1/100$. Crucially, the real-axis estimates for M_1 and μ in Lemma 6.4.2 hold without any assumption on kt_0. They give

$$M_1(t_0) + k\mu(t_0) \le 16500 + 103\sqrt{7n/5} < n.$$

At $n = 50000$ the last inequality follows by squaring positive sides; $(n - 16500)/\sqrt{n}$ increases afterwards. The strict monotonicity of the saddle mean implies $t < t_0$. This step is not circular.

At the actual saddle, put $u = kt$. Now $nt^2 \le 33/20 + (103/100)u$, and (6.6.1) gives

$$u^2 \le \frac{7}{5}\left(\frac{33}{20} + \frac{103}{100}u\right).$$

The right side is smaller than u^2 at $u = 3$, and its difference from u^2 decreases for $u \ge 3$. Hence $u < 3$. The hypotheses (6.5.1) are satisfied for every required k. Section 6.5 proves the strict inequalities. Since $K_*(n) \le K(n)$, Lemmas 6.2.1 and 6.2.2 then give unimodality. □

6.7 The exact finite certificate

6.7.1 A one-sided recurrence with a two-sided invariant

Restore the row index and put

$$B_{n,k} = k!A_{n,k} = [q^n]e^{F(q)}F(q)^k.$$

Then $B_{n,0} = p(n)$, the ordinary partition number, and

$$B_{n,k+1} = \sum_{j=1}^{n} \frac{\sigma_1(j)}{j} B_{n-j,k}. \tag{6.7.1}$$

For $n < k$ the coefficient is zero. For $n \ge k \ge 1$, $F(q) \ge q/(1-q)$ coefficientwise implies $B_{n,k} \ge \binom{n-1}{k-1} \ge 1$; also $B_{n,0} = p(n) \ge 1$. These coefficients need not be integers.

Fix $S = 2^b$ and compute only integers:

$$f_j = \left\lfloor \frac{S\sigma_1(j)}{j} \right\rfloor, \quad \ell_{n,0} = Sp(n), \quad \ell_{n,k+1} = \left\lfloor \frac{1}{S}\sum_{j=1}^{n} f_j \ell_{n-j,k} \right\rfloor. \tag{6.7.2}$$

Lemma 6.7.1 (Exact enclosure)**.** *For every computed column k, if $S > 2k$, then*

$$(S - 2k)B_{n,k} \le \ell_{n,k} \le SB_{n,k}, \qquad \frac{\ell_{n,k}}{S} \le B_{n,k} \le \frac{\ell_{n,k}}{S - 2k}. \tag{6.7.3}$$

Proof. The assertion is exact at $k = 0$. Write $\alpha_j = \sigma_1(j)/j \ge 1$. Then $f_j \le S\alpha_j$ and $f_j \ge S\alpha_j - 1 \ge (S-1)\alpha_j$. Positivity and the induction hypothesis give the upper enclosure in the next column. Before the last floor, the lower bound there is

$$\frac{(S-1)(S-2k)}{S} B_{n,k+1} = \left(S - 2k - 1 + \frac{2k}{S}\right) B_{n,k+1} \ge (S - 2k - 1)B_{n,k+1}.$$

Taking the floor loses less than one. On nonzero support, $B_{n,k+1} \ge 1$, so this loss is at most one additional copy of $B_{n,k+1}$. Off support both sides are zero. This proves the next lower bound $(S - 2k - 2)B_{n,k+1}$ and completes the induction. □

Strict log-concavity is equivalent to $(k+1)B_{n,k}^2 > kB_{n,k-1}B_{n,k+1}$. By the enclosure lemma it is certified whenever the following integer-only comparison holds:

$$(k+1)\ell_{n,k}^2(S-2k+2)(S-2k-2) > k\ell_{n,k-1}\ell_{n,k+1}S^2. \tag{6.7.4}$$

All factors cleared from denominators are positive. A failure of this sufficient test is reported as *inconclusive*, not automatically as a counterexample. No tolerance or floating-point slack enters it.

6.7.2 Exact packed convolution and the carry bound

The program computes the divisor sums by an integer sieve. It computes $p(0), \ldots, p(N)$ from Euler's generalized pentagonal recurrence

$$p(0) = 1, \quad p(n) = \sum_{j\geq 1}(-1)^{j-1}\left[p\Big(n - \frac{j(3j-1)}{2}\Big) + p\Big(n - \frac{j(3j+1)}{2}\Big)\right],$$

where negative-index terms are zero. This recurrence follows by multiplying the Euler product for $\sum p(n)q^n$ by its reciprocal pentagonal expansion. All its arithmetic is in GMP integers.

Each step of (6.7.2) uses Kronecker substitution for its nonnegative integer convolution. Suppose a and b are the largest bit lengths in the two coefficient arrays, truncated at degree N. Choose

$$W \geq a + b + \lceil \log_2(N+1) \rceil, \tag{6.7.5}$$

rounded up to a whole number of GMP limbs. Every coefficient of their full product, including degrees above N, is a sum of at most $N+1$ products each strictly below 2^{a+b}. It is therefore strictly below 2^W. Packing the arrays as base-2^W integers and multiplying in GMP produces exactly their convolution, with *no carry between slots.* High-degree slots cannot alter low-degree slots. Reading the first $N+1$ slots and shifting each right by b bits implements precisely the floor in (6.7.2); the scale exponent b here is not the operand bit length in (6.7.5). The code uses separate variables for these roles.

Three coefficient columns suffice: at iteration k, the stored previous, current, and newly computed columns have indices $k-1, k, k+1$. Thus (6.7.4) uses exactly the neighboring columns claimed.

The supported platform is explicitly checked: 8-bit bytes, 32-bit `int`, 64-bit `long`, pointers and `size_t`, 64-bit GMP limbs, and no GMP nails. Inputs satisfy $1 \leq N, K \leq 10^6$, $16 \leq b \leq 4096$, and $1 \leq$ `first_n` $\leq N$; full-consumption parsing precedes conversion. The guard $S > 2K$ ensures all enclosure denominators are positive. Slot-width sums and rounding are checked for overflow. Before allocation, the implementation checks

$$\texttt{limbs} \leq \frac{\texttt{SIZE_MAX}}{N+1}, \qquad (N+1)\texttt{limbs} \leq \frac{\texttt{INT_MAX}}{2}.$$

Both packed operands then fit within the allowed limb count, and their product fits GMP's signed-int limb-count field. Unsupported or oversized runs fail rather than silently overflowing. The bounded divisor sieve and pentagonal indices also fit their machine types on this platform; for instance $\sigma_1(n) \leq n(1+\log n) < 15 \cdot 10^6$ for $n \leq 10^6$.

6.7.3 Exact cap calculation, coverage, and observed result

The finite scan checks all pairs

$$1 \leq n \leq 50000, \qquad 1 \leq k \leq \min(n-1, K_*(n)). \tag{6.7.6}$$

The cap is calculated by the exact conditions

$$125^2K_*^2 \geq 142^2n, \qquad 125^2(K_*-1)^2 < 142^2n.$$

A floating-point square root supplies an initial guess only; integer loops correct it in both directions before any inclusion decision. The audit independently uses integer square roots. All products in the cap comparisons are safe within the declared input range. The floating-point relative-slack and timing fields are diagnostics only.

The complete run used $N = 50000$, $K = 256$, $b = 64$, `first_n` $= 1$. It completed with exit status 0, checked

$$\sum_{n=1}^{50000} \min(n-1, K_*(n)) = 8\,492\,344 \tag{6.7.7}$$

strict inequalities, and reported zero inconclusive tests. All 256 computed columns were present, including the initial no-test column. For $1 \leq k \leq 255$, the expected count in checked column k is exactly

$$\max\left(0, 50000 - \max\left(k+1, \left\lfloor \frac{125^2(k-1)^2}{142^2} \right\rfloor + 1\right) + 1\right). \tag{6.7.8}$$

Indeed, $K_*(n) \geq k$ is equivalent to $142^2n > 125^2(k-1)^2$. The audit verifies each of these column counts, compares their total with the independent row-wise sum (6.7.7), checks zero inconclusive entries and the final coverage flag, and verifies the source hash in the start record. The last checked index, $k = 255$, has seven comparisons. The analytic cutoff is 50000, so neither an n nor a required k is missed at the finite/infinite join.

The source SHA-256 is

```
8145fcafa5dba3d8d15aa60263779d05ce6a
    1dff49fa361ffd91195afe3cb64c
```

The two displayed lines concatenate to the 64-character hash; the unbroken value is embedded in the reproduction manifest. The completed JSONL run has SHA-256

```
fdcfcec8fdc6675fa663362b1279353ef93e
    caecdaee01fb189d8a343c63ab58.
```

The original run took approximately 181 seconds on its execution machine. This timing is not an input to the proof. A fresh replay will usually have a different JSONL hash because elapsed-time diagnostics change. The reproduction driver therefore checks all mathematical coverage and provenance fields, not equality with the timing-bearing log's historical hash.

Proposition 6.7.2 (Finite prefix)**.** *For every pair in* (6.7.6)*,* $A_{n,k}^2 > A_{n,k-1}A_{n,k+1}$.

Proof. The enclosed fixed-point integers were computed by (6.7.2), using the exact packed convolution just proved. Every corresponding integer comparison (6.7.4) passed. The completed output and the index audit establish precisely the coverage (6.7.6); Lemma 6.7.1 transfers each comparison to the exact rational coefficient inequality. The full source and replay procedure appear in the appendix. □

Proof of Theorem 6.1.1. For $n \geq 50000$, Theorem 6.6.1 applies. For $2 \leq n < 50000$, Proposition 6.7.2 supplies strict log-concavity through $\min(n-1, K_*(n))$ and Lemma 6.2.1 supplies the common nonincreasing tail. Lemma 6.2.2 proves unimodality. The remaining case is $Q_1(z) = 1 + z$. □

6.8 Reproducibility and scope of verification

The source listings below are complete, not pseudocode. The two analytic checkers have distinct extracted paths. Their proof comparisons use Python's exact `Fraction` arithmetic. The determinant checker retains a historical metadata string describing the finite certificate as pending; it is frozen analytic-source metadata, not the status of the completed finite scan. The definitive finite status is supplied by the completed scan and the reproduction coverage audit.

The small independent Python test uses an Euler-product partition expansion, rather than the C++ pentagonal recurrence, exact rational divisor sums, and schoolbook convolution. It verifies 2226 interval positions in three parameter sets. The C++ `naive` option also compares every packed convolution with schoolbook GMP convolution. An incomplete-column run is explicitly tested to return exit code 3. These small tests corroborate implementation correctness; they do not replace either the invariant proof or the complete finite run.

Separate internal reviews rederived the analytic estimates and the finite arithmetic invariant, inspected the implementation and platform guards, and audited the completed finite/infinite coverage join. These were computational research checks, not external peer review. The manuscript does not rely on a theorem about a different denominator polynomial or on transferring another problem's analytic hypotheses.

To reproduce from this single TEX file, save it with its original LF newlines. Copy the first appendix listing to `reproduce.py`; alternatively extract that listing directly by its marker lines. Run, without Python optimization:

```
python3 reproduce.py manuscript.tex replay --full
```

Here `manuscript.tex` is this file under any chosen name. The machine needs Python 3 standard-library modules, a C++17 compiler with unsigned 128-bit integer support, and GMP development headers and libraries; the engine checks the declared LP64/no-nails platform. No network access, SageMath, SymPy, external data file, or precomputed certificate is needed. The driver extracts and hash-verifies the source listings, runs the exact analytic checks, compiles the frozen engine, runs the small independent tests, executes the full finite scan, and audits every required coverage field.

Omitting `-full` performs only the quick checks and explicitly does not assert full finite coverage. The option `-audit-log PATH` validates an already completed scan instead of launching another one; `-extract-only` extracts the files. The output directory must not contain conflicting source files, and `-full` refuses to overwrite an existing full-run log. The engine returns 1 on an error, 2 if any test is inconclusive, 3 for insufficient column coverage, and 0 only for fully covered, all-passing declared ranges. The driver additionally requires the exact full-run parameters, the complete terminal record, the expected counts, and the finite/analytic endpoint match.

6.A Complete extraction and reproduction driver

This listing bootstraps extraction of itself and the four frozen source files below. Its source-hash checks apply to the original bytes between the listing delimiters, not to typography copied from the PDF.

```
#!/usr/bin/env python3
"""Extract and reproduce the single-TeX shifted-unimodality certificate.

Copy this listing to reproduce.py, then run (without python -O):
  python3 reproduce.py manuscript.tex replay --full
The default runs extraction, analytic checkers and small independent audits.
Use --audit-log PATH instead of --full to audit an existing completed log.
"""
import argparse
import hashlib
import json
import math
import os
from pathlib import Path
import re
import subprocess
import sys
from fractions import Fraction

EXPECTED = {
    "verify_h100_constants.py":
    "a98bd0725c8fd1d1d1df35d8187f46b49ff84660bc05efd706455d7e70b195d7",
    "eta_derivatives/verify_constants.py":
    "9c35412a4f5a08938dfd38f4848b5bc3d053743986d91cb779f8e4710bd3cebb",
    "finite_verifier/fixedpoint_verify.cpp":
    "8145fcafa5dba3d8d15aa60263779d05ce6a1dff49fa361ffd91195afe3cb64c",
    "finite_verifier/test_fixedpoint_verify.py":
    "1e6342a38e54dfa069158bf57a27f27214c7678555b1462d1d3fbd281974c103",
}


def require(ok, message):
    if not ok:
        raise RuntimeError(message)


def sha(data):
    return hashlib.sha256(data).hexdigest()


def extract(tex, root):
    data = tex.read_bytes()
    # Newline normalization is intentionally forbidden: hashes are byte-exact.
    require(b"\r" not in data, "Use the original LF-newline TeX source")
    pattern = re.compile(
        rb"^% BEGIN_FILE ([A-Za-z0-9_./-]+)\n"
        rb"\\begin\{lstlisting\}(?:\[[^\n]*\])?\n"
        rb"(.*?)^\\end\{lstlisting\}\n% END_FILE(?: [^\n]*)?\n",
        re.M | re.S)
    blocks = {}
    for match in pattern.finditer(data):
        name = match.group(1).decode("ascii")
        require(name not in blocks, "Duplicate listing: " + name)
        require(name in EXPECTED or name == "reproduce.py",
                "Unexpected extracted path: " + name)
        blocks[name] = match.group(2)
```

```
    require(set(EXPECTED) <= set(blocks), "Missing required source listing")
    for name, expected in EXPECTED.items():
        require(sha(blocks[name]) == expected, "Source hash mismatch: " + name)
    root.mkdir(parents=True, exist_ok=True)
    for name, content in blocks.items():
        target = root / name
        require(root in target.resolve().parents,
                "Extracted path escapes output directory")
        target.parent.mkdir(parents=True, exist_ok=True)
        if target.exists():
            require(target.read_bytes() == content,
                    "Refusing to overwrite different file: " + str(target))
        else:
            target.write_bytes(content)
    manifest = {name: sha(content) for name, content in sorted(blocks.items())}
    (root / "extracted_sources_sha256.json").write_text(
        json.dumps(manifest, indent=2) + "\n", encoding="utf-8")
    return manifest


def run_json(script, output, env):
    text = subprocess.check_output([sys.executable, str(script)],
                                   text=True, env=env)
    value = json.loads(text)
    output.write_text(json.dumps(value, indent=2) + "\n", encoding="utf-8")
    return value


def cap(n):
    target = 142 * 142 * n
    k = math.isqrt(target // (125 * 125))
    if 125 * 125 * k * k < target:
        k += 1
    require(125 * 125 * k * k >= target, "Cap upper bound failed")
    require(k == 0 or 125 * 125 * (k - 1) ** 2 < target,
            "Cap minimality failed")
    return k


def audit_log(path, root):
    records = [json.loads(line) for line in path.read_text().splitlines()
               if line.strip()]
    require(len(records) == 258, "Expected start, 256 columns, complete")
    start, end = records[0], records[-1]
    require(start.get("type") == "start" and end.get("type") == "complete",
            "Missing terminal completion or initial provenance record")
    require((start["N"], start["K"], start["b"], start["first_n"])
            == (50000, 256, 64, 1), "Unexpected finite-run parameters")
    source = "finite_verifier/fixedpoint_verify.cpp"
    require(start["source_sha256"] == EXPECTED[source]
            == sha((root / source).read_bytes()), "Wrong engine provenance")
    N, K, b = start["N"], start["K"], start["b"]
    require(K == cap(N) + 1 and 2 ** b > 2 * K,
            "Insufficient coefficient columns or nonpositive enclosure")
    total = 0
    column_counts = []
    for k, row in enumerate(records[1:-1]):
        require(row.get("type") == "column", "Non-column interior record")
        require((row["computed_k"], row["checked_k"]) == (k + 1, k),
                "Skipped, duplicated or reordered coefficient column")
        # cap(n)>=k iff 142^2*n > 125^2*(k-1)^2, for k>=1.
        first = max(k + 1, (125 * 125 * (k - 1) ** 2) // (142 * 142) + 1)
        expected = 0 if k == 0 else max(0, N - first + 1)
```

```
        require(row["checks"] == expected, "Column coverage count mismatch")
        require(row["inconclusive"] == 0 and row["first_inconclusive_n"] == 0,
                "A strict integer comparison was not certified")
        total += expected
        column_counts.append(expected)
    # Independent row-wise sum, with integer-only ceiling calculation.
    require(total == sum(min(n - 1, cap(n)) for n in range(1, N + 1))
            == 8492344, "Total required-index count mismatch")
    require(end["checks"] == total and end["inconclusive"] == 0,
            "Completion aggregate mismatch")
    require(end["full_tail_reduced_coverage"] is True
            and end["needed_K"] == K, "Incomplete tail-reduced coverage")
    require(Fraction(142, 125) ** 2 > Fraction(355, 113) ** 2 / 3 - 2,
            "Finite tail cap too small")
    require(Fraction(142, 125) ** 2 < Fraction(13, 10),
            "Finite and analytic caps incompatible")
    analytic = json.loads((root / "verified_h100_constants.json").read_text())
    eta = json.loads((root / "eta_derivatives/verified_constants.json").read_text())
    for report in (analytic, eta):
        require(report["status"] == "all exact rational assertions passed",
                "Analytic rational checks failed")
    require(analytic["n_cutoff"] == 50000 and N + 1 >= analytic["n_cutoff"],
            "Gap between finite and infinite ranges")
    report = {
        "status": "all required finite inequalities and coverage checks passed",
        "finite_n_range": [1, N], "analytic_n_range": [50000, "infinity"],
        "computed_columns": K, "fractional_bits": b,
        "strict_integer_comparisons": total, "column_checks": column_counts,
        "source_sha256": EXPECTED[source],
        "finite_log_path": str(path), "finite_log_sha256": sha(path.read_bytes()),
        "note": "Log hash records this replay only; timings differ across runs."
    }
    (root / "reproduction_coverage.json").write_text(
        json.dumps(report, indent=2) + "\n", encoding="utf-8")
    print(json.dumps(report, indent=2))


def main():
    require(sys.flags.optimize == 0, "Do not run with Python optimization (-O)")
    parser = argparse.ArgumentParser(description=__doc__)
    parser.add_argument("tex", type=Path)
    parser.add_argument("directory", type=Path)
    mode = parser.add_mutually_exclusive_group()
    mode.add_argument("--full", action="store_true")
    mode.add_argument("--audit-log", type=Path)
    mode.add_argument("--extract-only", action="store_true")
    parser.add_argument("--cxx", default="g++")
    args = parser.parse_args()
    root = args.directory.resolve()
    manifest = extract(args.tex.resolve(), root)
    print("Extracted and hash-verified", len(manifest), "source listings.")
    if args.extract_only:
        return
    env = dict(os.environ)
    env.pop("PYTHONOPTIMIZE", None)
    eta = run_json(root / "eta_derivatives/verify_constants.py",
                   root / "eta_derivatives/verified_constants.json", env)
    analytic = run_json(root / "verify_h100_constants.py",
                        root / "verified_h100_constants.json", env)
    for value in (eta, analytic):
        require(value["status"] == "all exact rational assertions passed",
                "Analytic scalar checker did not pass")
    source = root / "finite_verifier/fixedpoint_verify.cpp"
```

```
    binary = root / "finite_verifier/fixedpoint_verify"
    definition = '-DCERT_SOURCE_SHA256="' + sha(source.read_bytes()) + '"'
    subprocess.run([args.cxx, "-O3", "-std=c++17", definition, str(source),
                    "-lgmpxx", "-lgmp", "-o", str(binary)], check=True, env=env)
    test = run_json(root / "finite_verifier/test_fixedpoint_verify.py",
                    root / "finite_verifier/test_fixedpoint_verify.json", env)
    require(test["all_passed"] is True, "Independent small-range audit failed")
    print("Analytic checks, compilation and independent small audits passed.")
    if args.full:
        log = root / "full_N50000.jsonl"
        require(not log.exists(), "Refusing to overwrite existing full-run log")
        print("Running the complete finite scan; wait for its completion.", flush=True)
        with log.open("w", encoding="utf-8") as out:
            subprocess.run([str(binary), "50000", "256", "64", "1"],
                           stdout=out, check=True, env=env)
        audit_log(log, root)
    elif args.audit_log:
        audit_log(args.audit_log.resolve(), root)
    else:
        print("No full finite claim: add --full or --audit-log COMPLETED_LOG.")


if __name__ == "__main__":
    main()
```

6.B Exact eta-derivative scalar checker

```
#!/usr/bin/env python3
"""Exact rational verification of the elementary eta-derivative constants.
No floating-point number enters an assertion. Run from any directory.
"""
from fractions import Fraction as Q
from math import factorial
import json


def atan_bounds(x, terms=60):
    s=sum(((-1)**j*x**(2*j+1)/Q(2*j+1) for j in range(terms)),Q(0))
    nxt=(-1)**terms*x**(2*terms+1)/Q(2*terms+1)
    return min(s,s+nxt),max(s,s+nxt)


def exp_lower(x, terms=100):
    return sum((x**j/Q(factorial(j)) for j in range(terms+1)),Q(0))

lo5,hi5=atan_bounds(Q(1,5))
lo239,hi239=atan_bounds(Q(1,239))
pilo=16*lo5-4*hi239
pihi=16*hi5-4*lo239
Clo=Q(411,250)
Chi=Q(329,200)
assert pilo**2/6>Clo
assert pihi**2/6<Chi
assert 24*pilo**2/Q(17)>13  # modular remainder on the Cauchy disk
assert exp_lower(Q(129,20))>200*pihi
assert exp_lower(Q(121,25))>40*pihi
# e > 2.7 follows by five terms of the exponential series.
assert exp_lower(Q(1),5)>Q(27,10)
# 2*4!*4^4 * 20 * exp(-260) < 10^-100.
assert Q(2*factorial(4)*4**4*20,1)*10**100 < Q(27,10)**260

# All E-remnant contributions are < 10^-100; use larger epsilon below.
eps=Q(1,10**90)
results=[]
for T,logupper,avec,muupper,vlo,vhi,L3,L4 in [
    (Q(1,100),Q(129,20),[Q(1,40),Q(1,40),Q(1,300),Q(1,300),Q(1,150)],Q(51,50),Q(199,200),Q
    ↪ (201,200),Q(401,200),Q(601,100)),
    (Q(1,20),Q(121,25),[Q(1,10),Q(1,10),Q(2,125),Q(2,125),Q(31,1000)],Q(213,200),Q(24,25),Q
    ↪ (26,25),Q(203,100),Q(121,20)),
]:
    raw=[
        Q(9,16)*T/Clo*(logupper+Q(1,2))+Q(81,1536)*T*T/Clo+eps,
        T/(2*Clo)*(logupper+Q(3,2))+Q(9,96)*T*T/Clo+eps,
        T/(2*Clo)+T*T/(12*Clo)+eps,
        T/(2*Clo)+eps,
        T/Clo+eps,
    ]
    assert all(x<y for x,y in zip(raw,avec))
    a0,a1,a2,a3,a4=avec
    m=1-a0
    e2=a2/m+a1*a1/m**2
    e3=a3/m+3*a1*a2/m**2+2*a1**3/m**3
    e4=a4/m+(4*a1*a3+3*a2*a2)/m**2+12*a1*a1*a2/m**3+6*a1**4/m**4
    assert 1-e2>vlo and 1+e2<vhi
    assert 2+e3<L3
    assert 6+e4<L4
    flo=Clo-T*logupper/2-T*T/24
```

```
    ftarget=Q(161,100) if T==Q(1,100) else Q(38,25)
    assert flo>ftarget
    delta=T*(logupper-1)/2+T*T/12
    assert 1+delta/flo<muupper
    m2lo=2*Clo-T/2-eps
    m2target=Q(82,25) if T==Q(1,100) else Q(13,4)
    assert m2lo>m2target
    assert pihi*pihi/3+eps<Q(329,100)
    assert 6*Chi+eps<10
    assert 24*Chi+eps<40
    results.append({
        'T':str(T),'F_t_lower':str(ftarget),'mean_t_upper':str(muupper),
        'variance_t2_lower':str(vlo),'variance_t2_upper':str(vhi),
        'log_derivative3_t3_bound':str(L3),'log_derivative4_t4_bound':str(L4),
        'M2_t3_lower':str(m2target),'M2_t3_upper':'329/100',
        'M3_t4_upper':'10','M4_t5_upper':'40',
        'raw_E_bounds':[str(x) for x in raw],
        'round_E_bounds':[str(x) for x in avec],
        'log_correction2':str(e2),'log_correction3':str(e3),'log_correction4':str(e4),
    })

# A wider angular strip, useful for intermediate Fourier tails.
assert 24*pilo*pilo/Q(53)>4
assert Q(2*factorial(4)*4**4*20)*10**28 < Q(27,10)**80
assert Q(11,7)-Q(11,24)<Q(9,8)
T=Q(1,20)
lu=Q(121,25)
epswide=Q(1,10**25)
wide_raw=[
    Q(9,8)*T/Clo*(lu+Q(9,8))+Q(81,384)*T*T/Clo+epswide,
    T/(2*Clo)*(lu+Q(17,8))+Q(3,16)*T*T/Clo+epswide,
    T/(2*Clo)+T*T/(12*Clo)+epswide,
    T/(2*Clo)+epswide,
    T/Clo+epswide,
]
wide=[Q(21,100),Q(11,100),Q(2,125),Q(2,125),Q(31,1000)]
assert all(x<y for x,y in zip(wide_raw,wide))
a0,a1,a2,a3,a4=wide
m=1-a0
wb2=a2/m+a1*a1/m**2
wb3=a3/m+3*a1*a2/m**2+2*a1**3/m**3
wb4=a4/m+(4*a1*a3+3*a2*a2)/m**2+12*a1*a1*a2/m**3+6*a1**4/m**4
assert wb2<Q(1,25)
assert wb3<Q(7,200)
assert wb4<Q(3,50)
print(json.dumps({'status':'all exact rational assertions passed','ranges':results,
    'wide_strip':{'T':'1/20','theta_over_t_max':'2',
    'E_bounds':[str(x) for x in wide],
    'H_second_t2_bound':str(wb2),'H_third_t3_bound':str(wb3),
    'H_fourth_t4_bound':str(wb4),
    'log_derivative3_t3_bound':'407/200','log_derivative4_t4_bound':'303/50',
    'remainder_scaled_bound':'t*10^-28'}},indent=2))
```

6.C Exact determinant and cutoff scalar checker

```
#!/usr/bin/env python3
"""Exact rational scalar checks for direct_determinant_h100.md."""
from fractions import Fraction as F
import json
from pathlib import Path

def exp_lower(x, terms):
    x=F(x); s=z=F(1)
    for j in range(1,terms+1):
        z=z*x/j; s+=z
    return s

a=F(7,8); b=F(9,8); a2=F(37,50)
# Rounded real-axis inputs are weaker than eta_derivatives' certified values.
assert F(51,50)<F(103,100)
assert F(199,200)>F(19,20)
assert F(201,200)<F(11,10)
assert F(329,100)<F(33,10)
assert F(401,200)<F(5,2) and F(601,100)<8
# Core quartic comparisons and monotonicity in v=lt>=0.
assert 8*F(82,25)-40*F(19,20)<0
assert F(40)/F(82,25)==F(500,41)
assert F(125,1107)<F(1,8)
assert F(125,492)<F(13,50)
# c^2/sigma^6 decreases with u: derivative numerator has negative coefficients.
assert 5*F(82,25)-3*F(19,20)*10<0
assert 5*F(19,20)-3*F(19,20)*F(5,2)<0
assert F(100)/(36*F(82,25)**3)<F(2,25)
assert F(2,25)*F(1,100)==F(1,1250)
# sigma_l delta>6.
assert F(82,25)/(9*F(1,100))>36
# Exact positive exponential lower bounds.
assert exp_lower(13,100)>400000
assert exp_lower(19,150)>150000000
assert exp_lower(49,250)>10**21
assert exp_lower(F(81,4),150)>10**8 # lower-Gaussian tail e^{-20.25}
# Elementary sin and sinh envelopes on [0,1/200].
x=F(1,200)
assert 1-x*x/6>F(99,100)
# sinh(x)/x <= sum (x^2)^j = 1/(1-x^2).
assert 1/(1-x*x)<F(101,100)
assert F(9801,40804+9801)>F(19,100)
assert F(9801,10201+9801)>F(49,100)
# Variance upper powers.
assert F(33,5)<F(13,5)**2
assert F(33,5)**3<18**2
# All t^{-p} exp(-c/t) used are increasing on (0,1/100].
for p,c in [(F(1,2),F(19,100)),(F(3,2),F(19,100)),
            (F(3,2),F(49,100)),(F(9,2),F(49,100))]:
    assert F(1,100)<c/p
# Near tail after x>=6, using exponent >=13 and standard Gaussian bounds.
near0=2/(a2*6)/400000
near2=2*(6/a2+1/(a2*a2*6))/400000
assert near0<F(1,500000)
assert near2<F(1,20000)
mid0=F(26,150000000); mid2=F(18000,150000000)
far0=F(44,7)*F(13,5)*1000/F(10**21)
far2=F(2,3)*F(22,7)**3*18*10**9/F(10**21)
tail0=near0+mid0+far0; tail2=near2+mid2+far2
assert tail0<F(1,100000)
```

```
assert tail2<F(1,5000)
# Gaussian full moments with pi<22/7.
assert (F(27,10)-F(1,100000))**2>F(44,7)/a
assert (F(31,10)-F(1,5000))**2>F(44,7)/a**3
# Lower Gaussian tail at sigma*delta>=6 is below 1e-8.
assert 2/(b*6)/10**8<F(1,10**8)
# Elementary square root enclosures used for I.
assert F(5,2)**2<F(628,100) # 2pi>6.28, follows eta checker Machin enclosure.
assert F(51,20)**2>F(44,7)
assert F(47,50)**2<1/b
assert (1/a)**7<F(8,5)**2
phase=F(15,2)*F(51,20)*F(1,1250)*F(8,5)
I_lower=F(5,2)*F(47,50)-phase-F(1,10**8)-F(1,100000)
assert I_lower>F(23,10)
# Final ratio R(u) increases: derivative of u(A+Bu)/(C+Du)^2.
A=F(33,10); B=F(11,10); C=F(6541,2000); D=F(19,20)
assert A*C>0 and 2*B*C-A*D>0
ratio3=3*F(103,100)**2*(A+3*B)/(C+3*D)**2
assert ratio3<F(9,16)
assert F(837,100)*F(9,16)<F(529,100)
# Coverage: sqrt(1.3n)+1<=sqrt(1.4n) at n>=N.
N=50000
# It is enough to show n/10-1 >= 2sqrt(13n/10), increasing relatively thereafter.
assert F(N,10)>1
assert (F(N,10)-1)**2>4*F(13*N,10)
# (n/10-1)/sqrt n increases for positive n.
assert (N-16500)**2>103**2*F(7*N,5)
assert N>16500
# u>=3 contradicts u^2 <= (7/5)(33/20+(103/100)u).
assert 9>F(7,5)*(F(33,20)+3*F(103,100))
assert 6>F(7,5)*F(103,100)
result={"status":"all exact rational assertions passed","n_cutoff":N,
        "t_max":"1/100","kt_max":"3","I_sigma_lower":"23/10",
        "J0_sigma_upper":"27/10","J2_sigma3_upper":"31/10",
        "tail0_sigma_upper":str(tail0),"tail2_sigma3_upper":str(tail2),
        "computed_I_sigma_lower":str(I_lower),"final_ratio_at_u3":str(ratio3),
        "final_ratio_upper":"9/16","finite_certificate_status":"pending independent audit and
    ↪  full scan"}
path=Path(__file__).with_name('verified_h100_constants.json')
path.write_text(json.dumps(result,indent=2)+'\n')
print(json.dumps(result,indent=2))
```

6.D Complete fixed-point finite certificate engine

```
// Exact fixed-point certificate for shifted Nekrasov--Okounkov rows.
// See README.md for proof of interval invariant. GMP integers only enter proofs.
#include <gmpxx.h>
#include <algorithm>
#include <chrono>
#include <cmath>
#include <climits>
#include <cstdint>
#include <cstdlib>
#include <iomanip>
#include <iostream>
#include <limits>
#include <stdexcept>
#include <string>
#include <vector>

#ifndef CERT_SOURCE_SHA256
#define CERT_SOURCE_SHA256 "unrecorded"
#endif

using Poly = std::vector<mpz_class>;
using Clock = std::chrono::steady_clock;
static double seconds(Clock::time_point t) {
  return std::chrono::duration<double>(Clock::now()-t).count();
}
static size_t bits(const mpz_class& x) {
  return x == 0 ? 0 : mpz_sizeinbase(x.get_mpz_t(),2);
}
static unsigned parse_unsigned(const char* s,unsigned lo,unsigned hi) {
  if(!s || !*s || *s=='-' || *s=='+') throw std::runtime_error("invalid unsigned argument");
  size_t used=0; auto v=std::stoull(s,&used,10);
  if(used!=std::string(s).size() || v<lo || v>hi)
    throw std::runtime_error("argument outside supported range");
  return static_cast<unsigned>(v);
}
static unsigned ceillog2(size_t n) {
  unsigned b=0; size_t v=1; while(v<n) {v<<=1; ++b;} return b;
}
static uint64_t coefficient_digest(const Poly& p) {
  constexpr uint64_t prime=(uint64_t(1)<<61)-1, base=1000003;
  uint64_t h=0;
  for(auto& x:p) h=(static_cast<__uint128_t>(h)*base+mpz_fdiv_ui(x.get_mpz_t(),prime))%prime;
  return h;
}
static void dump_values(const Poly& p,unsigned k,const mpz_class& S) {
  std::cout<<"{\"type\":\"fixedpoint_values\",\"k\":"<<k<<",\"S\":\""<<S<<"\",\"values\":[";
  for(size_t n=0;n<p.size();++n) {if(n)std::cout<<',';std::cout<<'"'<<p[n]<<'"';}
  std::cout<<"]}"<<std::endl;
}
static unsigned tail_bound(unsigned n) {
  // ceil((142/125)*sqrt(n)); 142/125 > sqrt(2*(zeta(2)-1)).
  uint64_t target=uint64_t(142)*142*n;
  unsigned k=static_cast<unsigned>(std::sqrt(double(target)/(125*125)));
  while(uint64_t(125)*125*k*k < target) ++k;
  while(k && uint64_t(125)*125*(k-1)*(k-1)>=target) --k;
  return k;
}
static Poly partitions(unsigned N) {
  Poly p(N+1); p[0]=1;
  for(unsigned n=1;n<=N;++n) {
```

```
    for(unsigned j=1;;++j) {
      unsigned g1=j*(3*j-1)/2; if(g1>n) break;
      if(j&1) p[n]+=p[n-g1]; else p[n]-=p[n-g1];
      unsigned g2=j*(3*j+1)/2;
      if(g2<=n) { if(j&1) p[n]+=p[n-g2]; else p[n]-=p[n-g2]; }
    }
  }
  return p;
}
static size_t checked_packed_size(size_t coefficients,size_t limbs) {
  if(!coefficients || !limbs || limbs>SIZE_MAX/coefficients)
    throw std::runtime_error("packed-size multiplication overflow");
  size_t total=coefficients*limbs;
  // GMP stores its limb count in a signed int. Both packed operands have at
  // most total limbs, so their product needs at most 2*total limbs.
  if(total>static_cast<size_t>(INT_MAX)/2)
    throw std::runtime_error("packed product exceeds GMP signed-int limb limit");
  return total;
}
static void check_supported_platform() {
  if(CHAR_BIT!=8 || sizeof(int)!=4 || sizeof(long)!=8 ||
     sizeof(unsigned long)!=8 || sizeof(void*)!=8 || sizeof(size_t)!=8 ||
     sizeof(mp_limb_t)!=8 || GMP_NUMB_BITS!=64 || GMP_NAIL_BITS!=0)
    throw std::runtime_error("supported platform is LP64, 8-bit bytes, 64-bit GMP limbs, no
    ↪ nails");
}
static Poly exact_convolution_trunc(const Poly& a,const Poly& b,
                                    unsigned N,unsigned shift,
                                    size_t& slot_bits) {
  size_t ma=0,mb=0;
  for(unsigned i=0;i<=N;++i) {ma=std::max(ma,bits(a[i]));mb=std::max(mb,bits(b[i]));}
  // Every convolution coefficient < (N+1)*2^(ma+mb) <= 2^slot_bits.
  size_t extra=ceillog2(N+1);
  if(ma>SIZE_MAX-mb || ma+mb>SIZE_MAX-extra)
    throw std::runtime_error("convolution slot-size overflow");
  slot_bits=ma+mb+extra;
  size_t limbs=slot_bits/GMP_NUMB_BITS+(slot_bits%GMP_NUMB_BITS!=0);
  if(limbs>SIZE_MAX/GMP_NUMB_BITS)
    throw std::runtime_error("rounded convolution slot-size overflow");
  slot_bits=limbs*GMP_NUMB_BITS;
  if(GMP_NAIL_BITS != 0) throw std::runtime_error("GMP nails unsupported");
  size_t total=checked_packed_size(size_t(N)+1,limbs);
  std::vector<mp_limb_t> packed(total,0);
  mpz_class aa,bb,cc;
  auto pack=[&](const Poly& p,mpz_class& out) {
    std::fill(packed.begin(),packed.end(),0);
    for(unsigned i=0;i<=N;++i) {
      if(p[i]<0) throw std::runtime_error("negative coefficient");
      size_t m=mpz_size(p[i].get_mpz_t());
      if(m>limbs) throw std::runtime_error("coefficient spills slot");
      if(m) std::copy_n(mpz_limbs_read(p[i].get_mpz_t()),m,packed.data()+size_t(i)*limbs);
    }
    mpz_import(out.get_mpz_t(),total,-1,sizeof(mp_limb_t),0,0,packed.data());
  };
  pack(a,aa);pack(b,bb);
  mpz_mul(cc.get_mpz_t(),aa.get_mpz_t(),bb.get_mpz_t());
  // Only coefficients through degree N are needed; higher limbs cannot borrow.
  const mp_limb_t* cp=mpz_limbs_read(cc.get_mpz_t()); size_t cn=mpz_size(cc.get_mpz_t());
  Poly result(N+1);
  for(unsigned i=0;i<=N;++i) {
    size_t offset=size_t(i)*limbs;
    if(offset<cn) {
      size_t take=std::min(limbs,cn-offset);
```

```
      mpz_import(result[i].get_mpz_t(),take,-1,sizeof(mp_limb_t),0,0,cp+offset);
      mpz_fdiv_q_2exp(result[i].get_mpz_t(),result[i].get_mpz_t(),shift);
    }
  }
  return result;
}
static void compare_naive(const Poly& f,const Poly& old,const Poly& next,
                          unsigned N,unsigned b) {
  for(unsigned n=0;n<=N;++n) {
    mpz_class c=0;
    for(unsigned j=0;j<=n;++j) c+=f[j]*old[n-j];
    mpz_fdiv_q_2exp(c.get_mpz_t(),c.get_mpz_t(),b);
    if(c!=next[n]) throw std::runtime_error("Kronecker vs naive mismatch");
  }
}
int main(int argc,char**argv) {
 try {
  check_supported_platform();
  unsigned N=argc>1?parse_unsigned(argv[1],1,1000000):1000;
  unsigned K=argc>2?parse_unsigned(argv[2],1,1000000):tail_bound(N)+1;
  unsigned b=argc>3?parse_unsigned(argv[3],16,4096):64;
  unsigned start=argc>4?parse_unsigned(argv[4],1,N):1;
  bool naive=false,dump=false;
  for(int i=5;i<argc;++i) {
    if(std::string(argv[i])=="naive")naive=true;
    else if(std::string(argv[i])=="dump")dump=true;
    else throw std::runtime_error("unknown option");
  }
  if(N<1 || N>1000000 || K>1000000 || b<16 || start<1 || start>N) throw std::runtime_error("
    ↪ bad parameter");
  mpz_class S=mpz_class(1)<<b;
  if(S<=2*K) throw std::runtime_error("interval denominator not positive");
  auto all=Clock::now();
  std::vector<unsigned long> sigma(N+1,0);
  for(unsigned d=1;d<=N;++d) for(unsigned n=d;n<=N;n+=d) sigma[n]+=d;
  Poly f(N+1);
  for(unsigned n=1;n<=N;++n) f[n]=(S*sigma[n])/n;
  Poly current=partitions(N),previous;
  for(auto& c:current) c*=S;
  if(dump)dump_values(current,0,S);
  uint64_t count=0, failures=0;
  std::cout<<"{\"type\":\"start\",\"source_sha256\":\""<<CERT_SOURCE_SHA256<<"\",\"N\":"<<N<<
    ↪ ",\"K\":"<<K<<",\"b\":"<<b<<",\"first_n\":"<<start<<",\"setup_seconds\":"<<seconds(
    ↪ all)<<"}"<<std::endl;
  for(unsigned k=0;k<K;++k) {
    auto t=Clock::now();size_t width;
    Poly next=exact_convolution_trunc(f,current,N,b,width);
    double mult_seconds=seconds(t);
    if(dump)dump_values(next,k+1,S);
    if(naive) compare_naive(f,current,next,N,b);
    uint64_t checks=0,bad=0;
    unsigned first_bad=0,min_n=0;double min_slack=1;
    if(k>=1) {
      mpz_class qminus=S-2*(k-1),qplus=S-2*(k+1);
      for(unsigned n=std::max(start,k+1);n<=N;++n) {
        if(k>tail_bound(n)) continue;
        mpz_class lhs=(k+1)*current[n]*current[n]*qminus*qplus;
        mpz_class rhs=k*previous[n]*next[n]*S*S;
        ++checks;
        if(lhs<=rhs) {++bad;if(!first_bad)first_bad=n;}
        // Diagnostics only: proof comparison above is exact integer arithmetic.
        if(lhs>0) {
          mpz_class gap=lhs-rhs;
```

```
          long ge=0,le=0;
          double gm=mpz_get_d_2exp(&ge,gap.get_mpz_t());
          double lm=mpz_get_d_2exp(&le,lhs.get_mpz_t());
          double slack=std::ldexp(gm/lm,int(ge-le));
          if(slack<min_slack){min_slack=slack;min_n=n;}
        }
      }
    }
    count+=checks;failures+=bad;
    size_t maxbits=0;for(auto&v:next)maxbits=std::max(maxbits,bits(v));
    std::cout<<std::setprecision(10)<<"{\"type\":\"column\",\"computed_k\":"<<k+1<<",\"
    ↪ checked_k\":"<<k<<",\"slot_bits\":"<<width<<",\"max_coefficient_bits\":"<<maxbits<<"
    ↪ ,\"coefficient_digest_mod_2p61_minus_1\":"<<coefficient_digest(next)<<",\"
    ↪ multiply_seconds\":"<<mult_seconds<<",\"column_seconds\":"<<seconds(t)<<",\"checks\":
    ↪ "<<checks<<",\"inconclusive\":"<<bad<<",\"first_inconclusive_n\":"<<first_bad<<",\"
    ↪ min_relative_slack_diagnostic\":"<<min_slack<<",\"min_slack_n\":"<<min_n<<"}"<<std::
    ↪ endl;
    previous=std::move(current);current=std::move(next);
  }
  unsigned need=tail_bound(N)+1;
  bool coverage=K>=need;
  std::cout<<"{\"type\":\"complete\",\"checks\":"<<count<<",\"inconclusive\":"<<failures<<"
    ↪ ,\"full_tail_reduced_coverage\":"<<(coverage?"true":"false")<<",\"needed_K\":"<<need
    ↪ <<",\"seconds\":"<<seconds(all)<<"}"<<std::endl;
  if(failures) return 2;
  return coverage?0:3; // A partial run is never reported as overall success.
 }catch(std::exception&e){std::cerr<<"ERROR: "<<e.what()<<std::endl;return 1;}
}
```

6.E Independent small-range arithmetic checks

```
#!/usr/bin/env python3
"""Independent Fraction and naive convolution audits of the C++ verifier."""
from fractions import Fraction
from pathlib import Path
import subprocess,json,math,hashlib
ROOT=Path(__file__).resolve().parent
P=(1<<61)-1

def partitions(N):
    # Product expansion, independent from the C++ pentagonal recurrence.
    p=[1]+[0]*N
    for j in range(1,N+1):
        for n in range(j,N+1):p[n]+=p[n-j]
    return p

def conv(a,b,N):
    return [sum(a[j]*b[n-j] for j in range(n+1)) for n in range(N+1)]

def digest(a):
    h=0
    for x in a:h=(1000003*h+x)%P
    return h

def audit(N,K,b):
    S=1<<b;p=partitions(N)
    f=[Fraction(0)]+[sum((Fraction(1,d) for d in range(1,n+1) if n%d==0),Fraction(0)) for n
    ↪ in range(1,N+1)]
    fd=[int(x*S) for x in f]
    exact=list(map(Fraction,p));lower=[x*S for x in p]
    got=[json.loads(x) for x in subprocess.check_output([str(ROOT/'fixedpoint_verify'),str(N)
    ↪ ,str(K),str(b),'1','naive'],text=True).splitlines()]
    cols={x['computed_k']:x for x in got if x['type']=='column'}
    checks=0
    for k in range(1,K+1):
        exact=conv(f,exact,N)
        lower=[x//S for x in conv(fd,lower,N)]
        assert digest(lower)==cols[k]['coefficient_digest_mod_2p61_minus_1']
        for n,(v,l) in enumerate(zip(exact,lower)):
            assert Fraction(l,S)<=v<=Fraction(l,S-2*k),(n,k,l,v)
            assert (v==0 if n<k else v>=1)
            checks+=1
    return {'N':N,'K':K,'b':b,'exact_interval_checks':checks,'all_passed':True}

tail_gap=Fraction(142,125)**2-(Fraction(355,113)**2/Fraction(3)-2)
assert tail_gap>0
partial=subprocess.run([str(ROOT/'fixedpoint_verify'),'100','2','64','1'],text=True,
    ↪ capture_output=True)
assert partial.returncode==3
assert json.loads(partial.stdout.splitlines()[-1])['full_tail_reduced_coverage'] is False
results=[audit(30,16,16),audit(50,10,64),audit(60,20,128)]
report={'tests':results,'tail_cap_rational_gap':str(tail_gap),'incomplete_coverage_returns_3'
    ↪ :True,'source_sha256':hashlib.sha256((ROOT/'fixedpoint_verify.cpp').read_bytes()).
    ↪ hexdigest(),'all_passed':True}
print(json.dumps(report,indent=2))
```

Chapter 7

Uniform Log Concavity of Regular Partition Numbers

Chapter overview

Let $b_k(n)$ count partitions of n with no part divisible by k. We prove $b_k(n)^2 > b_k(n-1)b_k(n+1)$ for every $k \geq 30$ and $n \geq 26$, establishing the $k \geq 30$ assertion of the log-concavity conjecture stated by Heim and Neuhauser. The threshold 26 is exact. We also give a uniform analytic cutoff for every $k \geq 2$, determine all nonpositive determinants, and extend the large-modulus estimate to uniformly sparse sets of forbidden parts.

The analytic argument uses an exact factorization of the Fourier representation of the Turan determinant and estimates its weighted integrals using positive cumulants, a Gaussian core, and a uniform bound on the entire complementary arc. An elementary comparison with ordinary partitions reduces the finite verification to finitely many integer comparisons despite the unbounded parameter k. All verification code and a replay driver are embedded. The same Fourier estimates give uniform Hermite limits for Jensen polynomials and eventual stabilization of their hyperbolicity thresholds.

7.1 Statement and scope

For integers $k \geq 2$ define

$$P_k(q) = \prod_{j \geq 1} \frac{1-q^{kj}}{1-q^j} = \prod_{\substack{j \geq 1 \\ k \nmid j}} (1-q^j)^{-1} = \sum_{n \geq 0} b_k(n) q^n \qquad (|q| < 1). \tag{7.1.1}$$

Thus $b_k(n)$ is the number of partitions having no part divisible by k. We write $p(n)$ for the unrestricted partition function, put $p(0) = 1$, and set $p(m) = 0$ for negative integers m. The determinant of interest is

$$D_k(n) = b_k(n)^2 - b_k(n-1)b_k(n+1), \qquad n \geq 1.$$

Let N_k be the least positive integer such that $D_k(n) \geq 0$ for every $n \geq N_k$, when such an integer exists. Heim and Neuhauser conjectured in Section 4 of the original preprint [7.1] that $N_k = 26$ for $k \geq 30$. The published article is [7.2]. The principal result settles this assertion. Section 7.7 also gives a complete sign classification for every $k \geq 2$.

Theorem 7.1.1. *For every integer $k \geq 30$ and every integer $n \geq 26$,*

$$b_k(n)^2 > b_k(n-1)b_k(n+1). \tag{7.1.2}$$

Consequently $N_k = 26$ for every $k \geq 30$. The lower bound on k cannot be replaced by 29 while retaining the same conclusion.

The last two statements follow from exact small values, not from asymptotics. Since $b_k(m) = p(m)$ for $m < k$, for every $k \geq 30$ one has

$$D_k(25) = 1958^2 - 1575 \cdot 2436 = -2936,$$
$$D_k(26) = 2436^2 - 1958 \cdot 3010 = 40516.$$

For $k=29$, the three relevant coefficients are $b_{29}(28)=3718$, $b_{29}(29)=4564$, and $b_{29}(30)=5603$, so

$$D_{29}(29)=4564^2-3718\cdot 5603=-1858.$$

We do not assert that the analytic principles used here are historically new. The contribution of the argument is a systematic uniform estimate with explicit constants, combined with an exact finite covering reduction. No claim of external peer review or priority is made. Craig and Pun [7.3] established eventual higher-order Turan inequalities for every fixed modulus. Dong and Ji [7.4] proved the distinct-partition threshold $N_2=33$ and discussed $N_3=58$, $N_4=17$, and $N_5=42$. Liu and Zhang [7.5] also obtained log-concavity results for $2\le k\le 9$. These small-modulus values are not claimed as new here.

7.2 Uniform moment estimates

For $t>0$ define

$$F(t)=\log\prod_{m\ge 1}(1-e^{-mt})^{-1}=\sum_{r\ge 1}\sigma_{-1}(r)e^{-rt},\qquad H(t)=F(t)-F(kt),$$

where $\sigma_{-1}(r)=\sum_{d\mid r}d^{-1}$. We suppress the dependence of H on k. All series and their derivatives converge absolutely for $t>0$. Put

$$M_j(t)=\sum_{r\ge 1}\sigma_{-1}(r)r^j e^{-rt},\qquad K_j(t)=(-1)^jH^{(j)}(t)=M_j(t)-k^jM_j(kt).$$

Equivalently,

$$H(t)=\sum_{\substack{m\ge 1\\ k\nmid m}}\sum_{\ell\ge 1}\frac{e^{-m\ell t}}{\ell},\qquad K_j(t)=\sum_{\substack{m\ge 1\\ k\nmid m}}\sum_{\ell\ge 1}m^j\ell^{j-1}e^{-m\ell t}>0. \tag{7.2.1}$$

In particular, $0<K_j(t)\le M_j(t)$.

Lemma 7.2.1. *For every $t>0$ and every positive integer j,*

$$M_j(t)\le C_jt^{-j-1},\qquad C_j=\zeta(2)e^{-j}\left(j!\sum_{i=0}^{j}\frac{j^i}{i!}+j^{j+1}\right). \tag{7.2.2}$$

Moreover,

$$M_1(t)\le\frac{\zeta(2)}{t^2},\qquad C_2<4.009,\quad C_3<13.023,\quad C_4<55.678. \tag{7.2.3}$$

Proof. For $x\ge 0$ let $A(x)=\sum_{r\le x}\sigma_{-1}(r)$. Reversing the divisor sum gives

$$A(x)=\sum_{d\le x}\frac{1}{d}\left\lfloor\frac{x}{d}\right\rfloor\le\zeta(2)x.$$

Integration by parts with $f(x)=x^je^{-tx}$ yields $M_j(t)=-\int_0^\infty A(x)f'(x)\,\mathrm{d}x$. The integrand is nonpositive for $0<x<j/t$. Discard this interval and use $A(x)\le\zeta(2)x$ on the rest to obtain

$$M_j(t)\le\zeta(2)t^{-j-1}\int_j^\infty(y^{j+1}-jy^j)e^{-y}\,\mathrm{d}y.$$

The integral equals $\Gamma(j+1,j)+j^{j+1}e^{-j}$, and repeated integration by parts gives $\Gamma(j+1,j)=j!e^{-j}\sum_{i=0}^{j}j^i/i!$. For the first moment, expand by divisors once more:

$$M_1(t)=\sum_{d\ge 1}\sum_{\ell\ge 1}\ell e^{-d\ell t}=\sum_{d\ge 1}\frac{e^{-dt}}{(1-e^{-dt})^2}=\sum_{d\ge 1}\frac{1}{4\sinh^2(dt/2)}\le\sum_{d\ge 1}\frac{1}{d^2t^2}.$$

The three displayed numerical upper bounds follow from $\zeta(2) = \pi^2/6$ and the explicit formula for C_j. □

Lemma 7.2.2. *For every $t > 0$ one has*

$$\frac{2\zeta(2)}{t^3} - \frac{1}{t^2} \le M_2(t) \le \frac{2\zeta(2)}{t^3}. \tag{7.2.4}$$

If $k \ge 30$ and $0 < t \le 0.008$, then, writing $V = K_2(t)$,

$$\frac{3.17}{t^3} < V < \frac{3.29}{t^3}, \qquad K_3(t) < \frac{13.1}{t^4}, \qquad K_4(t) < \frac{56}{t^5}. \tag{7.2.5}$$

Proof. The function $h_2(y) = (y/(2\sinh(y/2)))^2$, extended by $h_2(0) = 1$, is positive and decreasing: $\sinh x/x$ is increasing for $x > 0$, as its power series has positive coefficients. Termwise integration of $e^{-y}/(1-e^{-y})^2 = \sum_{\ell\ge1} \ell e^{-\ell y}$ gives

$$\int_0^\infty h_2(y)\,\mathrm{d}y = \sum_{\ell\ge1} \ell \int_0^\infty y^2 e^{-\ell y}\,\mathrm{d}y = 2\zeta(2).$$

By lower and upper rectangle comparison for a decreasing function,

$$\frac{1}{t}\int_0^\infty h_2(y)\,\mathrm{d}y - 1 \le \sum_{m\ge1} h_2(mt) \le \frac{1}{t}\int_0^\infty h_2(y)\,\mathrm{d}y.$$

Since $M_2(t) = t^{-2}\sum_{m\ge1} h_2(mt)$, this proves (7.2.4). Therefore

$$V = M_2(t) - k^2 M_2(kt) \ge \frac{2\zeta(2)(1-1/k) - t}{t^3} \ge \frac{2\zeta(2)(29/30) - 0.008}{t^3} > \frac{3.17}{t^3}.$$

The upper bound for V follows from $V \le M_2(t)$ and $2\zeta(2) < 3.29$. The remaining bounds follow from Lemma 7.2.1 (with slightly relaxed constants). □

For every positive integer n, there is a unique $t > 0$ satisfying the saddle equation

$$K_1(t) = n. \tag{7.2.6}$$

Indeed $K_1' = -K_2 < 0$, its limit at infinity is zero, and its limit at zero is infinite (the allowed part $m = 1$ alone suffices). By (7.2.3),

$$t \le \sqrt{\frac{\zeta(2)}{n}} < 0.008 \quad \text{if } n \ge 25703, \tag{7.2.7}$$

since $\zeta(2)/(0.008)^2 < 25703$. All ensuing estimates use this exact saddle; neither kt nor the position of a root of unity is approximated.

7.3 A direct Fourier estimate for the determinant

Theorem 7.3.1. *For every integer $k \ge 30$ and every integer $n \ge 25703$, $D_k(n) > 0$.*

In this section t satisfies (7.2.6), $V = K_2(t)$, and $0 < t \le 0.008$. Define

$$\chi(\theta) = \exp\{H(t - i\theta) - H(t) - in\theta\}, \qquad -\pi \le \theta \le \pi.$$

The extension $H(t - i\theta)$ is given by the absolutely convergent series (7.2.1). In particular $|\chi(\theta)| \le 1$ and $\chi(-\theta) = \overline{\chi(\theta)}$.

7.3.1 An exact factorization

Let

$$J_0 = \int_{-\pi}^{\pi} \chi(\theta)\,\mathrm{d}\theta, \qquad C = \int_{-\pi}^{\pi} \chi(\theta)\cos\theta\,\mathrm{d}\theta,$$
$$is = \int_{-\pi}^{\pi} \chi(\theta)\sin\theta\,\mathrm{d}\theta, \qquad A_0 = J_0 - C, \qquad B_0 = J_0 + C.$$

Conjugate symmetry makes J_0, C, s, A_0, B_0 real. Fourier inversion gives

$$b_k(n+a) = \frac{e^{H(t)+(n+a)t}}{2\pi} J_a, \qquad J_a = \int_{-\pi}^{\pi} \chi(\theta) e^{-ia\theta}\,\mathrm{d}\theta \quad (a = -1, 0, 1).$$

In particular $J_1 = C + s$ and $J_{-1} = C - s$, so

$$(2\pi)^2 e^{-2H(t)-2nt} D_k(n) = J_0^2 - J_{-1}J_1 = A_0B_0 + s^2. \tag{7.3.1}$$

The sign before s^2 is positive. It is therefore sufficient to prove $A_0 > 0$ and $B_0 > 0$; the sine integral need not be estimated. Equivalently,

$$A_0 = \int_{-\pi}^{\pi} (1 - \cos\theta)\,\mathrm{Re}\,\chi(\theta)\,\mathrm{d}\theta, \quad B_0 = \int_{-\pi}^{\pi} (1 + \cos\theta)\,\mathrm{Re}\,\chi(\theta)\,\mathrm{d}\theta. \tag{7.3.2}$$

7.3.2 A uniform bound away from the core

For $u > 0$ and real x define

$$S(u, x) = \sum_{m\ge 1} e^{-mu}(1 - \cos mx) = \frac{1}{2}\coth(u/2)\frac{\sin^2(x/2)}{\sinh^2(u/2) + \sin^2(x/2)}. \tag{7.3.3}$$

The identity follows from geometric series. The unrestricted logarithmic modulus drop is

$$F(t) - \mathrm{Re}\,F(t - i\theta) = \sum_{\ell\ge 1} \frac{S(\ell t, \ell\theta)}{\ell}.$$

Deleting the factors whose part sizes are multiples of k removes a nonnegative amount of this drop. Its size is at most

$$\sum_{m\ge 1} \log\frac{1 + e^{-kmt}}{1 - e^{-kmt}} \le \frac{1}{kt}\int_0^{\infty} \log\coth(y/2)\,\mathrm{d}y$$
$$= \frac{\pi^2}{4kt} < \frac{0.08225}{t}. \tag{7.3.4}$$

Here the summand is decreasing. To evaluate the integral, use

$$\log\coth(y/2) = 2\sum_{r\ge 0} \frac{e^{-(2r+1)y}}{2r+1}.$$

Integrating termwise gives $2\sum_{r\ge 0}(2r+1)^{-2} = \pi^2/4$. The pointwise bound before summation follows from $|1 - e^{-kmt+ikm\theta}| \le 1 + e^{-kmt}$.

Suppose first that $t/2 \le |\theta| \le t$, and retain the first ten harmonics $\ell = 1, \dots, 10$. All arguments $\ell t/2$ and $\ell|\theta|/2$ lie in $[0, 0.04]$, on which

$$\sin x \ge 0.999x, \qquad \sinh x \le 1.001x.$$

These estimates follow from their elementary power series (for example, $\sin x/x \geq 1 - x^2/6$ and $\sinh x/x \leq 1 + (x^2/6)/(1 - x^2/20)$). The squared ratio in (7.3.3) is therefore at least

$$\frac{0.999^2}{4 \cdot 1.001^2 + 0.999^2} > 0.199.$$

Also $\frac{1}{2}\coth(\ell t/2) \geq 1/(\ell t)$ and $\sum_{\ell=1}^{10} \ell^{-2} > 1.549$. The unrestricted drop exceeds $0.199(1.549)/t = 0.308251/t$. Subtracting (7.3.4) leaves a drop greater than $0.226001/t > 0.225/t$.

If $t \leq |\theta| \leq \pi$, retain only $\ell = 1$. Monotonicity of $\sin(|\theta|/2)$ gives a squared ratio at least

$$\frac{0.999^2}{1.001^2 + 0.999^2} > 0.499.$$

After removal the drop exceeds $(0.499 - 0.08225)/t > 0.225/t$. Thus, for the entire complementary arc,

$$|\chi(\theta)| \leq e^{-0.225/t} \qquad (t/2 \leq |\theta| \leq \pi). \tag{7.3.5}$$

This estimate is uniform even when $k\theta$ lies near a multiple of 2π; no competing arc is omitted.

7.3.3 A real part bound on the core

Put $h = t/2$ and $\alpha = 0.63$. Write $\chi(\theta) = M(\theta)e^{i\varphi(\theta)}$, using the phase given by the positive logarithmic expansion. Taylor inequalities give

$$e^{-V\theta^2/2} \leq M(\theta) \leq \exp\left(-\frac{V\theta^2}{2} + \frac{K_4\theta^4}{24}\right),$$
$$|\varphi(\theta)| \leq K_3|\theta|^3/6.$$

For $|\theta| \leq h$, the quartic loss relative to the quadratic exponent is at most

$$\frac{K_4 h^2}{12V} < \frac{56}{48 \cdot 3.17} < 0.37.$$

Writing $G(\theta) = e^{-V\theta^2/2}$ and $U(\theta) = e^{-\alpha V\theta^2/2}$, we have $G \leq M \leq U$ on the core. Using $\cos v \geq 1 - v^2/2$, in the order $M\cos v = M - M(1 - \cos v)$, gives the valid lower bound

$$\operatorname{Re}\chi(\theta) \geq G(\theta) - \frac{K_3^2\theta^6}{72}U(\theta) \qquad (|\theta| \leq h). \tag{7.3.6}$$

This derivation remains valid where the lower bound for $\cos v$ is negative. Put

$$G_0 = \sqrt{\frac{2\pi}{V}}, \qquad r = \frac{K_3^2}{V^3}.$$

The moment bounds imply

$$r < \frac{13.1^2}{3.17^3}t < 0.044, \qquad \frac{1}{4V} < 10^{-6}. \tag{7.3.7}$$

7.3.4 Positivity of the two weighted integrals

The elementary inequalities

$$\frac{\theta^2}{2} - \frac{\theta^4}{24} \leq 1 - \cos\theta \leq \frac{\theta^2}{2}, \qquad 2 - \frac{\theta^2}{2} \leq 1 + \cos\theta \leq 2$$

combine with (7.3.6). Enlarge all subtracted nonnegative Gaussian integrals to the real line. For each nonnegative integer j,

$$\int_{\mathbb{R}} \theta^{2j} e^{-\beta V\theta^2/2}\,\mathrm{d}\theta = G_0 \frac{(2j-1)!!}{V^j \beta^{j+1/2}}, \qquad \beta > 0.$$

It follows that the core contributions satisfy

$$\begin{aligned} \frac{A_0(\text{core})}{G_0/(2V)} &\geq 1 - \frac{1}{4V} - r\frac{105}{72}\alpha^{-9/2} - \epsilon_2, \\ \frac{B_0(\text{core})}{2G_0} &\geq 1 - \frac{1}{4V} - r\frac{15}{72}\alpha^{-7/2} - \epsilon_0. \end{aligned} \tag{7.3.8}$$

Here ϵ_2 is the standard-normal second-moment tail outside $|Z| \leq u_0$, and ϵ_0 is the standard-normal mass tail there, where

$$u_0 = h\sqrt{V} > \sqrt{\frac{3.17}{4(0.008)}} > 9.95.$$

Integration by parts gives

$$\epsilon_2 \leq \sqrt{\frac{2}{\pi}}(u_0 + u_0^{-1})e^{-u_0^2/2}, \qquad \epsilon_0 \leq \sqrt{\frac{2}{\pi}}u_0^{-1}e^{-u_0^2/2}.$$

The right sides decrease in this range, and both are less than $11e^{-49} < 10^{-19}$. The constants in (7.3.7) yield

$$0.044\frac{105}{72}\,0.63^{-9/2} < 0.514, \qquad 0.044\frac{15}{72}\,0.63^{-7/2} < 0.047.$$

Consequently

$$A_0(\text{core}) > 0.485\frac{G_0}{2V}, \qquad B_0(\text{core}) > 0.952(2G_0). \tag{7.3.9}$$

On the tail $h \leq |\theta| \leq \pi$, both weights $1 \pm \cos\theta$ have absolute value at most two. Equation (7.3.5) gives a loss at most $4\pi e^{-0.225/t}$ for each integral. Since $V < 3.29/t^3$,

$$\begin{aligned} \frac{|A_0(\text{tail})|}{G_0/(2V)} &\leq \frac{8\pi}{\sqrt{2\pi}} 3.29^{3/2} t^{-9/2} e^{-0.225/t} < 0.101, \\ \frac{|B_0(\text{tail})|}{2G_0} &\leq \frac{2\pi}{\sqrt{2\pi}}\sqrt{3.29}\,t^{-3/2} e^{-0.225/t} < 10^{-8}. \end{aligned} \tag{7.3.10}$$

Both upper bounds increase for $0 < t \leq 0.008$, since $0.225 > (9/2)(0.008) > (3/2)(0.008)$; checking $t = 0.008$ suffices. The strict endpoint comparisons are verified with rational arithmetic in Appendix 7.C. Combining (7.3.9) and (7.3.10) yields

$$A_0 > 0.384\frac{G_0}{2V} > 0, \qquad B_0 > 0.951(2G_0) > 0.$$

The positive factorization (7.3.1) proves Theorem 7.3.1.

Remark 7.3.2. Every terminating decimal used as a bound denotes an exact rational number. Appendix 7.C verifies all 26 numerical comparisons using only Python's exact `Fraction` arithmetic. It encloses π by Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$ and alternating series bounds. It bounds positive exponentials from below by finite Taylor sums. Half-integer powers are checked after squaring positive quantities. Thus no floating-point approximation is used as a proof assumption.

7.4 An exact reduction of the finite range

The unbounded range of k cannot be replaced by an arbitrary experimental cutoff. This section proves the finite covering principle actually used by the program. Fix $N \geq 30$, and assume the exact partition values satisfy

$$D(n) := p(n)^2 - p(n-1)p(n+1) > 0 \qquad (26 \leq n \leq N). \tag{7.4.1}$$

The program verifies this assumption, rather than invoking an asymptotic result about $p(n)$. It also checks the 27 exact inequalities

$$p(m)p(27)^{27-m} \leq p(27)p(26)^{27-m} \qquad (0 \leq m < 27). \tag{7.4.2}$$

Lemma 7.4.1. *Under* (7.4.1) *and* (7.4.2)*, for* $27 \leq n \leq N$ *and* $0 \leq m \leq n$ *one has*

$$p(m) \leq p(n)r_n^{n-m}, \qquad r_n = \frac{p(n-1)}{p(n)} \in (0,1). \tag{7.4.3}$$

Proof. Strict increase of $p(n)$ for $n \geq 2$ follows by adjoining a part 1, with partitions having no part 1 showing strictness. Equation (7.4.1) implies $r_j < r_{j+1}$ for $26 \leq j \leq N$. For $27 \leq m < n$, write $p(m)/p(n) = \prod_{i=m+1}^{n} r_i$ and bound each factor by r_n. For $m < 27$, (7.4.2) first gives $p(m) \leq p(27)r_{27}^{27-m}$; combine this with the same product estimate from 27 to n and $r_{27} \leq r_n$. The case $m = n$ is equality. □

Lemma 7.4.2. *Under the hypotheses of Lemma 7.4.1, if* $27 \leq n \leq N$ *and*

$$2\frac{r_n^k}{1-r_n^k}\,p(n)^2 < D(n), \tag{7.4.4}$$

then $D_k(n) > 0$.

Proof. Every partition counted by $p(n) - b_k(n)$ contains a part jk for some $j \geq 1$. For a fixed j, removing one such part injects this set into the partitions of $n - jk$. A union bound and (7.4.3) give

$$0 \leq p(n) - b_k(n) \leq \sum_{j\geq 1} p(n-jk) \leq p(n)\frac{r_n^k}{1-r_n^k}.$$

Put $d = r_n^k/(1-r_n^k)$. Hypothesis (7.4.4) implies $d < D(n)/(2p(n)^2) < 1/2$. Hence

$$b_k(n)^2 \geq (1-d)^2 p(n)^2 \geq (1-2d)p(n)^2 > p(n-1)p(n+1) \geq b_k(n-1)b_k(n+1).$$

□

For each $27 \leq n \leq N$, choose an integer $L = L_n \geq 2$ with

$$Lp(n-1) \leq (L-1)p(n). \tag{7.4.5}$$

Such an integer exists because $p(n-1) < p(n)$. The program starts at $\lceil\sqrt{n}\rceil$ and increments until (7.4.5) holds; this starting choice affects speed only. Thus $r_n \leq 1 - 1/L$. Since $(1-1/L)^L < e^{-1} < 3/8$, for every positive integer j and every $k \geq jL$,

$$\frac{r_n^k}{1-r_n^k} < \frac{3^j}{8^j - 3^j}. \tag{7.4.6}$$

Choose the least positive $j = j_n$ satisfying the exact integer test

$$2 \cdot 3^j p(n)^2 < D(n)(8^j - 3^j), \tag{7.4.7}$$

and define

$$c_{26} = 30, \qquad c_n = \min\{n+2, j_n L_n\} \ (27 \le n \le N), \qquad C = \max\{30, c_{26}, \dots, c_N\}. \tag{7.4.8}$$

The j search terminates because $(3/8)^j \to 0$ and $D(n) > 0$. If $k \ge j_n L_n$, Lemma 7.4.2 applies. If $k \ge n+2$, then the three regular partition values equal the unrestricted values, so (7.4.1) applies directly. At $n = 26$, this equality holds for all $k \ge 30$. Therefore only the pairs

$$27 \le n \le N, \qquad 30 \le k < c_n \tag{7.4.9}$$

remain to be compared explicitly. All have $k < C$.

Proposition 7.4.3. *If the program in Appendix 7.A returns* `PASS` *for an input N, then $D_k(n) > 0$ for every $k \ge 30$ and every $26 \le n \le N$.*

Proof. The preceding lemmas cover all pairs outside (7.4.9). The program obtains $p(0), \dots, p(N+1)$ by Euler's pentagonal recurrence:

$$p(n) = \sum_{r \ge 1} (-1)^{r-1} \left[p\left(n - \frac{r(3r-1)}{2}\right) + p\left(n - \frac{r(3r+1)}{2}\right) \right].$$

The same pentagonal identity for the numerator of (7.1.1) gives the exact coefficients

$$b_k(n) = p(n) + \sum_{r \ge 1} (-1)^{r} \left[p\left(n - k\frac{r(3r-1)}{2}\right) + p\left(n - k\frac{r(3r+1)}{2}\right) \right]. \tag{7.4.10}$$

Every sum is finite under the convention $p(m) = 0$ for $m < 0$. For each k, the program sets ℓ to the least $n \ge 27$ requiring an explicit comparison. It computes (7.4.10) for every index from $\ell - 1$ to $N + 1$, which includes both neighbors of every index tested. All coefficient arithmetic, powers, and determinant comparisons use GMP arbitrary-precision integers. The floating-point square root only selects an initial integer L; the exact test (7.4.5) is always performed. Floating-point timing has no mathematical role. A `PASS` result therefore certifies every pair in (7.4.9), and the reduction certifies all the other pairs. □

7.5 Verification record and completion of the proof

The complete run with $N = 25703$ returned `PASS`. It made 43,931,660 explicit regular-partition determinant comparisons, with maximum exclusive parameter bound $C = 2737$. In addition, it checked all ordinary-partition inequalities (7.4.1) and all 27 base inequalities (7.4.2). This is a finite exact computation, not a sample of parameter values. The output record is

```
N=25703
all_k_ge=30
exact_checks=43931660
max_k_exclusive=2737
status=PASS
```

A redundant longer run through $N = 200000$ also returned `PASS`, with 1,146,293,779 explicit comparisons and $C = 8960$. It is not needed for the proof or the shorter reproduction below.

The independent Python audit in Appendix 7.B uses ordinary coin-change dynamic programming as its coefficient oracle, rather than Euler's recurrence. Its run at $N = 350$ compared 352 unrestricted coefficients, 123,904 regular-partition coefficients, and 104,975 direct log-concavity pairs. It also checked 61,074 geometric-envelope pairs, all cutoff inequalities, and the optimized coefficient window used by the C++ program. The 33,497 explicit comparisons and exclusive bound $C = 209$ agreed exactly between the independent audit and the C++ verifier. This small audit is an implementation cross-check, not a substitute for the full finite certificate or the uniform analytic theorem.

Proof of Theorem 7.1.1. The successful exact run and Proposition 7.4.3 prove

$D_k(n) > 0$ for $k \geq 30$ and $26 \leq n \leq 25703$. Theorem 7.3.1 proves the same statement for $n \geq 25703$. The common endpoint is 25703, so all $n \geq 26$ are covered. The exact negative values at $(k, n) = (k, 25)$ for $k \geq 30$ and $(29, 29)$, computed in Section 7.1, prove the two sharpness statements. □

7.6 Extension to uniformly sparse forbidden parts

The analytic estimate depends on a deletion-density bound, not on periodicity. This permits the following additional conclusion.

Theorem 7.6.1. *Let E be a finite or infinite subset of the positive integers satisfying*

$$E(x) := \#\{e \in E : e \leq x\} \leq \frac{x}{30} \qquad \text{for every real } x \geq 0.$$

Define

$$\prod_{\substack{m \geq 1 \\ m \notin E}} (1 - q^m)^{-1} = \sum_{n \geq 0} b_E(n) q^n.$$

Then $b_E(n)^2 > b_E(n-1) b_E(n+1)$ for every $n \geq 25703$.

Proof. If $e_1 < e_2 < \cdots$ enumerates E, its counting bound implies $e_j \geq 30j$. For any nonnegative decreasing integrable function h,

$$\sum_{e \in E} h(te) \leq \sum_{j \geq 1} h(30tj) \leq \frac{1}{30t} \int_0^\infty h(y)\, \mathrm{d}y. \tag{7.6.1}$$

For finite E, extend only the comparison sum to infinity. This also allows an integrable singularity at zero. Let H_E be the logarithm of the generating function at e^{-t} and $K_j = (-1)^j H_E^{(j)}(t)$. Its retained-parts expansion is positive. The first moment and the upper bounds for all cumulants are therefore bounded by the unrestricted moments, as before. The counting hypothesis ensures that part 1 is retained, so the saddle exists uniquely and satisfies (7.2.7).

Apply (7.6.1) to the decreasing function h_2 from Lemma 7.2.2. The removed variance is at most

$$\frac{1}{t^2} \sum_{e \in E} h_2(te) \leq \frac{2\zeta(2)}{30t^3}.$$

Thus exactly the cumulant bounds (7.2.5) hold. Apply (7.6.1) again to $g(y) = \log \coth(y/2)$, whose integral is $\pi^2/4$. The removed logarithmic modulus drop is at most $\pi^2/(120t) < 0.08225/t$. Consequently the same ten-harmonic and one-harmonic estimates prove (7.3.5) for the new characteristic function.

All remaining steps depend only on these cumulant and tail bounds, the positive retained-parts expansion, and Fourier inversion. The factorization (7.3.1), the core bounds (7.3.9), and the relative tail bounds (7.3.10) therefore apply without change and prove the result. □

Remark 7.6.2. The hypothesis is a bound at every x, not merely a bound on a limiting density. The finite certificate in Section 7.4 applies only to $E = k\mathbb{N}$. No threshold 26 is asserted for arbitrary sets E.

7.7 All moduli and the complete sign classification

Theorem 7.7.1. *For every integer $k \geq 2$ and every integer $n \geq 65798$, one has $D_k(n) > 0$.*

The same Fourier factorization proves this result once the removed cumulants and harmonics are estimated relative to the unrestricted ones. We give the necessary estimates, including their dependence on k.

7.7.1 Reduced cumulants for every modulus

Put $A = \zeta(2)$, $\beta = 1 - 1/k$, and $v = 2A\beta$, so $A \leq v < 2A$. For $j = 2, 3, 4$ define

$$h_j(y) = y^j \sum_{\ell \geq 1} \ell^{j-1} e^{-\ell y}.$$

Then $M_j(t) = t^{-j} \sum_{m \geq 1} h_j(mt)$ and $\int_0^\infty h_j(y)\,\mathrm{d}y = j!A$, by termwise integration. For a nonnegative decreasing integrable function h with $h(0) = c$,

$$\int_0^\infty h - ct \leq t \sum_{m \geq 1} h(mt) \leq \int_0^\infty h.$$

For a nonnegative integrable unimodal h, the layer-cake formula gives

$$\left| t \sum_{m \geq 1} h(mt) - \int_0^\infty h \right| \leq t \max h,$$

since each interval superlevel set has a positive-lattice-point count within one of its length divided by t.

The kernel h_2 decreases from 1 to 0, as already proved. Writing $x = y/2$, one has $h_3(y) = 2x^3 \cosh x / \sinh^3 x$, which decreases from 2 to 0. Indeed its derivative has the sign of $3 \sinh x \cosh x - x(3 + 2\sinh^2 x)$. The negative of this expression is

$$x(2 + \cosh 2x) - \tfrac{3}{2} \sinh 2x \geq 0 :$$

the coefficients of x and x^3 vanish, and the coefficient of x^{2m+1} for $m \geq 2$ is $2^{2m}(2m-2)/(2m+1)! > 0$.

The identity $h_4(y) = 6h_2(y)^2 + y^2 h_2(y)$ and $\sinh x \geq x + x^3/6$ give

$$h_4(y) \leq 6 + \frac{4x^2}{(1 + x^2/6)^2} \leq 12.$$

This kernel is unimodal. Its derivative has the sign of

$$T(x) = 2 \sinh 3x + 6 \sinh x - x \cosh 3x - 11x \cosh x.$$

The coefficient of x^{2m+1} is $((5-2m)9^m - 22m - 5)/(2m+1)!$. It is zero for $m = 0, 1$, positive for $m = 2$, and negative for $m \geq 3$. Thus $T(x)/x^5$ strictly decreases on $(0, \infty)$ from a positive value to a negative value, with exactly one zero: already its negative x^2 term forces its limit to be negative. This proves unimodality.

The Riemann-sum bounds consequently imply, for every $u > 0$,

$$\begin{aligned} 2A/u^3 - 1/u^2 \leq M_2(u) \leq 2A/u^3, \\ 6A/u^4 - 2/u^3 \leq M_3(u) \leq 6A/u^4, \\ 24A/u^5 - 12/u^4 \leq M_4(u) \leq 24A/u^5 + 12/u^4. \end{aligned}$$

Apply these at both t and kt, with no smallness assumption on kt:

$$\frac{v-t}{t^3} \le V \le \frac{v+t}{t^3}, \qquad K_3 \le \frac{3v+2t}{t^4}, \qquad K_4 \le \frac{12v+24t}{t^5}. \tag{7.7.1}$$

For example, the sharper upper variance is $V \le (2A\beta + t/k)t^{-3} \le (v+t)t^{-3}$; the relaxed error terms above are convenient uniform bounds.

At the unique saddle $K_1(t) = n$, the bound $K_1(t) \le A/t^2$ gives $t < 0.005$ for $n \ge 65798$. For $0 < t \le 0.005$, (7.7.1) gives

$$V < 3.3t^{-3}, \qquad \frac{t^2 K_4}{48V} < 0.253, \qquad r = \frac{K_3^2}{V^3} < 0.028, \qquad \frac{1}{4V} < 10^{-6}. \tag{7.7.2}$$

Here $(3v+2t)^2/(v-t)^3$ decreases in v, as does $(12v+24t)/(v-t)$; their largest bounds occur at $v = A$, $t = 0.005$. The resulting rational comparisons are certified in Appendix 7.E.

7.7.2 A uniform complementary arc bound

For the function S in (7.3.3), use $|\sin(kx/2)| \le k|\sin(x/2)|$ and $\sinh(ku/2) \ge k\sinh(u/2)$ to obtain

$$S(ku, kx) \le \frac{\coth(ku/2)}{\coth(u/2)} S(u,x) \le (1/k + u/2)S(u,x). \tag{7.7.3}$$

The first sine inequality follows by induction from the addition formula; the hyperbolic inequality follows from convexity. The final comparison uses $\coth z \le 1 + 1/z$ and $\coth(u/2) \ge 2/u$. The logarithmic modulus drop is exactly

$$-\log|\chi(\theta)| = \sum_{\ell \ge 1} \frac{S(\ell t, \ell\theta) - S(k\ell t, k\ell\theta)}{\ell}.$$

Each bracket is nonnegative, since it is the sum over retained part sizes. For $t/2 \le |\theta| \le t$, retain $1 \le \ell \le 10$ and apply the same sine estimates as before. They give $S(\ell t, \ell\theta) > 0.199/(\ell t)$. By (7.7.3), $k \ge 2$, and $t \le 0.005$,

$$-\log|\chi(\theta)| > \frac{0.199}{t}\left(\frac{1}{2}\sum_{\ell=1}^{10}\ell^{-2} - \frac{t}{2}\sum_{\ell=1}^{10}\ell^{-1}\right) > \frac{0.15}{t}.$$

Here $\sum \ell^{-2} > 1.549$ and $\sum \ell^{-1} < 2.93$. For $t \le |\theta| \le \pi$, retain only $\ell = 1$ and use $S(t,\theta) \ge S(t,t) > 0.499/t$. The drop then exceeds $(0.5 - t/2)(0.499)/t > 0.15/t$. Hence

$$|\chi(\theta)| < e^{-0.15/t} \qquad (t/2 \le |\theta| \le \pi, k \ge 2). \tag{7.7.4}$$

7.7.3 Completion of the uniform argument

The core estimate (7.3.6) now holds with $\alpha = 0.747$, by (7.7.2). The standardized endpoint satisfies

$$(t/2)\sqrt{V} \ge \sqrt{\frac{A-t}{4t}} > 9.$$

The normal mass and second-moment tail losses are therefore less than 10^{-15}. Reuse the Gaussian formulas (7.3.8), now with $r < 0.028$ and $\alpha = 0.747$. The phase losses are less than 0.152 and 0.017, respectively, so

$$A_0(\text{core}) > 0.847\frac{G_0}{2V}, \qquad B_0(\text{core}) > 0.982(2G_0).$$

By (7.7.4), each tail integral has absolute value at most $4\pi e^{-0.15/t}$. Relative to the same normalizations, the losses are bounded by

$$\sqrt{32\pi}\,3.3^{3/2}t^{-9/2}e^{-0.15/t} < 0.128,$$
$$\sqrt{2\pi\cdot 3.3}\,t^{-3/2}e^{-0.15/t} < 2\cdot 10^{-9}.$$

Both bounds increase for $0 < t \le 0.005$; the endpoint inequalities, and all other numerical comparisons in this section, are among the 24 exact checks in Appendix 7.E. Consequently

$$A_0 > 0.719\frac{G_0}{2V} > 0, \qquad B_0 > 0.981(2G_0) > 0.$$

The factorization (7.3.1) proves Theorem 7.7.1.

7.7.4 The complete finite classification

For $2 \le k \le 29$, the exact program in Appendix 7.D classifies every determinant for $1 \le n \le 65798$. It uses the same Euler recurrence and numerator identity (7.4.10), without any cutoff optimization. It computes through $N+1$, then records every negative and zero determinant separately. The completed run classified $28 \cdot 65798 = 1,842,344$ determinants and found 389 negative and three zero values. Its full replay agreed exactly. An independent coin-change oracle through $n = 500$ compared 14,056 coefficients and 14,000 determinant classifications; Appendix 7.F contains that audit. Combined with Theorem 7.7.1, the finite record proves the complete classification in Table 7.1. For $k \ge 30$ the earlier theorem covers $n \ge 26$, while the remaining indices are unrestricted partition values; direct evaluation gives exactly the odd indices through 25 as failures and no equalities.

Corollary 7.7.2. *For every $k \ge 2$, $D_k(n) < 0$ precisely at the indices in Table 7.1. Equality occurs precisely at $(k,n) = (2,1),(4,5),(4,7)$. At every other positive index the determinant is strictly positive. The table gives the exact value of N_k.*

Table 7.1: Complete sign classification. Here $O_m = \{1,3,\dots,m\}$ for odd m.

Modulus k	Indices with $D_k(n) < 0$	N_k
2	$\{2,4,8,11,13,14,16,17,19,20,23,26,29,32\}$	33
3	O_{57}	58
4	$\{1,4,8,11,13,16\}$	17
5	O_{41}	42
6	O_{11}	12
7	O_{35}	36
8	$\{1,3,5,7,11,13,17,19\}$	20
9	O_{33}	34
10	O_{19}	20
11 and 13	O_{31}	32
12	O_{21}	22
14	$O_{19} \cup \{23\}$	24
15, 17, 19, and 21	O_{29}	30
16, 18, 20, 22, 24, 25, 26, 28, and all $k \ge 30$	O_{25}	26
23 and 29	$O_{25} \cup \{29\}$	30
27	O_{27}	28

7.8 Uniform Hermite limits and threshold stabilization

For a positive sequence $a(n)$ define its Jensen polynomials by

$$J_a^{d,n}(X) = \sum_{j=0}^{d} \binom{d}{j} a(n+j) X^j.$$

A real polynomial is called hyperbolic if all its roots are real. Fixed-modulus Hermite limits and eventual hyperbolicity for regular partitions were established by Craig and Pun [7.3]; see also O'Sullivan [7.6] for the log-polynomial framework and its regular partition application. Broader recent asymptotic frameworks include [7.7]. Here the estimates above give a limit uniform over the entire family, with an elementary cancellation inside the exact Fourier integral.

Let $\mathcal{F}$ consist of all sequences b_k, $k \geq 2$, together with all sequences b_E from Theorem 7.6.1. For $a \in \mathcal{F}$, write $H_a(t) = \log \sum_{m \geq 0} a(m) e^{-mt}$, choose the saddle $-H_a'(t_a(n)) = n$, and put $V_a(n) = H_a''(t_a(n))$. The probabilists' Hermite polynomials are normalized by

$$e^{Ys - s^2/2} = \sum_{d \geq 0} \mathrm{He}_d(Y) \frac{s^d}{d!}.$$

Theorem 7.8.1. *For every fixed nonnegative integer d and every fixed $R > 0$,*

$$\frac{V_a(n)^{d/2}}{a(n)} J_a^{d,n}\left(e^{-t_a(n)} \left(-1 + \frac{Y}{\sqrt{V_a(n)}} \right) \right) = \mathrm{He}_d(Y) + O_{d,R}(n^{-1/4}) \tag{7.8.1}$$

as $n \to \infty$, uniformly over $a \in \mathcal{F}$ and complex $|Y| \leq R$. Consequently, for each fixed degree d, a single integer $M(d)$ exists such that every $J_a^{d,n}$ has d distinct real roots for $a \in \mathcal{F}$ and $n \geq M(d)$.

Proof. We suppress the subscript a and put $L(\theta) = H(t - i\theta) - H(t) - in\theta$, $\chi = e^L$. The previous estimates give absolute constants, independent of the family member, such that for $0 < t \leq 0.005$,

$$\begin{aligned} 1.63 t^{-3} \leq V \leq 3.3 t^{-3}, \quad K_3/V^{3/2} \leq C\sqrt{t}, \quad t \leq \sqrt{\zeta(2)/n}, \\ |\chi(\theta)| \leq e^{-0.63 V \theta^2/2} \quad (|\theta| \leq t/2), \\ |\chi(\theta)| \leq e^{-0.15/t} \quad (t/2 \leq |\theta| \leq \pi). \end{aligned} \tag{7.8.2}$$

For regular partitions use (7.7.1)–(7.7.4); for sparse deletions use the stronger bounds of Theorem 7.6.1. The bound $1.63 < \zeta(2) - 0.005$ supplies a common lower variance constant.

The positive logarithmic expansion and the integral remainder of e^{ix} give, for real u,

$$\left| L(u/\sqrt{V}) + \frac{u^2}{2} \right| \leq \frac{K_3 |u|^3}{6 V^{3/2}} \leq \frac{C}{6} \sqrt{t}\, |u|^3.$$

On the core $|u| \leq U = t\sqrt{V}/2$, exponential interpolation $e^z - e^w = (z - w) \int_0^1 e^{(1-s)w + sz}\, \mathrm{d}s$ and (7.8.2) imply the integrable error bound

$$|\chi(u/\sqrt{V}) - e^{-u^2/2}| \leq \frac{C}{6} \sqrt{t}\, |u|^3 e^{-0.63 u^2/2}. \tag{7.8.3}$$

Set

$$Q_V(u, Y) = \sqrt{V}(1 - e^{-iu/\sqrt{V}}) + Y e^{-iu/\sqrt{V}}.$$

Fourier inversion for $a(n + j)$, followed by the binomial theorem, gives the exact identity

$$\frac{V^{d/2}}{a(n)} J_a^{d,n}\left(e^{-t}(-1 + Y/\sqrt{V}) \right) = \frac{\displaystyle\int_{-\pi\sqrt{V}}^{\pi\sqrt{V}} \chi(u/\sqrt{V}) Q_V(u, Y)^d \, \mathrm{d}u}{\displaystyle\int_{-\pi\sqrt{V}}^{\pi\sqrt{V}} \chi(u/\sqrt{V}) \, \mathrm{d}u}. \tag{7.8.4}$$

The denominator is positive by coefficient inversion and $a(n) > 0$. For real u and complex $|Y| \le R$,

$$|Q_V(u,Y)| \le |u| + R, \qquad |Q_V(u,Y) - (iu+Y)| \le V^{-1/2}(u^2/2 + R|u|).$$

Consequently, for $d \ge 1$,

$$|Q_V(u,Y)^d - (iu+Y)^d| \le dV^{-1/2}(u^2/2 + R|u|)(|u|+R)^{d-1}.$$

Multiply (7.8.3) by $(|u|+R)^d$ and integrate. The Gaussian moments are finite, so the numerator on the core differs from $\int_{-U}^{U} e^{-u^2/2}(Y+iu)^d\,\mathrm{d}u$ by $O_{d,R}(\sqrt{t}) + O_{d,R}(V^{-1/2}) = O_{d,R}(\sqrt{t})$.

Since $U \ge \sqrt{1.63}/(2\sqrt{t})$, extending this Gaussian integral to $\mathbb{R}$ has error $o_{d,R}(\sqrt{t})$, uniformly in the family: absorb the fixed polynomial weight into $e^{u^2/4}$ and apply a Gaussian tail bound. On the complementary Fourier arcs use $|Q_V| \le 2\sqrt{V} + R$ and (7.8.2). Their entire numerator contribution has absolute value at most

$$2\pi\sqrt{V}(2\sqrt{V}+R)^d e^{-0.15/t} = O_{d,R}\Big(t^{-3(d+1)/2}e^{-0.15/t}\Big) = o_{d,R}(\sqrt{t}).$$

The same argument with $d = 0$ shows that the denominator in (7.8.4) equals $\sqrt{2\pi} + O(\sqrt{t})$. Thus its ratio equals

$$\frac{1}{\sqrt{2\pi}}\int_{\mathbb{R}} e^{-u^2/2}(Y+iu)^d\,\mathrm{d}u + O_{d,R}(\sqrt{t}) = \mathrm{He}_d(Y) + O_{d,R}(\sqrt{t}).$$

The last equality follows by taking the exponential generating function: for a standard normal Z, $\mathbb{E}e^{s(Y+iZ)} = e^{Ys - s^2/2}$. The uniform bound on t proves (7.8.1); degree zero is immediate.

For $d \ge 1$, choose $d+1$ real points interlacing the distinct real zeros of He_d, on which its nonzero signs alternate. The uniform estimate gives the same signs for every normalized Jensen polynomial once $n \ge M(d)$, with $M(d)$ independent of a. The intermediate value theorem yields d distinct real roots. The polynomial has degree exactly d, and its real affine normalization preserves reality and distinctness of roots. □

Corollary 7.8.2. *Let $N_k(d)$ and $N_p(d)$ be the least nonnegative shifts after which all degree-d Jensen polynomials of b_k and p, respectively, are hyperbolic. For each fixed d there is an integer $M(d)$ such that*

$$N_k(d) = N_p(d) \qquad (k \ge M(d) + d).$$

The same conclusion holds for the least eventual strict-hyperbolicity shifts, where all roots are required to be distinct.

Proof. Choose the common $M(d)$ from Theorem 7.8.1; it covers ordinary partitions because the empty forbidden set belongs to the sparse family. If $k \ge M(d)+d$, $0 \le n < M(d)$, and $0 \le j \le d$, then $n + j \le M(d) - 1 + d < k$, so $b_k(n+j) = p(n+j)$. The Jensen polynomials are therefore identical at every earlier shift, and both families are strictly hyperbolic at every later shift. Their exceptional-shift sets and minimal eventual thresholds coincide. □

Remark 7.8.3. This establishes the eventual stabilization suggested in [7.3, Remark 1.3] by proving uniformity in k; it does not claim fixed-k eventual hyperbolicity as a new result. These are Jensen *shift* thresholds. Degree two tests log-concavity at the central index $n+1$, so $N_k(2) = N_k - 1$ with the matching weak or strict convention. The proof provides a threshold computable in principle from Gaussian moments and sign-test points, but no optimized numerical $M(d)$ is asserted for general d.

7.9 Reproduction instructions

This manuscript has no external TeX inputs or bibliography database. Appendix 7.G provides an automatic extraction and replay driver; the manual commands below are equivalent. Compile the single source file twice with a standard `pdflatex` installation. To reproduce the finite certificate, save the complete listing in Appendix 7.A as `certify.cpp`. With a C++17 compiler and GMP development libraries installed, run

```
g++ -O3 -std=c++17 certify.cpp -lgmpxx -lgmp -o certify
./certify 25703 > cert25703.json 2> cert25703.progress
```

A nonzero exit code or any message beginning `FAIL` is not a certificate. Require exit code zero and a final JSON record with `"status":"PASS"` and `"N":25703`. Progress lines are not final results. Runtime depends on the machine; no particular elapsed time is part of the proof.

Save Appendix 7.B as `audit_independent.py` in the same directory as `certify.cpp`, and run

```
python3 audit_independent.py
```

The Python audit uses only the standard library, compiles a separate copy of the C++ executable, and writes its own JSON record. The source hash in that record identifies the precise C++ file audited; changes in whitespace alone may change the hash without changing the program.

Save Appendix 7.C as `certify_fourier_constants.py` and run it with `python3`. All 26 exact rational comparisons must pass. This verifies the numerical constants in the analytic proof; the integral identities and inequalities are established in the text.

7.A Complete exact C++ verifier

The following is the complete source used for the finite computation.

```
#include <gmpxx.h>
#include <vector>
#include <iostream>
#include <cmath>
#include <chrono>
#include <algorithm>
int main(int argc,char**argv){
 int N=argc>1?std::stoi(argv[1]):10000;if(N<30||N>300000)return 1;
 auto start=std::chrono::steady_clock::now();
 std::vector<mpz_class> p(N+2);p[0]=1;
 std::vector<std::pair<int,int>> pent;
 for(int r=1;;r++){int a=r*(3*r-1)/2,b=r*(3*r+1)/2;if(a>N+1)break;pent.push_back({a,r
    %2?1:-1});if(b<=N+1)pent.push_back({b,r%2?1:-1});}
 for(int n=1;n<=N+1;n++)for(auto [g,s]:pent){if(g>n)break;if(s==1)p[n]+=p[n-g];else p[n]-=p[
    n-g];}
 // Geometric dominance base at 27, exact integer powers.
 for(int m=0;m<27;m++){mpz_class a,b;mpz_pow_ui(a.get_mpz_t(),p[27].get_mpz_t(),27-m);a*=p[m
    ];mpz_pow_ui(b.get_mpz_t(),p[26].get_mpz_t(),27-m);b*=p[27];if(a>b){std::cerr<<"base
    fail";return 2;}}
 std::vector<int> cutoff(N+1);int maxk=30;
 for(int n=26;n<=N;n++){
  mpz_class square=p[n]*p[n],D=square-p[n-1]*p[n+1];if(D<=0){std::cerr<<"p LC fail "<<n;
    return 2;}
  if(n==26){cutoff[n]=30;continue;}
  int L=(int)std::ceil(std::sqrt(n));while(L*p[n-1]>(L-1)*p[n])L++;
  // (1-1/L)^L < 3/8, via e>8/3. If k>=jL, deficit <=3^j/(8^j-3^j).
  int j=1;mpz_class a=3,b=8;
```

```
  while(2*a*square>=D*(b-a)){a*=3;b*=8;j++;}
  cutoff[n]=std::min(n+2,j*L);maxk=std::max(maxk,cutoff[n]);
 }
 unsigned long long checks=0;
 for(int k=30;k<maxk;k++){
  int lo=N+1;for(int n=27;n<=N;n++)if(k<cutoff[n]){lo=n;break;}if(lo>N)continue;
  std::vector<mpz_class>b=p;
  for(auto [g,s]:pent){long long shift=1LL*g*k;if(shift>N+1)break;for(int n=std::max(lo-1,(
    int)shift);n<=N+1;n++){if(s==1)b[n]-=p[n-shift];else b[n]+=p[n-shift];}}
  for(int n=lo;n<=N;n++)if(k<cutoff[n]){checks++;if(b[n]*b[n]<=b[n-1]*b[n+1]){std::cout<<"
    FAIL k="<<k<<" n="<<n<<std::endl;return 3;}}
  if(k%250==0)std::cerr<<"k="<<k<<"/"<<maxk<<" checks="<<checks<<std::endl;
 }
 double secs=std::chrono::duration<double>(std::chrono::steady_clock::now()-start).count();
 std::cout<<"{\"N\":"<<N<<",\"all_k_ge\":30,\"exact_checks\":"<<checks<<",\"max_k_exclusive
    \":"<<maxk<<",\"seconds\":"<<secs<<",\"status\":\"PASS\"}"<<std::endl;
}
```

7.B Independent small range Python audit

This audit checks the arithmetic and covering implementation against independent coin-change coefficients. Its fixed bound is intentionally small enough to reproduce quickly.

```
"""Independent exact audit of certify.cpp; coin-change is the oracle."""
import json, math, subprocess, pathlib, hashlib, time
from fractions import Fraction
DIR=pathlib.Path(__file__).parent
N=350
start=time.time()

def coin_partitions(M,k=None):
 a=[0]*(M+1);a[0]=1
 for part in range(1,M+1):
  if k is not None and part%k==0:continue
  for n in range(part,M+1):a[n]+=a[n-part]
 return a
p=coin_partitions(N+1)
assert [n for n in range(1,26) if p[n]**2 < p[n-1]*p[n+1]] == list(range(1,26,2))
assert all(p[n]**2 != p[n-1]*p[n+1] for n in range(1,26))
# Generate generalized pentagonal numbers by signed r, then sort independently.
pent=sorted((r*(3*r-1)//2,(-1)**(abs(r)-1)) for r in range(-N,N+1) if r and r*(3*r-1)//2<=N
    +1)
p_euler=[1]+[0]*(N+1)
for n in range(1,N+2):p_euler[n]=sum(sign*p_euler[n-g] for g,sign in pent if g<=n)
assert p==p_euler
assert all(p[n]*p[n]>p[n-1]*p[n+1] for n in range(26,N+1))
base_margins=[]
for m in range(27):
 lhs=p[m]*p[27]**(27-m);rhs=p[27]*p[26]**(27-m)
 assert lhs<=rhs;base_margins.append(rhs-lhs)
# Verify geometric envelope directly, not through the inductive argument.
envelope_checks=0
for n in range(27,N+1):
 for m in range(n):
  assert p[m]*p[n]**(n-m)<=p[n]*p[n-1]**(n-m)
  envelope_checks+=1
cutoff={26:30};details={};maxk=30
for n in range(27,N+1):
 square=p[n]**2;D=square-p[n-1]*p[n+1]
 L=math.isqrt(n)
 if L*L<n:L+=1
```

```
 while L*p[n-1]>(L-1)*p[n]:L+=1
 assert 8*(L-1)**L<3*L**L
 j=1;a=3;b=8
 while 2*a*square>=D*(b-a):j+=1;a*=3;b*=8
 cutoff[n]=min(n+2,j*L);maxk=max(maxk,cutoff[n]);details[n]=(L,j,D)
 # Independent direct rational cutoff inequality when transfer is used.
 k=j*L
 u=p[n-1]**k;v=p[n]**k
 assert 2*u*square<D*(v-u)

numerator_checks=0;lc_checks=0;engine_queries=0
for k in range(1,N+3):
 b=coin_partitions(N+1,k)
 # Euler numerator formula, independent of original's lo optimization.
 e=[p[n]-sum(sign*p[n-k*g] for g,sign in pent if k*g<=n) for n in range(N+2)]
 assert b==e,(k,'Euler numerator mismatch')
 numerator_checks+=N+2
 if k<30:continue
 lo=next((n for n in range(27,N+1) if k<cutoff[n]),N+1)
 optimized=p.copy()
 for g,sign in pent:
  shift=k*g
  if shift>N+1:break
  for n in range(max(lo-1,shift),N+2):optimized[n]-=sign*p[n-shift]
 for n in range(26,N+1):
  assert b[n]**2>b[n-1]*b[n+1],(k,n,'LC failure')
  lc_checks+=1
  if n>=27 and k<cutoff[n]:
   assert optimized[n-1:n+2]==b[n-1:n+2],(k,n,'optimized neighbor mismatch')
   engine_queries+=1
  elif k>=n+2 or n==26:
   assert b[n-1:n+2]==p[n-1:n+2]
  else:
   L,j,D=details[n]
   assert k>=j*L
   a=3**j;c=8**j
   # Direct actual deficit check against the claimed upper bound.
   assert (p[n]-b[n])*(c-a)<=a*p[n]

binary=DIR/'audit_certify'
subprocess.run(['g++','-O2','-std=c++17',str(DIR/'certify.cpp'),'-lgmpxx','-lgmp','-o',str(
    binary)],check=True)
r=subprocess.run([str(binary),str(N)],capture_output=True,text=True,check=True)
engine=json.loads(r.stdout)
assert engine['exact_checks']==engine_queries,(engine,engine_queries)
assert engine['max_k_exclusive']==maxk
report={
 'N':N,'ordinary_partition_values_compared':N+2,'regular_partition_values_compared':
    numerator_checks,
 'geometric_envelope_pairs_compared':envelope_checks,'direct_LC_pairs_compared':lc_checks,
 'engine_exact_queries_compared':engine_queries,'engine_max_k_exclusive':maxk,
 'n25_difference':p[25]**2-p[24]*p[26], 'n26_difference':p[26]**2-p[25]*p[27],
 'base27_all_pass':True, 'cutoff_inequalities_exact_pass':True,
 'source_sha256':hashlib.sha256((DIR/'certify.cpp').read_bytes()).hexdigest(),
 'status':'PASS','seconds':time.time()-start,
}
assert report['n25_difference']<0<report['n26_difference']
(DIR/'audit_independent_results.json').write_text(json.dumps(report,indent=2)+'\n')
print(json.dumps(report,indent=2))
```

7.C Exact rational certificate for the analytic constants

The following standard-library program performs no floating-point assertion. A failed assertion prevents a successful certificate.

```
#!/usr/bin/env python3
"""Exact rational certificate for the 25703 Fourier proof constants.

No floating-point assertion is used. Pi is enclosed using Machin's identity
and alternating arctangent series; exponential upper bounds use reciprocals
of positive Taylor partial sums. See the analytic proof in this manuscript.
"""
from fractions import Fraction as Q
from math import factorial
from pathlib import Path
import json


def atan_enclosure(x, terms=40):
    s = sum(((-1) ** j * x ** (2*j+1) / (2*j+1)
             for j in range(terms)), Q(0))
    snext = s + (-1) ** terms * x ** (2*terms+1) / (2*terms+1)
    return min(s, snext), max(s, snext)


def exp_lower(x, terms=250):
    return sum((x**j / factorial(j) for j in range(terms)), Q(0))


a, b = atan_enclosure(Q(1, 5))
c, d = atan_enclosure(Q(1, 239))
pi_lo, pi_hi = 16*a-4*d, 16*b-4*c
zeta_lo, zeta_hi = pi_lo*pi_lo/6, pi_hi*pi_hi/6
t = Q(1, 125)
alpha = Q(63, 100)
r = Q(44, 1000)
checks = {}


def check(name, statement):
    checks[name] = bool(statement)
    assert statement, name


check('pi enclosure nonempty', pi_lo < pi_hi)
check('M3 constant < 13.1', zeta_hi*159/exp_lower(Q(3)) < Q(131, 10))
check('M4 constant < 56', zeta_hi*1848/exp_lower(Q(4)) < 56)
check('variance lower > 3.17/t^3', 2*zeta_lo*Q(29, 30)-t > Q(317, 100))
check('variance upper < 3.29/t^3', 2*zeta_hi < Q(329, 100))
check('saddle threshold < 25703', zeta_hi/t**2 < 25703)
check('squared sine ratio > .199', Q(999,1000)**2 /
      (4*Q(1001,1000)**2+Q(999,1000)**2) > Q(199,1000))
check('squared sine ratio > .499 in outer region', Q(999,1000)**2 /
      (Q(1001,1000)**2+Q(999,1000)**2) > Q(499,1000))
check('H10 > 1.549', sum((Q(1,j*j) for j in range(1,11)), Q(0)) > Q(1549,1000))
check('removed factor exponent < .08225/t', pi_hi*pi_hi/120 < Q(8225,100000))
check('transition exponent > .225/t', Q(199,1000)*Q(1549,1000)-Q(8225,100000) > Q(225,1000))
check('outer exponent > .225/t', Q(499,1000)-Q(8225,100000) > Q(225,1000))
check('quartic loss < .37', Q(56)/(48*Q(317,100)) < Q(37,100))
check('skew ratio r < .044', Q(131,10)**2/Q(317,100)**3*t < r)
check('1/(4V) < 1e-6', t**3/(4*Q(317,100)) < Q(1,10**6))
check('Gaussian cutoff > 9.95', Q(317,100)/(4*t) > Q(995,100)**2)
check('Gaussian tail epsilon < 1e-19', 11/exp_lower(Q(49)) < Q(1,10**19))
```

```
check('A moment loss < .514', (r*Q(105,72))**2 * alpha**(-9) < Q(514,1000)**2)
check('B moment loss < .047', (r*Q(15,72))**2 * alpha**(-7) < Q(47,1000)**2)
check('A core > .485', 1-Q(1,10**6)-Q(514,1000)-Q(1,10**19) > Q(485,1000))
check('B core > .952', 1-Q(1,10**6)-Q(47,1000)-Q(1,10**19) > Q(952,1000))
exp_sq_upper = 1/exp_lower(Q(225,4))
tail_a_sq_upper = 32*pi_hi*Q(329,100)**3*t**(-9)*exp_sq_upper
tail_b_sq_upper = 2*pi_hi*Q(329,100)*t**(-3)*exp_sq_upper
check('A tail relative loss < .101', tail_a_sq_upper < Q(101,1000)**2)
check('B tail relative loss < 1e-8', tail_b_sq_upper < Q(1,10**8)**2)
check('A tail bound increases with t', Q(225,1000) > Q(9,2)*t)
check('B tail bound increases with t', Q(225,1000) > Q(3,2)*t)
check('B total > .951', Q(952,1000)-Q(1,10**8) > Q(951,1000))

result = {
    'method': 'exact Fraction arithmetic; Machin pi enclosure; positive exp Taylor partial sums',
    'passed': sum(checks.values()),
    'total': len(checks),
    'checks': checks,
    'pi_enclosure_width_denominator': (pi_hi-pi_lo).denominator,
    'analytic_threshold_n': 25703,
    'analytic_range_k': 'all integer k >= 30',
}
out = Path(__file__).with_name('fourier_constants_certificate.json')
out.write_text(json.dumps(result, indent=2)+'\n')
print(json.dumps({'passed': result['passed'], 'total': result['total'], 'output': str(out)}))
```

7.D Complete small modulus classifier

The following source records every nonpositive determinant for $2 \le k \le 29$ and $1 \le n \le 65798$ using exact GMP arithmetic.

```
#include <gmpxx.h>
#include <vector>
#include <iostream>
int main(){const int N=65798;std::vector<mpz_class>p(N+2);p[0]=1;std::vector<std::pair<int,int>> pent;
for(int r=1;;r++){int a=r*(3*r-1)/2,b=r*(3*r+1)/2;if(a>N+1)break;pent.push_back({a,r%2?1:-1});if(b<=N+1)pent.push_back({b,r%2?1:-1});}
for(int n=1;n<=N+1;n++)for(auto[g,s]:pent){if(g>n)break;if(s==1)p[n]+=p[n-g];else p[n]-=p[n-g];}
for(int k=2;k<30;k++){std::vector<mpz_class>b=p;for(auto[g,s]:pent){int shift=g*k;if(shift>N+1)break;for(int n=shift;n<=N+1;n++){if(s==1)b[n]-=p[n-shift];else b[n]+=p[n-shift];}}
std::cout<<"{\"k\":"<<k<<",\"N\":"<<N<<",\"negative\":[";bool first=true;for(int n=1;n<=N;n++)if(b[n]*b[n]<b[n-1]*b[n+1]){if(!first)std::cout<<",";first=false;std::cout<<n;}std::cout<<"],\"zero\":[";first=true;for(int n=1;n<=N;n++)if(b[n]*b[n]==b[n-1]*b[n+1]){if(!first)std::cout<<",";first=false;std::cout<<n;}std::cout<<"]}"<<std::endl;
}}
```

7.E Exact constants for the all modulus cutoff

This is the complete 24-check rational certificate for Section 7.7.

```
#!/usr/bin/env python3
"""Exact rational numeric checks for the all-modulus theorem in this manuscript."""
from fractions import Fraction as Q
from math import factorial
from pathlib import Path
import json

def atan_enclosure(x, terms=40):
    s=sum(((-1)**j*x**(2*j+1)/(2*j+1) for j in range(terms)),Q(0))
    z=s+(-1)**terms*x**(2*terms+1)/(2*terms+1)
    return min(s,z),max(s,z)
def exp_lower(x,terms=250):
    return sum((x**j/factorial(j) for j in range(terms)),Q(0))
a,b=atan_enclosure(Q(1,5));c,d=atan_enclosure(Q(1,239))
plo,phi=16*a-4*d,16*b-4*c
Alo,Ahi=plo**2/6,phi**2/6
t=Q(1,200);alpha=Q(747,1000);r=Q(28,1000)
checks={}
def check(name,statement):
    checks[name]=bool(statement)
    assert statement,name
check('saddle threshold <65798',Ahi/t**2<65798)
check('variance upper <3.3/t^3',2*Ahi+t<Q(33,10))
check('quartic loss <.253',(12*Ahi+24*t)/(48*(Alo-t))<Q(253,1000))
check('skew ratio <.028',(3*Ahi+2*t)**2/(Alo-t)**3*t<r)
check('1/(4V)<1e-6',t**3/(4*(Alo-t))<Q(1,10**6))
check('standardized core endpoint >9',(Alo-t)/(4*t)>81)
# Normal second moment bound sqrt(2/pi)*(9+1/9)*exp(-81/2)<1e-15.
check('both Gaussian tail losses <1e-15',Q(82,9)/exp_lower(Q(81,2))<Q(1,10**15))
check('sin x >=.999x for x<=.04',1-Q(4,100)**2/6>Q(999,1000))
# sinh(x)/x <=1+(x^2/6)/(1-x^2/20), by ratios in the tail series.
check('sinh x <=1.001x for x<=.04',1+(Q(4,100)**2/6)/(1-Q(4,100)**2/20)<Q(1001,1000))
check('near tail S ratio >.199',Q(999,1000)**2/(4*Q(1001,1000)**2+Q(999,1000)**2)>Q
    (199,1000))
check('far tail S ratio >.499',Q(999,1000)**2/(Q(1001,1000)**2+Q(999,1000)**2)>Q(499,1000))
H2=sum((Q(1,j*j) for j in range(1,11)),Q(0));H1=sum((Q(1,j) for j in range(1,11)),Q(0))
check('H10 second >1.549',H2>Q(1549,1000))
check('H10 first <2.93',H1<Q(293,100))
check('near tail exponent >.15',Q(199,1000)*(H2/2-t*H1/2)>Q(15,100))
check('far tail exponent >.15',(Q(1,2)-t/2)*Q(499,1000)>Q(15,100))
check('A phase loss <.152',(r*Q(105,72))**2*alpha**(-9)<Q(152,1000)**2)
check('B phase loss <.017',(r*Q(15,72))**2*alpha**(-7)<Q(17,1000)**2)
check('A core >.847',1-Q(1,10**6)-Q(152,1000)-Q(1,10**15)>Q(847,1000))
check('B core >.982',1-Q(1,10**6)-Q(17,1000)-Q(1,10**15)>Q(982,1000))
expsq=1/exp_lower(2*Q(15,100)/t)
check('A tail loss <.128',32*phi*Q(33,10)**3*t**(-9)*expsq<Q(128,1000)**2)
check('B tail loss <2e-9',2*phi*Q(33,10)*t**(-3)*expsq<Q(2,10**9)**2)
check('both tail bounds increase with t',Q(15,100)>Q(9,2)*t)
check('A final >.719',Q(847,1000)-Q(128,1000)>=Q(719,1000))
check('B final >.981',Q(982,1000)-Q(2,10**9)>Q(981,1000))
result={'method':'exact Fraction arithmetic, Machin pi enclosure, positive exponential
    Taylor partial sums','passed':sum(checks.values()),'total':len(checks),'checks':checks,'
    analytic_cutoff':65798,'moduli':'all integers k>=2'}
p=Path(__file__).with_name('constants_certificate.json');p.write_text(json.dumps(result,
    indent=2)+'\n');print(json.dumps(result,indent=2))
```

7.F Independent small modulus audit

The expected JSONL record is generated and checked against Table 7.1 by the replay driver before this audit runs. This script then compares coin-change coefficients and performs a fresh complete C++ replay.

```
"""Independent coin-change oracle and fresh full-source replay for k=2,...,29."""
import hashlib,json,pathlib,subprocess,time
D=pathlib.Path(__file__).parent
src=D/'check_small_moduli.cpp';data=D/'small_moduli_65798.jsonl'
start=time.time();N=500
records=[json.loads(line) for line in data.read_text().splitlines() if line.strip()]
assert [r['k'] for r in records]==list(range(2,30))
assert all(r['N']==65798 for r in records)
for r in records:
 for field in ['negative','zero']:
  assert r[field]==sorted(set(r[field]))
  assert all(1<=n<=65798 for n in r[field])
 assert not set(r['negative'])&set(r['zero'])

def coin(k=None):
 b=[1]+[0]*(N+1)
 for part in range(1,N+2):
  if k is not None and part%k==0:continue
  for n in range(part,N+2):b[n]+=b[n-part]
 return b
p=coin()
# Independent signed-index pentagonal generation.
gpent=sorted((j*(3*j-1)//2,(-1)**abs(j)) for j in range(-N,N+1) if j and j*(3*j-1)//2<=N+1)
table=[];values=0;neg=0;zero=0
for r in records:
 k=r['k'];b=coin(k)
 e=[p[n]+sum(s*p[n-k*g] for g,s in gpent if k*g<=n) for n in range(N+2)]
 assert b==e,(k,'b coefficient mismatch')
 negative=[n for n in range(1,N+1) if b[n]**2<b[n-1]*b[n+1]]
 zeros=[n for n in range(1,N+1) if b[n]**2==b[n-1]*b[n+1]]
 assert negative==[n for n in r['negative'] if n<=N],(k,'negative list mismatch')
 assert zeros==[n for n in r['zero'] if n<=N],(k,'zero list mismatch')
 assert not [n for n in r['negative']+r['zero'] if n>N],(k,'unexpected late exception')
 cutoff=max(negative+zeros)+1
 assert b[cutoff-1]**2<=b[cutoff-2]*b[cutoff]
 assert b[cutoff]**2>b[cutoff-1]*b[cutoff+1]
 table.append({'k':k,'finite_last_nonpositive_plus_one':cutoff,'negative':negative,'zero':
    zeros})
 values+=N+2;neg+=len(negative);zero+=len(zeros)
# Recompile the untouched source; its hard-coded N remains 65798.
binary=D/'audit_small_moduli'
subprocess.run(['g++','-O2','-std=c++17','-Wall','-Wextra',str(src),'-lgmpxx','-lgmp','-o',
    str(binary)],check=True)
r=subprocess.run([str(binary)],capture_output=True,text=True,check=True)
replay=[json.loads(line) for line in r.stdout.splitlines() if line.strip()]
assert replay==records,'fresh full-source output differs from recorded artifact'
(D/'audit_small_moduli_replay_65798.jsonl').write_text(r.stdout)
report={'status':'PASS','oracle_N':N,'full_replay_N':65798,'moduli':28,
'coefficient_values_compared':values,'oracle_determinants_compared':28*N,
'full_replay_determinants':28*65798,'negative_count':neg,'zero_count':zero,
'source_sha256':hashlib.sha256(src.read_bytes()).hexdigest(),
'data_sha256':hashlib.sha256(data.read_bytes()).hexdigest(),
'full_replay_exit_code':r.returncode,'seconds':time.time()-start,'finite_threshold_table':
    table,
'all_n_status':'NOT ESTABLISHED BY THIS FINITE AUDIT; requires separately verified analytic
    tail'}
(D/'audit_small_moduli_results.json').write_text(json.dumps(report,indent=2)+'\n')
```

```
print(json.dumps({k:v for k,v in report.items() if k!='finite_threshold_table'},indent=2))
print('threshold table:',[(x['k'],x['finite_last_nonpositive_plus_one']) for x in table])
```

7.G Single source extraction and replay

Save this short driver as `replay.py`. It extracts the six tagged source listings from the single manuscript, checks that none is missing, compiles the exact verifier, and runs all certificates in a new directory. For example,

```
python3 replay.py KEIO_Regular_Partition_Log_Concavity_2026.tex replay
```

The directory `replay` must not already exist. No files from the research workspace are used. The finite $N = 25703$ run took about 10.4 seconds in the recorded environment; compilation, the independent audit, and machine differences add to the total elapsed time. The small-modulus scan and its full replay add further computation, typically tens of seconds. The driver verifies the complete negative and zero lists against the table rather than interpreting exit code zero as a sign result.

```
#!/usr/bin/env python3
import pathlib, re, subprocess, sys

if len(sys.argv) != 3:
    raise SystemExit("usage: replay.py MANUSCRIPT.tex NEW_DIRECTORY")
source = pathlib.Path(sys.argv[1]).read_text()
dest = pathlib.Path(sys.argv[2]).resolve()
dest.mkdir(parents=True, exist_ok=False)
pattern = (r"\\begin\{lstlisting\}\[label=src:(\w+),[^\n]*\]\n"
           r"(.*?)\n\\end\{lstlisting\}")
blocks = dict(re.findall(pattern, source, flags=re.S))
files = {"cpp": "certify.cpp", "oracle": "audit_independent.py",
         "constants": "certify_fourier_constants.py",
         "smallcpp": "check_small_moduli.cpp",
         "allkconstants": "certify_all_k_constants.py",
         "smallaudit": "audit_small_moduli.py"}
for tag, name in files.items():
    if tag not in blocks:
        raise SystemExit("missing source listing: " + tag)
    (dest / name).write_text(blocks[tag] + "\n")

def run(args):
    subprocess.run(args, cwd=dest, check=True)

run(["g++", "-O3", "-std=c++17", "certify.cpp",
     "-lgmpxx", "-lgmp", "-o", "certify"])
with (dest / "cert25703.json").open("w") as out:
    subprocess.run([str(dest / "certify"), "25703"],
                   cwd=dest, stdout=out, check=True)
import json
record = json.loads((dest / "cert25703.json").read_text())
assert record["N"] == 25703 and record["status"] == "PASS"
assert record["exact_checks"] == 43931660
assert record["max_k_exclusive"] == 2737
run([sys.executable, "certify_fourier_constants.py"])
run([sys.executable, "audit_independent.py"])
run(["g++", "-O3", "-std=c++17", "check_small_moduli.cpp",
     "-lgmpxx", "-lgmp", "-o", "check_small_moduli"])
with (dest / "small_moduli_65798.jsonl").open("w") as out:
    subprocess.run([str(dest / "check_small_moduli")],
                   cwd=dest, stdout=out, check=True)
```

```
records = [json.loads(line) for line in
           (dest / "small_moduli_65798.jsonl").read_text().splitlines()]
O = lambda m: list(range(1, m+1, 2))
expected = {2: [2,4,8,11,13,14,16,17,19,20,23,26,29,32],
            3: O(57), 4: [1,4,8,11,13,16], 5: O(41), 6: O(11),
            7: O(35), 8: [1,3,5,7,11,13,17,19], 9: O(33),
            10: O(19), 11: O(31), 12: O(21), 13: O(31),
            14: O(19)+[23], 23: O(25)+[29], 27: O(27),
            29: O(25)+[29]}
for k in (15,17,19,21): expected[k] = O(29)
for k in (16,18,20,22,24,25,26,28): expected[k] = O(25)
assert [z["k"] for z in records] == list(range(2,30))
for z in records:
    k = z["k"]
    assert z["N"] == 65798 and z["negative"] == expected[k]
    assert z["zero"] == ({2: [1], 4: [5,7]}.get(k, []))
run([sys.executable, "certify_all_k_constants.py"])
run([sys.executable, "audit_small_moduli.py"])
print("PASS: both finite certificates, all constants, both audits")
print("Reproduction records:", dest)
```

Part III

Involutions and Partial Log Concavity

The final chapter retains partial results on inversion-parity log concavity for involutions, including infinite edge strips, an eventual linear-window theorem, formal identities and exact obstructions. It does not resolve the full Dukes conjecture. Unproved candidates remain explicitly distinguished from theorems.

Chapter 8

Partial Results on Dukes Inversion Parity Log Concavity

Chapter overview

Let $a_{n,k}$ count involutions of S_n with k inversions. We establish $a_{n,k}^2 \geq a_{n,k-2}a_{n,k+2}$ for every $n \geq 4$ in both edge strips $2 \leq k \leq 100$ and $2 \leq \binom{n}{2} - k \leq 100$, with exactly four equality pairs within $2 \leq k \leq \binom{n}{2} - 2$. The lower strip uses stable coefficient polynomials and an exact nonnegative Newton-basis certificate; the upper strip uses a finite base and reversed convolution propagation. For every fixed $0 < a < b < \infty$ and positive integer h, we also prove strict h-step log concavity uniformly for $an \leq k \leq bn$ when n is sufficiently large. Its threshold is existential, not numerically explicit. We give conditional likelihood-ratio preservation, exact formal identities, and obstructions to several stronger induction claims. Bernstein and four-step strong-minor assertions are recorded only as unproved candidates. The full all-n, all-k Dukes conjecture is not resolved. A self-contained exact replay program for the two infinite edge strips is included in this single source file.

8.1 The problem and the scope of the results

For $n \geq 0$, set

$$A_n(q) = \sum_{\pi \in S_n,\, \pi^2 = 1} q^{\mathrm{inv}(\pi)} = \sum_{k \geq 0} a_{n,k} q^k, \qquad N_n = \binom{n}{2}.$$

All coefficient sequences are extended by zero outside their support. Write $[m]_t = 1 + t + \cdots + t^{m-1}$ for $m \geq 1$. The basic recurrence is

$$A_0 = A_1 = 1, \qquad A_n = A_{n-1} + q[n-1]_{q^2} A_{n-2} \quad (n \geq 2). \tag{8.1.1}$$

Indeed, an involution either fixes n, or pairs n with some $i < n$. Deleting the pair and standardizing leaves an involution on $n-2$ points; the removed entries contribute $2(n-i)-1$ inversions. Summing over i gives (8.1.1). This recurrence and the conjecture below appear in Dukes [8.1].

Dukes' inversion-parity conjecture asks for

$$D_{n,k} := a_{n,k}^2 - a_{n,k-2}a_{n,k+2} \geq 0 \qquad (n \geq 4,\ 2 \leq k \leq N_n - 2). \tag{8.1.2}$$

Equivalently, both parity sections in

$$A_n(q) = E_n(t) + qO_n(t), \qquad t = q^2, \tag{8.1.3}$$

are log concave. Here a nonnegative sequence v is log concave when $v_j^2 \geq v_{j-1}v_{j+1}$ for all integers j. Hart and Rowley [8.2], Conjecture 6.1(a), restate the corresponding type-A length-profile question. Their counterexample for a type-B unimodality question is not a counterexample to (8.1.2).

The principal conclusions of this paper are the following.

1. For fixed k, $a_{n,k}$ is a polynomial in n for every $n \geq k$, with degree k and leading coefficient $1/k!$.

2. For every $n \geq 4$, (8.1.2) holds whenever $2 \leq k \leq 100$ or $2 \leq N_n - k \leq 100$. In the original admissible domain, equality in either strip occurs exactly at

$$(n,k) = (4,2), (4,4), (6,3), (6,5). \tag{8.1.4}$$

3. For fixed $0 < a < b < \infty$ and fixed $h \geq 1$, there is an integer $n_0(a,b,h)$ such that

$$a_{n,k}^2 > a_{n,k-h}a_{n,k+h} \quad (n \geq n_0(a,b,h),\ an \leq k \leq bn).$$

No numerical value of n_0 is claimed.

The proofs of the infinite edge statements have finite exact computational parts. Appendix 8.A contains all code needed to regenerate them, without auxiliary files or third-party Python packages. The other proved statements are algebraic or analytic. No finite experiment in this paper is substituted for an unproved infinite assertion.

The compact-saddle method has established precedents, notably De Angelis [8.3], building on earlier work of Odlyzko and Richmond. The finite q-Hermite expansion used below is classical [8.4]. Neither the general method nor publication priority for this application is claimed here.

8.2 Stable coefficient polynomials

Theorem 8.2.1. *For each integer $k \geq 0$, there is a polynomial $P_k(X)$ of degree k and leading coefficient $1/k!$ such that*

$$a_{n,k} = P_k(n) \qquad (n \geq k).$$

More precisely, for nonnegative integers $c_{k,d,m}$ independent of n,

$$P_k(X) = \sum_{0 \leq d,m \leq k} c_{k,d,m} \binom{X-d}{m}. \tag{8.2.1}$$

For $k = 0$, the only term is $c_{0,0,0} = 1$.

Proof. The inversion graph of a permutation has one vertex for each position and one edge for each inversion. If an inversion edge joins $i < j$, every intermediate position t is adjacent to at least one endpoint: either $\pi(t) < \pi(i)$, or $\pi(t) > \pi(i) > \pi(j)$. Consequently each connected component is a position interval. Distinct components have no inversions between them, so their value intervals occur in the same order. Thus the components are precisely the unique direct-sum indecomposable blocks of the permutation.

For an involution each such position interval is also its value interval, so each block is itself an involution. An indecomposable block of length ℓ has a connected inversion graph and hence at least $\ell - 1$ inversions.

Delete all singleton blocks, and fix the resulting ordered list of m non-singleton blocks. Let its total length be L, its inversion count be k, and put $d = L - m$. The preceding bound gives $d \leq k$; since each non-singleton block has at least one inversion, also $m \leq k$. There are only finitely many such lists, because $L \leq k + m \leq 2k$.

For $n \geq L$, inserting the $n - L$ singleton blocks into the $m + 1$ gaps can be done in

$$\binom{n-L+m}{m} = \binom{n-d}{m}$$

ways. For $k \leq n < L$, the top entry satisfies $0 \leq n - d < m$, so the same ordinary binomial coefficient is zero, as required. Counting the ordered lists with given (k,d,m) proves (8.2.1) for every $n \geq k$, including the boundary $n = k$.

The degree is at most k. A degree-k contribution requires $m = k$; then each block has exactly one inversion and must be 21. There is exactly one such ordered list, giving $\binom{X-k}{k}$. All other terms have smaller degree. This proves the stated degree and leading coefficient. □

Corollary 8.2.2. *For every fixed $k \geq 2$, $D_{n,k} > 0$ for all sufficiently large n.*

Proof. For $n \geq k+2$, $D_{n,k}$ agrees with the degree-$2k$ polynomial $P_k(n)^2 - P_{k-2}(n)P_{k+2}(n)$. Its leading coefficient is

$$\frac{1}{(k!)^2} - \frac{1}{(k-2)!(k+2)!} = \frac{2(2k+1)}{(k+1)(k+2)(k!)^2} > 0. \tag{8.2.2}$$

□

This fixed-k conclusion alone gives no threshold uniform in growing k.

8.3 Exact infinite edge strips

8.3.1 The lower strip

Theorem 8.3.1. *For every integer $n \geq 4$ and $2 \leq k \leq 100$, $D_{n,k} \geq 0$. Within $2 \leq k \leq N_n - 2$, its only zeros in this strip are* (8.1.4).

Proof. For each $k = 2, \ldots, 100$, write the polynomial of Corollary 8.2.2 in Newton's basis at $n = k + 3$:

$$D_{k+3+x,k} = \sum_{j=0}^{2k} c_{k,j} \binom{x}{j}, \qquad c_{k,j} = \Delta^j D_{k+3,k}. \tag{8.3.1}$$

Here Δ acts on the first index. The degree bound proves the identity, rather than merely predicting it from extrapolation. The exact computation in Appendix 8.A establishes that all

$$\sum_{k=2}^{100} (2k+1) = 10\,197$$

integers $c_{k,j}$ are nonnegative. Hence (8.3.1) is nonnegative for every integer $x \geq 0$.

The finite remainder $4 \leq n \leq k+2$ is checked directly from (8.1.1). It comprises 4 950 checks including out-of-support centers, of which 4 279 are in the original admissible domain. Two independently coded recurrence evaluations agree, and three additional values beyond each interpolation interval agree with (8.3.1). These are implementation checks; the polynomiality proof justifies the infinite tail.

The computed $c_{k,0}$ is positive except for $k = 3$. In that case $c_{3,0} = 0$ and $c_{3,1} > 0$, so its only tail zero is $n = 6$. Enumeration of the finite remainder gives the other three pairs in (8.1.4). All equality claims are restricted to the original admissible domain. □

The replay program also verifies the exact top coefficient

$$c_{k,2k} = \binom{2k}{k} - \binom{2k}{k-2},$$

as follows independently from (8.2.2). It reproduces the archived lower-strip JSON certificate with SHA-256

```
ad3217b1c51edfdceb79419b85d2a21a412d
ba1e559f8f7cc2e2280e2a30c947.
```

The line break is typographical; the digest is one 64-character string.

8.3.2 Convolution and reversed prefixes

We use two elementary facts about nonnegative sequences. A sequence has *interval support* if its positive entries form an integer interval; the zero sequence is allowed when discussing prefixes.

Lemma 8.3.2. *Convolution preserves log concavity and interval support for nonnegative sequences. If v, h have these properties, are supported on nonnegative indices, and $h_0 = 1$, then for every $j \geq 0$,*

$$(v * h)_j^2 - (v * h)_{j-1}(v * h)_{j+1} \geq v_j^2 - v_{j-1}v_{j+1}. \tag{8.3.2}$$

Proof. For an interval-supported sequence v, adjacent log concavity implies that the ratios v_{j+1}/v_j decrease on its positive support. Multiplying these ratio inequalities shows that every 2×2 minor of the Toeplitz matrix $T(v)_{i,j} = v_{j-i}$ is nonnegative; minors involving zeros have the same sign by interval support. Conversely its consecutive row and column minors are the log-concavity inequalities. Thus log concavity with interval support is equivalent to Toeplitz total positivity of order two.

The convolution identity is $T(v * h) = T(v)T(h)$. Cauchy–Binet expresses each 2×2 minor of the product as a sum of nonnegative products of minors, proving log concavity. The support of a convolution is the sum of the two support intervals. For rows $0, 1$ and columns $j, j+1$, the intermediate pair $j, j+1$ contributes exactly

$$(v_j^2 - v_{j-1}v_{j+1}) \det \begin{pmatrix} h_0 & h_1 \\ h_{-1} & h_0 \end{pmatrix} = v_j^2 - v_{j-1}v_{j+1}.$$

All remaining terms are nonnegative, proving (8.3.2). For finitely many prescribed coefficients all matrices can be truncated to a sufficiently large finite index set, so no infinite summation issue occurs. □

Define the reversed polynomial

$$B_n(q) = q^{N_n} A_n(q^{-1}) = \sum_{d \geq 0} r_{n,d} q^d.$$

Reversal of (8.1.1) gives

$$B_n = q^{n-1} B_{n-1} + [n-1]_{q^2} B_{n-2}. \tag{8.3.3}$$

Indeed $N_n - N_{n-1} = n - 1$, $N_n - N_{n-2} = 2n - 3$, and the top degree of $q[n-1]_{q^2}$ is $2n - 3$.

Lemma 8.3.3. *Fix $D \geq 2$. Suppose the parity prefixes of B_n through degree $D + 2$ have interval support and are log concave for $4 \leq n \leq D + 3$. Then the same assertions hold for every $n \geq 4$.*

Proof. When $n > D + 3$, the first summand of (8.3.3) vanishes through degree $D + 2$. Each relevant parity prefix is therefore the corresponding prefix of the convolution of the parity prefix of B_{n-2} with an all-ones sequence of length $n - 1$. Lemma 8.3.2 applies; truncation preserves both properties. The two terminal base rows start the two parity classes of n. □

Theorem 8.3.4. *For every $n \geq 4$ and $2 \leq d \leq 100$,*

$$r_{n,d}^2 \geq r_{n,d-2} r_{n,d+2}.$$

Equivalently, (8.1.2) *holds for $2 \leq N_n - k \leq 100$. Its admissible equality pairs are exactly* (8.1.4)*.*

Proof. The exact base in Appendix 8.A checks all 9 900 inequalities with $4 \leq n \leq 103$, $2 \leq d \leq 100$, including out-of-support centers, and both complete parity-prefix support conditions through degree 102. The even prefix has interior centers $2, 4, \ldots, 100$; the odd prefix has interior centers

$3, 5, \ldots, 99$. Thus all required interior inequalities are present. Lemma 8.3.3 proves the infinite claim.

For strictness, (8.3.2) shows that after $n = 103$ the defect at deficiency d is at least that at the same deficiency in row $n - 2$. All tested defects in terminal rows $102, 103$ are strictly positive. The only admissible base zeros are

$$(n, d) = (4, 2), (4, 4), (6, 10), (6, 12).$$

Translating $k = N_n - d$ gives (8.1.4). □

Corollary 8.3.5. *If n is the least row in which (8.1.2) fails, every failing deficiency $d = N_n - k$ satisfies $d \geq n - 3$.*

Proof. For $d \leq n - 4$, the reversed recurrence through degree $d + 2$ is the convolution of row $n - 2$ with the uniform kernel. That row is log concave by minimality. Its parity sections have interval support: this holds in rows $2, 3$, and the recurrences

$$E_n = E_{n-1} + t[n-1]_t O_{n-2}, \qquad O_n = O_{n-1} + [n-1]_t E_{n-2}$$

preserve it. The odd summands start at degree zero; the even supports start at zero and one and meet. Reversal preserves interval support. Lemma 8.3.2 excludes a failure at such d. □

8.4 Strict log concavity on fixed linear windows

Theorem 8.4.1. *Fix real numbers $0 < a < b < \infty$ and an integer $h \geq 1$. There exists $n_0(a, b, h)$ such that*

$$a_{n,k}^2 > a_{n,k-h} a_{n,k+h} \qquad (n \geq n_0(a, b, h),\ an \leq k \leq bn). \tag{8.4.1}$$

The threshold is existential. In particular $h = 1$ gives ordinary log concavity and $h = 2$ gives Dukes' parity log concavity in these windows.

The proof includes a global arc bound, not only a local saddle expansion. The final determinant sign uses an exact Fourier factorization, avoiding subtraction of two leading asymptotic equivalents.

8.4.1 The exact expansion and its amplitude

For $|q| < 1$, put

$$w(q) = \frac{q}{1 - q^2}, \quad \Delta(q) = \sqrt{1 + 4w(q)}, \quad L(q) = \frac{1 + \Delta(q)}{2}, \quad M(q) = 1 - L(q),\ z(q) = \frac{M(q)}{L(q)}.$$

Near any compact subinterval of $(0, 1)$ the branches are fixed by $\Delta(r) > 0$. Then $L + M = 1$, $LM = -w$ and $|z(r)| < 1$. Use $(a; t)_j = \prod_{i=0}^{j-1}(1 - at^i)$ and $\begin{bmatrix} n \\ j \end{bmatrix}_t = (t; t)_n / ((t; t)_j (t; t)_{n-j})$.

Lemma 8.4.2. *On these neighborhoods,*

$$A_n(q) = \sum_{j=0}^{n} \begin{bmatrix} n \\ j \end{bmatrix}_{q^2} L(q)^{n-j} M(q)^j. \tag{8.4.2}$$

If $I \Subset (0, 1)$ is compact, there is a complex neighborhood of I on which, uniformly,

$$A_n(q) = L(q)^n (F(q) + O(\tau^n)), \qquad 0 < \tau < 1, \tag{8.4.3}$$

where

$$F(q) = \sum_{j \geq 0} \frac{z(q)^j}{(q^2; q^2)_j} = \frac{1}{(z(q); q^2)_\infty}, \qquad F(r) > 0 \quad (r \in I). \tag{8.4.4}$$

Proof. The classical Rogers–Szegő, equivalently finite q-Hermite, polynomial

$$R_n(X,Y;t)=\sum_j \begin{bmatrix} n \\ j \end{bmatrix}_t X^{n-j}Y^j$$

satisfies

$$R_n=(X+Y)R_{n-1}-XY(1-t^{n-1})R_{n-2}.$$

This follows coefficientwise from Gaussian Pascal recursion and

$$(1-t^{n-1})\begin{bmatrix} n-2 \\ j-1 \end{bmatrix}_t=(1-t^{n-j})\begin{bmatrix} n-1 \\ j-1 \end{bmatrix}_t.$$

Substitution $(X,Y,t)=(L,M,q^2)$ gives (8.1.1) with the correct initial values, proving (8.4.2). See also the classical finite expansion in [8.4, Eq. 18.28.16].

Shrink the neighborhood so that $|q|\le r_0<1$ and $|z(q)|\le z_0<1$. The positive number

$$p_0=\prod_{\ell\ge 1}(1-r_0^{2\ell})$$

bounds all denominator products away from zero. For $j\le n/2$,

$$\begin{bmatrix} n \\ j \end{bmatrix}_{q^2}=\frac{\prod_{\ell=n-j+1}^{n}(1-q^{2\ell})}{(q^2;q^2)_j}=\frac{1+O(r_0^n)}{(q^2;q^2)_j}$$

uniformly. All Gaussian binomials are uniformly bounded, by bounding their numerator products with $\prod_{\ell\ge1}(1+r_0^{2\ell})$ and their denominators with p_0. After division by L^n, the tails $j>n/2$ in both sums are $O(z_0^{n/2})$. Summing the first error against $|z|^j$ proves (8.4.3), for any $\tau>\max(r_0,\sqrt{z_0})$ below one.

For $S(z)=\sum_{j\ge0}z^j/(t;t)_j$, the identity $(1-z)S(z)=S(tz)$ gives the infinite product by iteration. All products converge uniformly and are nonzero. On the real interval, $z(r)<0$ and

$$F(r)=\prod_{i\ge0}(1+|z(r)|r^{2i})^{-1}>0.$$

In particular F has a positive minimum on I. □

On a slightly smaller complex neighborhood, Cauchy's estimates also bound every fixed derivative of the error in (8.4.3) by $C_m\tau^n$. Analytic derivatives of L,z,F are uniformly bounded there.

8.4.2 A compact saddle interval

Set

$$\nu(r)=r\frac{L'(r)}{L(r)},\qquad V(r)=r\nu'(r).$$

Differentiating $L^2-L=w$ yields

$$\nu(r)=\frac{1+r^2}{2(1-r^2)}\left(1-\frac{1}{\Delta(r)}\right). \tag{8.4.5}$$

Both positive factors are strictly increasing on $(0,1)$, with positive derivatives. Thus $V(r)>0$, while $\nu(r)\to0$ as $r\to0$ and $\nu(r)\to\infty$ as $r\to1$. For $\alpha=k/n\in[a,b]$, the unique saddle $r=\nu^{-1}(\alpha)$ lies in the compact interval

$$I=[\nu^{-1}(a),\nu^{-1}(b)]\Subset(0,1).$$

On I, V is bounded above and below by positive constants.

With $s = \log r$ and $\phi(s) = \log L(e^s)$, one has $\phi' = \nu$ and $\phi'' = V$. Uniform Taylor expansion in a common complex neighborhood gives a fixed $\delta > 0$ and $c > 0$ such that

$$\Re\{\phi(s + i\theta) - \phi(s) - i\nu(r)\theta\} \le -c\theta^2 \qquad (|\theta| \le \delta,\ r \in I). \tag{8.4.6}$$

For bounded real x, uniformly in $r \in I$,

$$n\left\{\phi\left(s + \frac{ix}{\sqrt{nV(r)}}\right) - \phi(s) - \frac{i\nu(r)x}{\sqrt{nV(r)}}\right\} \longrightarrow -\frac{x^2}{2}. \tag{8.4.7}$$

8.4.3 The global complementary arc

Lemma 8.4.3. *For every fixed $\delta > 0$, there are $C_\delta < \infty$ and $\eta_\delta < 1$ such that*

$$|A_n(re^{i\theta})| \le C_\delta L(r)^n \eta_\delta^n \quad (r \in I,\ \delta \le |\theta| \le \pi). \tag{8.4.8}$$

Proof. For $q = re^{i\theta}$, the limiting transfer matrix is

$$T(q) = \begin{pmatrix} 1 & w(q) \\ 1 & 0 \end{pmatrix}.$$

Each eigenvalue ξ satisfies $\xi^2 = \xi + w(q)$, hence

$$|\xi|^2 \le |\xi| + |w(q)| \le |\xi| + w(r),$$

so $|\xi| \le L(r)$. Equality in $|w(q)| \le w(r)$ is possible only at $q = r$ or $q = -r$. At $q = -r$, equality in the first triangle inequality would force ξ to be negative real, whereas $\xi^2 = \xi - w(r) < 0$ would be impossible. Thus the limiting spectral radius is strictly below $L(r)$ on the complementary arc, including its negative-real point.

To pass from limiting spectra to finite products, normalize the state:

$$Y_n = \begin{pmatrix} A_n(q)/L(r)^n \\ A_{n-1}(q)/L(r)^{n-1} \end{pmatrix}, \qquad Y_n = N_n(q, r)Y_{n-1},$$

$$N_n(q, r) = \begin{pmatrix} L(r)^{-1} & q[n-1]_{q^2} L(r)^{-2} \\ 1 & 0 \end{pmatrix}.$$

Uniformly on the compact arc set, these matrices converge geometrically to $N_\infty(q, r)$. The eigenvalues of N_∞ are $\xi/L(r)$; compactness bounds their moduli by a common $\rho < 1$.

There is a uniform block length ℓ with $\|N_\infty^\ell\| < 1/2$ on the whole compact set. Indeed, Cayley–Hamilton gives a uniform bound $C(\ell + 1)\rho^{\ell-1}$ for these powers. It remains valid at repeated eigenvalues: the divided difference

$$\frac{\lambda_1^\ell - \lambda_2^\ell}{\lambda_1 - \lambda_2} = \sum_{j=0}^{\ell-1} \lambda_1^{\ell-1-j}\lambda_2^j$$

has modulus at most $\ell\rho^{\ell-1}$, including its limiting value. Because ℓ is fixed and $N_n \to N_\infty$ uniformly, every sufficiently late product of ℓ consecutive actual matrices has norm below $3/4$. The finitely many early factors and the leftover factors are uniformly bounded. Multiplying blocks proves (8.4.8). This uses convergence of the matrices, not an invalid inference about arbitrary products from spectral radii alone. □

8.4.4 Weighted Fourier positivity and the determinant

At the saddle for k/n, put

$$\chi(\theta) = \frac{A_n(re^{i\theta})}{F(r)L(r)^n} e^{-ik\theta}.$$

It obeys $\chi(-\theta) = \overline{\chi(\theta)}$. Equations (8.4.3), (8.4.6), and (8.4.7) give Gaussian convergence under $\theta = x/\sqrt{nV(r)}$, uniformly on bounded x and in the prescribed window. On the small arc,

$$|\chi(\theta)| \le Ce^{-cn\theta^2} + C\tau^n. \tag{8.4.9}$$

Lemma 8.4.3 gives exponential smallness on the complementary arc. Define

$$U_h = \int_{-\pi}^{\pi} (1 - \cos(h\theta)) \Re\chi(\theta)\, d\theta,$$
$$W_h = \int_{-\pi}^{\pi} (1 + \cos(h\theta)) \Re\chi(\theta)\, d\theta, \qquad G_0 = \sqrt{\frac{2\pi}{nV(r)}}.$$

Then, uniformly in the window,

$$U_h = \frac{h^2 G_0}{2nV(r)}(1 + o(1)), \qquad W_h = 2G_0(1 + o(1)). \tag{8.4.10}$$

For completeness, in the first assertion use $1 - \cos(h\theta) \le h^2\theta^2/2$ after Gaussian rescaling. Its normalized integrand is dominated by $Cx^2e^{-c'x^2}$. First truncate to a common large bounded x interval, then use uniform Gaussian convergence there. The common Gaussian tail can be made arbitrarily small. The exponential errors in (8.4.9) and (8.4.8) still vanish after the $n^{3/2}$ normalization. The same argument without the quadratic weight proves the assertion for W_h. Both quantities are therefore strictly positive for all sufficiently large n, uniformly in k/n.

Let

$$J_j = \int_{-\pi}^{\pi} \chi(\theta) e^{-ij\theta}\, d\theta, \quad C_h = \int_{-\pi}^{\pi} \chi(\theta)\cos(h\theta)\, d\theta, \quad is_h = \int_{-\pi}^{\pi} \chi(\theta)\sin(h\theta)\, d\theta.$$

Conjugate symmetry makes J_j, C_h, s_h real. Since $J_h = C_h + s_h$ and $J_{-h} = C_h - s_h$, the exact identity is

$$J_0^2 - J_{-h}J_h = (J_0 - C_h)(J_0 + C_h) + s_h^2 = U_hW_h + s_h^2 > 0. \tag{8.4.11}$$

Finally Cauchy's coefficient formula gives

$$J_j = \frac{2\pi r^{k+j} a_{n,k+j}}{F(r)L(r)^n}.$$

All normalization factors are positive, and the powers of r cancel in the neighboring product. This proves Theorem 8.4.1. For sufficiently large n, fixed a, b, h also ensure $k - h \ge 0$ and $k + h \le N_n$.

The compactness is essential. No estimates above are uniform as the saddle approaches 1 or 0. In particular the theorem does not cover the quadratic regime $k = \Theta(n^2)$, arbitrary $k/n \to \infty$, or a transition with $k/n \to 0$ and $k \to \infty$.

8.5 Conditional likelihood ratio preservation

This section records a proved auxiliary theorem, not an induction proof of log concavity. Fix a scalar $c > 0$, and put $U_r(t) = [r]_t$. Define

$$\begin{aligned} E_n &= E_{n-1} + ctU_{n-1}O_{n-2}, \\ O_n &= O_{n-1} + U_{n-1}E_{n-2}, \end{aligned} \qquad E_0 = E_1 = 1, \quad O_0 = O_1 = 0. \tag{8.5.1}$$

At $c = 1$ these are (8.1.3). In this section c is numerical, not an indeterminate in a coefficientwise ordered ring.

For nonzero nonnegative interval-supported sequences, write $a \leq_{\mathrm{LR}} b$ when

$$a_i b_j - a_j b_i \geq 0 \qquad \text{for every } i < j. \tag{8.5.2}$$

This is the full likelihood-ratio order. Adjacent minors alone do not suffice on arbitrary separated supports. With ordered support endpoints, adjacent ratio comparisons on the positive overlap do imply (8.5.2): multiplication gives the distant comparisons, and the zeros outside the overlap have the required signs.

We use the following elementary consequences of (8.5.2). The order is transitive and is unchanged by positive rescaling. Convolution by a log-concave interval-supported kernel preserves it: apply Cauchy–Binet to the two-row matrix with rows a, b and the Toeplitz kernel. Thus if $a \leq_{\mathrm{LR}} b$, $h \leq_{\mathrm{LR}} k$ and all four sequences are log concave with interval support, then $a*h \leq_{\mathrm{LR}} b*k$, by two successive convolutions and transitivity. If each $a_i \leq_{\mathrm{LR}} b_j$, then their positively weighted sums have the same comparison, by bilinearity of each minor. Also $U_r \leq_{\mathrm{LR}} U_s$ for $r \leq s$ and $\delta_0 \leq_{\mathrm{LR}} tU_rU_s$, so $a \leq_{\mathrm{LR}} tU_rU_sa$ for log-concave a. All nonzero rows in (8.5.1) have interval support, by the same support induction used after Theorem 8.3.4.

Define the auxiliary assertions

$$\begin{aligned} T_n &: \quad E_{n-1} \leq_{\mathrm{LR}} E_n, \quad O_{n-1} \leq_{\mathrm{LR}} O_n, \\ S_n &: \quad E_n \leq_{\mathrm{LR}} tU_{n+1}O_n, \quad O_n \leq_{\mathrm{LR}} U_{n+1}E_n, \\ R_m &: \quad E_m \leq_{\mathrm{LR}} tU_mU_{m+2}E_{m-1}, \quad O_m \leq_{\mathrm{LR}} tU_mU_{m+2}O_{m-1}. \end{aligned}$$

Lemma 8.5.1. *For $m \geq 4$, S_{m-2}, T_{m-1}, and log concavity of rows $m-2, m-1$ imply R_m. No log concavity of row m is needed.*

Proof. In $E_m = E_{m-1} + ctU_{m-1}O_{m-2}$, the first summand is LR-below $tU_mU_{m+2}E_{m-1}$ by the preceding facts. For the second,

$$tU_{m-1}O_{m-2} \leq_{\mathrm{LR}} tU_{m-1}^2E_{m-2} \leq_{\mathrm{LR}} tU_mU_{m+2}E_{m-1}.$$

The first comparison uses S_{m-2}; the second uses T_{m-1} and enlargement of the two uniform kernels. Positive rescaling and summation give the result. For the odd equation the corresponding comparison is

$$U_{m-1}E_{m-2} \leq_{\mathrm{LR}} tU_{m-1}^2O_{m-2} \leq_{\mathrm{LR}} tU_mU_{m+2}O_{m-1},$$

and the first summand is handled identically. □

Lemma 8.5.2. *For $m \geq 4$, assume S_{m-1}, S_m, T_m, R_m and log concavity of rows $m-1, m$. Then S_{m+1} holds.*

Proof. Write $U = U_m$, $V = U_{m+2}$. Compare

$$E_{m+1} = E_m + ctUO_{m-1}, \qquad tVO_{m+1} = tVO_m + tVUE_{m-1}.$$

Each left summand is LR-below each right summand. For E_m, the comparisons follow from S_m with kernel enlargement, and R_m, respectively. For $ctUO_{m-1}$, they follow from T_m and $U \leq_{\mathrm{LR}} V$, and from S_{m-1} enlarged to V and convolved by tU. Summation proves the even assertion.

For the odd assertion compare

$$O_{m+1} = O_m + UE_{m-1}, \qquad VE_{m+1} = VE_m + ctVUO_{m-1}.$$

The same four comparisons follow, respectively, from S_m, R_m, T_m, and S_{m-1} with kernel enlargement. The scalar c does not affect LR order. □

Lemma 8.5.3. *If S_m, T_m hold and rows $m-1, m$ are log concave, then T_{m+2} holds.*

Proof. In

$$E_{m+1} = E_m + ctU_mO_{m-1}, \qquad E_{m+2} = E_{m+1} + ctU_{m+1}O_m,$$

both summands of E_{m+1} are LR-below $ctU_{m+1}O_m$, by S_m and by T_m with kernel enlargement. Thus $E_{m+1} \leq_{\mathrm{LR}} E_{m+2}$. The odd argument is identical. □

Theorem 8.5.4. *Fix $c > 0$ and $N \geq 4$. If E_j, O_j are log concave for $2 \leq j \leq N-1$, then S_j, T_j hold for $3 \leq j \leq N$.*

Proof. The base rows are

$$E_2 = E_3 = 1, \quad O_2 = 1, \quad O_3 = 2 + t, \qquad E_4 = 1 + c(t + t^2 + t^3), \quad O_4 = 3 + 2t + t^2.$$

The nonzero adjacent base minors are listed below; omitted minors are zero. All base pairs have ordered support endpoints, so these checks establish their full LR orders.

Assertion	Even comparison	Odd comparison
T_3	none	1
T_4	c	$1, 1$
S_2	1	1
S_3	2	$1, 1$
S_4	$3, 2c, c, 6c$	$1+3c, 1+3c, 1+3c$

For example, the supports in the even S_4 comparison are $[0, 3]$ and $[1, 7]$. The other comparisons either have a point mass on the left, or begin together at zero with ordered right endpoints.

For $n \geq 5$, T_n follows from Lemma 8.5.3 at $m = n-2$. Lemma 8.5.1 gives R_{n-1} from S_{n-3}, T_{n-2} and log concavity through row $n-2$. Lemma 8.5.2 then gives S_n from earlier auxiliary assertions and log concavity through row $n-1$. Neither conclusion uses log concavity of row n. □

Thus the auxiliary comparisons are available one row ahead under the log-concavity induction hypothesis. They do not themselves imply log concavity of that new row. For $c < 1$, even row 4 need not be log concave, so the premise must not be suppressed. Nor does the proof automatically give coefficientwise statements in $\mathbb{R}[u]$: adjacent coefficientwise Turán inequalities do not imply all Toeplitz minors by a numerical ratio argument.

8.6 Exact obstructions to stronger induction claims

For sequences a, b, put

$$D_j(a) = a_j^2 - a_{j-1}a_{j+1}, \qquad M_j(a, b) = 2a_jb_j - a_{j-1}b_{j+1} - a_{j+1}b_{j-1}.$$

Then

$$D_j(a + cb) = D_j(a) + cM_j(a, b) + c^2D_j(b). \tag{8.6.1}$$

The following negative quantities refute stronger proposed properties, not the original conjecture.

8.6.1 Actual recurrence summands need not be synchronized

At $n = 6$, the equation $O_6 = O_5 + [5]_t E_4$ has summand triples $(4, 3, 2)$ and $(1, 2, 3)$ at index $j = 1$. Their mixed term is

$$2 \cdot 3 \cdot 2 - 4 \cdot 3 - 2 \cdot 1 = -2.$$

The two self-defects are each 1, so the total is exactly zero. A proof must account for this cancellation.

There are also infinitely many endpoint failures of same-row mixed nonnegativity. For odd $n \geq 3$, (8.3.3), starting with $B_3 = 1 + 2q^2 + q^3$, gives

$$[q^0]B_n = 1, \qquad [q^1]B_n = 0, \qquad [q^3]B_n = 1.$$

If $n \equiv 1 \pmod 4$, N_n is even and $M_{N_n/2-1}(E_n, O_n) = -1$. If $n \equiv 3 \pmod 4$, N_n is odd and $M_{(N_n-1)/2}(E_n, tO_n) = -1$. These values follow by inserting the displayed terminal coefficients and the zero coefficients beyond the support. Real-rootedness is also unavailable: $E_4 = 1 + t + t^2 + t^3$ has roots $i, -i$.

8.6.2 The available auxiliary cone does not preserve log concavity

Take abstract inputs at $m = 4$, $c = 1$,

$$E_{m-1} = E_m = 1, \qquad O_{m-1} = O_m = 1 + t = U_2.$$

They are log concave with interval support. The row comparison T_m holds by equality. The comparisons S_{m-1}, S_m hold because a point mass at zero is LR-below the shifted convolutions and $U_2 \leq_{\mathrm{LR}} U_r$ for $r \geq 2$. Both bridge comparisons R_m follow by convolution with tU_mU_{m+2}. Nevertheless the update gives

$$E_{m+1} = 1 + tU_4U_2 = 1 + t + 2t^2 + 2t^3 + 2t^4 + t^5, \qquad D_1(E_{m+1}) = -1.$$

In (8.6.1), $a = 1$, $b = tU_4U_2$ have $D_1(a) = 0$, $M_1(a, b) = -2$, $D_1(b) = 1$. These are not rows of the actual recurrence from its prescribed initial conditions. They prove that the presently available abstract cone needs additional recurrence-history information or a quantitative mixed-curvature bound.

8.6.3 Refinement does not remove every obstruction

Let $F_{n,m}(t)$ be defined by

$$Z_n(x, q) := \sum_{\pi^2=1} x^{\mathrm{fix}(\pi)} q^{\mathrm{inv}(\pi)} = \sum_m x^{n-2m} q^m F_{n,m}(q^2). \tag{8.6.2}$$

Then

$$F_{n,m} = F_{n-1,m} + [n-1]_t F_{n-2,m-1}.$$

At $(n, m) = (5, 2)$ the summands are $[3]_t$ and $[4]_t(2 + t)$; their mixed defect at index 3 is -1. Writing $F_{n,m,s} = [t^s]F_{n,m}(t)$, another exact minor is

$$F_{6,2,4}F_{5,1,4} - F_{5,2,4}F_{6,1,4} = 7 \cdot 0 - 1 \cdot 1 = -1.$$

Thus joint MTP2 in row size and number of transpositions is false, so a general MTP2 marginalization argument cannot be invoked.

For the scalar recurrence (8.5.1), the relevant odd coefficient pairs are

$$([t^5]O_7, [t^6]O_7) = (1 + 18c, 17c), \qquad ([t^5]O_6, [t^6]O_6) = (3c, 2c).$$

Their row-LR minor is $15c^2 - 2c$, negative for $0 < c < 2/15$. This does not contradict Theorem 8.5.4, whose earlier log-concavity premise already fails there, or the original case $c = 1$.

8.7 Formal identities and their limitations

8.7.1 A positive diagonal convolution identity

Let P_k be from Theorem 8.2.1, and form

$$S(X,q)=\sum_{k\geq 0}P_k(X)q^k.$$

For each coefficient, (8.1.1) holds for all sufficiently large integers X, hence as a polynomial identity:

$$S(X,q)=S(X-1,q)+\frac{q}{1-q^2}S(X-2,q).$$

Let $\lambda(q)\in\mathbb{R}[[q]]$ be the unique series with constant term one satisfying $\lambda^2-\lambda-q/(1-q^2)=0$. Then

$$S(X,q)=C(q)\lambda(q)^X,\qquad C(q)=S(0,q),\tag{8.7.1}$$

where $\lambda^X=\exp(X\log\lambda)$ has polynomial coefficients in X. To prove the identity, the first coefficient at which the two sides could differ would be a polynomial $R(X)$ satisfying $R(X)=R(X-1)$ and $R(0)=0$; it is therefore zero. Induction on the coefficient index proves (8.7.1).

Set $u=q/\lambda(q)$, let $q=Q(u)$ be its formal inverse, and put $T(u)=\lambda(Q(u))=Q(u)/u$. Define

$$\mathcal{B}_r(u)=\sum_{k\geq 0}a_{k+r,k}u^k\qquad(r\geq 0).$$

Formal residue substitution in (8.7.1) gives

$$\begin{aligned}P_k(k+r)&=[q^k]C(q)\lambda(q)^{k+r}\\&=[u^k]C(Q(u))T(u)^r\frac{uQ'(u)}{Q(u)}.\end{aligned}$$

Taking $r=0$ identifies the common prefactor, so

$$\mathcal{B}_r=\mathcal{B}_0T^r,\qquad T=1+\frac{u}{1-u^2T^2},\qquad Q=u+\frac{u^2}{1-Q^2}.\tag{8.7.2}$$

The equation $T=1+\sum_{j\geq 0}u^{2j+1}T^{2j}$ recursively determines a unique solution with nonnegative coefficients. Its first terms are

$$T=1+u+u^3+2u^4+2u^5+6u^6+13u^7+22u^8+54u^9+\cdots,$$

while

$$\mathcal{B}_0=1+u^3+u^4+2u^5+10u^6+17u^7+39u^8+119u^9+\cdots.$$

The generator T is not log concave: its u^2 coefficient is zero between two positive coefficients. Ordinary convolution closure therefore does not prove Dukes' inequalities from (8.7.2).

For completeness the associated forest kernel has the exact formula, for $r>0$, $n\geq r$, and $k=n-r$,

$$[u^n]Q(u)^r=\mathbf{1}_{k=0}+\sum_{\substack{1\leq b\leq\min(k,\lfloor n/2\rfloor)\\ b\equiv k\pmod 2}}\frac{r}{n-b}\binom{n-b}{b}\binom{(k+b)/2-1}{(k-b)/2}.\tag{8.7.3}$$

To derive it, first solve $W=z(a+b/(1-W^2))$ with independent type weights. Lagrange inversion at m vertices contributes r/m; choosing b vertices of the second type contributes $\binom{m}{b}$, and the

even total outdegree $2s = m - r$ contributes $\binom{b+s-1}{s}$. Give the two types weights u, u^2. Then $m = n - b$ and $s = (k - b)/2$, which gives (8.7.3); the all-first-type case is $k = 0$. Equivalently,

$$a_{n,k} = [u^n]\mathcal{B}_0(u)Q(u)^{n-k} \qquad (n \geq k).$$

The actual boundary factor matters. For the artificial factor $1 + 16u^3$, the coefficients $[u^6](1 + 16u^3)Q^r$ at $r = 2, 4, 6$ are $(38, 6, 1)$, giving $6^2 - 38 = -2$. This refutes arbitrary-boundary closure, not the result for the genuine $\mathcal{B}_0$, whose u^3 coefficient is 1.

8.7.2 Hard dimers and a continuant determinant

The recurrence for (8.6.2) is $Z_n = xZ_{n-1} + q[n-1]_{q^2}Z_{n-2}$, with $Z_0 = 1$, $Z_1 = x$. Classify tilings of a path with n vertices by their final tile, with monomer weight x and dimer weight $q[i]_t$ on edge i. The same recurrence and bases prove

$$F_{n,m}(t) = \sum_{\substack{S \subseteq \{1,\dots,n-1\} \\ |S|=m, S \text{ has no consecutive elements}}} \prod_{i \in S} [i]_t. \tag{8.7.4}$$

For general edge weights w_i, let $K_{a,b}$ be this path partition polynomial on vertices $a, \dots, b$, with an empty interval assigned value one. It is also the tridiagonal determinant with diagonal x, superdiagonal 1, and subdiagonal $-w_i$. Then, for $n \geq 2$,

$$K_{1,n}K_{2,n-1} - K_{1,n-1}K_{2,n} = (-1)^n \prod_{i=1}^{n-1} w_i. \tag{8.7.5}$$

Indeed the left side is w_1 at $n = 2$. Expanding both long intervals at the last vertex makes it $-w_{n-1}$ times the preceding left side, proving the formula by induction. Its shifted spatial intervals and alternating sign differ from the coefficient shifts in (8.1.2). An interval-uniform log-concavity claim is false: replacing the edge sequence by $[a]_t, [a+1]_t, \dots$ with $a \geq 2$ gives the even four-vertex polynomial $1 + t[a]_t[a+2]_t$, whose first coefficients are $(1, 1, 2)$.

8.8 Bernstein and four step candidates remain unproved

The definitions in this section are algebraic; the infinite positivity assertions are conjectural. For $p \in \{0, 1\}$ put $J = \lfloor (n - 2p)/4 \rfloor$. For a nonzero parity class define

$$H_{n,p}(t, u) = \sum_{j=0}^{J} t^j u^{J-j} (1 + u)^j F_{n,p+2j}(t). \tag{8.8.1}$$

Set $H_{0,1} = H_{1,1} = 0$. The coefficient of u^J is the original parity polynomial. Equivalently this is the Bernstein homogenization of the fixed-point variable x^4 under $x^4 = u/(1 + u)$. The transformed recurrence is

$$\begin{aligned} H_{n,0} &= c_{n,0}H_{n-1,0} + t(1 + u)[n - 1]_t H_{n-2,1}, \\ H_{n,1} &= c_{n,1}H_{n-1,1} + [n - 1]_t H_{n-2,0}, \end{aligned} \tag{8.8.2}$$

where $c_{n,p} = u$ if $n \equiv 2p \pmod 4$, and $c_{n,p} = 1$ otherwise. To check this, the fixed-point term gains a factor x^4 exactly when the residue of $n - 2p$ wraps from 3 to 0; homogenization turns that factor into u. Only the even transposition term increases J by one, giving its extra factor $1 + u$ and its factor t.

Write $h_{n,p,k}(u) = [t^k]H_{n,p}(t, u)$. The candidate assertion is

$$h_{n,p,k}^2 - h_{n,p,k-1}h_{n,p,k+1} \in \mathbb{N}[u]. \tag{8.8.3}$$

A proof would imply (8.1.2) by taking the leading u^{2J} coefficient. The completed exact finite screen through $n = 120$ found no negative coefficient in 287 567 Turán polynomials, comprising 13 522 422 integer coefficients. This is not an infinite certificate. The equivalent clean recurrence is (8.5.1) with $c = 1 + u$, after coefficient reversal with degree padding.

8.8.1 The exact four step identity

For $n \geq 5$, put $w_i = [n - i]_t$ for $1 \leq i \leq 4$ and $r \equiv n - 2p \pmod 4$, $0 \leq r < 4$. Define

$$\beta_0 = t(1 + u), \quad \beta_1 = 1, \qquad \alpha_r = \begin{cases} u & r = 0, 1, \\ 1 & r = 2, 3, \end{cases}$$

$$\gamma_r = \begin{cases} u & r = 0, \\ 1 & r = 1, 2, 3, \end{cases} \qquad \delta_r = \begin{cases} u & r = 0, 1, 2, \\ 1 & r = 3. \end{cases}$$

Then the following is an exact identity:

$$\begin{aligned} H_{n,p} = {} & (u + t(1 + u)w_1w_3)H_{n-4,p} \\ & + \beta_p\alpha_r(w_1 + w_2 + w_3)H_{n-4,1-p} \\ & + t(1 + u)\gamma_r w_4(w_1 + w_2)H_{n-5,p} \\ & + \beta_p\delta_r w_4 H_{n-5,1-p}. \end{aligned} \tag{8.8.4}$$

To verify it, expand the untransformed recurrence four times:

$$\begin{aligned} Z_n = {} & (x^4 + x^2q(w_1 + w_2 + w_3) + q^2w_1w_3)Z_{n-4} \\ & + (x^3qw_4 + xq^2w_4(w_1 + w_2))Z_{n-5}. \end{aligned}$$

Extract parity and apply (8.8.1); the four residues give exactly the displayed $\alpha_r, \gamma_r, \delta_r$. Although (8.8.4) proves $H_{n,p} - uH_{n-4,p} \in \mathbb{N}[t, u]$, its same-parity multiplier has first three t coefficients $(u, 1 + u, 2 + 2u)$. Their Turán difference is $1 - u^2$, not coefficientwise nonnegative. Entrywise positivity of the four-step transfer therefore does not close the log-concavity argument.

8.8.2 A sufficient strong minor invariant

Define

$$S_{n,p;i,j} = h_{n,p,i}h_{n,p,j} - h_{n,p,i-1}h_{n,p,j+1} \qquad (0 \leq i \leq j).$$

The proposed invariant is

$$S_{n,p;i,j} - u^2 S_{n-4,p;i,j} \in \mathbb{N}[u] \qquad (n \geq 4). \tag{FS}$$

This assertion is unproved. If true, it would imply all strong minors are coefficientwise nonnegative: the base nonzero polynomials for $n \leq 3$ are 1 and $2 + t$, and iteration at the same i, j gives, with $m = \lfloor n/4 \rfloor$,

$$S_{n,p;i,j} = \sum_{\ell=0}^{m-1} u^{2\ell}(S_{n-4\ell,p;i,j} - u^2 S_{n-4\ell-4,p;i,j}) + u^{2m} S_{n \bmod 4,p;i,j}.$$

Taking $i = j$ and then the leading u coefficient would prove Dukes' conjecture. Strong minors are specified because Cauchy–Binet preserves them coefficientwise; numerical adjacent-ratio reasoning is not a substitute over a polynomial ring.

The completed screen through $n = 30$ checked 306 631 strong-minor increments and 4 440 082 integer coefficients, finding none negative. It is only finite evidence for (FS). The analogous two-step assertion is false: at $(n, p, i, j) = (6, 1, 1, 1)$ the aligned increment is $1 - u^2$. No sign-reversing involution, coefficientwise positive expansion, or invariant induction establishing (FS) is supplied.

8.9 What remains unresolved

The two certified strips and each eventual fixed linear window are infinite statements, but they do not cover all of the bulk $100 < k < N_n - 100$. The analytic estimates are compact-saddle estimates; no numerical threshold or moving-saddle extension is inferred from them.

For the recurrence summands the exact unresolved sign is

$$D_j(a) + cM_j(a, b) + c^2 D_j(b) \geq 0.$$

Prior log concavity controls the two self-defects, and Section 8.5 supplies LR and smoothing comparisons one row ahead. Section 8.6 shows why these properties do not bound the negative mixed defect sufficiently. The exact identities in Sections 8.7 and 8.8 do not supply the missing cancellation. Every counterexample reported here concerns a stronger auxiliary property or artificial input, not an actual counterexample to (8.1.2).

Accordingly, this manuscript records partial results rather than a solution of the full conjecture. No claim of an exhaustive literature status determination or priority is made.

8.A Self contained exact edge certificate replay

The following Python 3 program uses only the standard library and is embedded verbatim in this single `.tex` file. It contains no third-party imports, data inputs, or certificate dependencies. Copy the listing to `dukes_edge_replay.py`, or extract it from the source with

```
python3 - <<'PY'
from pathlib import Path
s = Path('KEIO_Dukes_Partial_Results_2026.tex').read_text()
a = s.index('\n% BEGIN EXACT REPLAY\n')
a = s.index('\n', s.index(chr(92)+'begin{lstlisting}', a))+1
b = s.index(chr(92)+'end{lstlisting}', a)
Path('dukes_edge_replay.py').write_text(s[a:b])
PY
python3 dukes_edge_replay.py
```

All checks use explicit exceptions, so they remain enabled under `python3 -O`. The program writes `low_inversion_certificate_100.json` in its current working directory and checks the archived SHA-256 exactly. Its independently computed reversed prefixes also verify the upper-strip base and the strict terminal rows. The mathematical polynomiality and propagation proofs above, not the finite computations alone, justify all n.

```
import hashlib
import json
import math
from pathlib import Path

LIMIT = 100
DIGEST = ('ad3217b1c51edfdceb79419b85d2a21a412d'
          'ba1e559f8f7cc2e2280e2a30c947')
ZEROS = {(4, 2), (4, 4), (6, 3), (6, 5)}


def require(ok, message):
    if not ok:
        raise RuntimeError(message)


def forward(N, K, direct=False):
    rows = [[1] + [0]*K, [1] + [0]*K]
    for n in range(2, N+1):
        a, b = rows[-1], rows[-2]
        if direct:
            row = [a[k] + sum(b[k-1-2*j] for j in
                   range(min(n-2, (k-1)//2)+1))
                   for k in range(K+1)]
        else:
            row, sums = a.copy(), [0, 0]
            for k in range(K+1):
                if k >= 1:
                    sums[k % 2] += b[k-1]
                if k >= 2*n-1:
                    sums[k % 2] -= b[k-(2*n-1)]
                row[k] += sums[k % 2]
        rows.append(row)
    return rows


def newton(values):
    out = []
    while values:
```

```
        out.append(values[0])
        values = [b-a for a, b in zip(values, values[1:])]
    return out


def low_certificate():
    K = LIMIT
    Nmax = 3*K+3
    rows = forward(Nmax+3, K+2)
    require(rows == forward(Nmax+3, K+2, True),
            'Independent forward recurrences disagree')
    certificates = []
    small_checks = finite_all = newton_total = 0
    zeros = set()
    for k in range(2, K+1):
        N = k+3
        def D(n):
            return rows[n][k]**2-rows[n][k-2]*rows[n][k+2]
        values = [D(n) for n in range(N, N+2*k+1)]
        c = newton(values)
        require(len(c) == 2*k+1 and min(c) >= 0,
                f'Newton positivity failure at k={k}')
        require(c[-1] == math.comb(2*k, k)
                - math.comb(2*k, k-2), 'Leading coefficient')
        require((c[0] == 0) == (k == 3), 'Tail equality')
        if k == 3:
            require(c[1] > 0, 'Tail strictness')
            zeros.add((N, k))
        for x in range(2*k+1, 2*k+4):
            require(sum(v*math.comb(x, j)
                    for j, v in enumerate(c)) == D(N+x),
                    'Polynomial replay beyond interpolation')
        for n in range(4, N):
            require(D(n) >= 0, f'Finite lower base: {n},{k}')
            finite_all += 1
            if 2 <= k <= n*(n-1)//2-2:
                small_checks += 1
                if D(n) == 0:
                    zeros.add((n, k))
        newton_total += len(c)
        certificates.append({
            'k': k, 'base_n': N, 'degree_bound': 2*k,
            'newton_coefficients': c})
    require((newton_total, finite_all, small_checks)
            == (10197, 4950, 4279), 'Lower counts')
    require(zeros == ZEROS, 'Lower equality classification')
    artifact = {
        'method': 'exact nonnegative Newton-basis polynomial certificates',
        'claim': f'For every n>=4 and 2<=k<={K}, a[n,k]^2 >= a[n,k-2]*a[n,k+2].',
        'status': 'infinite n, bounded k; NOT the full Dukes conjecture',
        'maximum_k': K,
        'newton_coefficients': newton_total,
        'finite_remaining_in_domain_checks': small_checks,
        'two_recurrence_implementations_agree_through_n': Nmax+3,
        'certificates': certificates}
    data = (json.dumps(artifact, indent=2)+'\n').encode()
    digest = hashlib.sha256(data).hexdigest()
    require(digest == DIGEST, 'Archived SHA-256 mismatch')
    Path('low_inversion_certificate_100.json').write_bytes(data)
    return {'newton_coefficients': newton_total,
            'finite_checks_all': finite_all,
            'finite_checks_admissible': small_checks,
            'equality_pairs': sorted(zeros), 'sha256': digest}
```

```
def upper_certificate():
    D = LIMIT
    K = D+2
    rows = [[1]+[0]*K, [1]+[0]*K]
    checks = 0
    zeros = set()
    # Independently use the reversed recurrence, not reversed
    # coefficients from the lower-strip implementation.
    for n in range(2, D+4):
        a, b = rows[-1], rows[-2]
        row = []
        for d in range(K+1):
            shifted = a[d-n+1] if d >= n-1 else 0
            row.append(shifted + sum(b[d-2*j] for j in
                       range(min(n-2, d//2)+1)))
        rows.append(row)
        if n < 4:
            continue
        for p in (0, 1):
            support = [j for j, x in enumerate(row[p::2])
                       if x > 0]
            require(not support or support ==
                    list(range(support[0], support[-1]+1)),
                    f'Upper prefix support: {n},{p}')
        for d in range(2, D+1):
            defect = row[d]**2-row[d-2]*row[d+2]
            require(defect >= 0, f'Upper base: {n},{d}')
            checks += 1
            if n in (D+2, D+3):
                require(defect > 0, 'Terminal strictness')
            Nn = n*(n-1)//2
            if d <= Nn-2 and defect == 0:
                zeros.add((n, Nn-d))
    require(checks == 9900, 'Upper count')
    require(zeros == ZEROS, 'Upper equality classification')
    # Cross-check full forward rows through n=35.
    full = [[1], [1]]
    for n in range(2, 36):
        a = [0]*(n*(n-1)//2+1)
        a[:len(full[-1])] = full[-1]
        for shift in range(1, 2*n-2, 2):
            for k, x in enumerate(full[-2]):
                a[k+shift] += x
        full.append(a)
        reverse = a[::-1]
        for d in range(K+1):
            expected = reverse[d] if d < len(reverse) else 0
            require(rows[n][d] == expected,
                    f'Forward/reverse cross-check: {n},{d}')
    return {'base_n_through': D+3, 'inequalities': checks,
            'terminal_rows_strict': [D+2, D+3],
            'equality_pairs': sorted(zeros)}


def main():
    report = {'lower_strip': low_certificate(),
              'upper_strip': upper_certificate(),
              'status': 'PASS: both infinite width-100 edge strips',
              'scope': 'NOT the full Dukes conjecture; no linear-window threshold'}
    print(json.dumps(report, indent=2))
```

```
if __name__ == '__main__':
    main()
```

Expected checks and the role of computation

A successful run reports 10 197 nonnegative Newton coefficients, 4 950 lower finite checks (4 279 admissible), the stated digest, 9 900 upper finite inequalities, strict terminal rows 102, 103, and the same four admissible equality pairs in both strips. It prints a `PASS` only after every check succeeds. The fixed width 100 is deliberate: no larger strip, empirical Bernstein screen, four-step invariant, or numerical linear-window threshold is certified by this program. The linear-window theorem is proved analytically in Section 8.4 and does not depend on numerical samples.

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